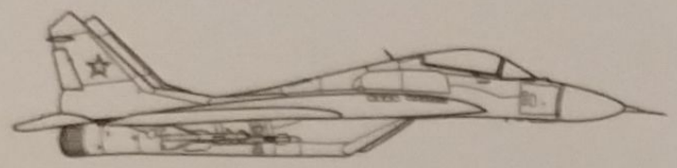
ELECTRONIC BATTLEFIELD SERIES®

MiG-29

DEADLY ADVERSARY OF FALCON® 3.0

Spectrum HoloByte

IBM Flight Manual



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MiG-29

DEADLY ADVERSARY OF FALCON® 3.0

FLIGHT MANUAL

Spectrum HoloByte

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You can post and read messages in the "Flight Simulators" topic on the Game Club bulletin board (located in the Game Center area). You can also send electronic mail to Customer Support at TKNJ33A.

If you are having problems with *MiG-29*, we can best help you if (1) you are at your computer when you call, and (2) you have the following information handy:

- Version number or serial number of *MiG-29*
- Your computer's brand and model
- Your computer's BIOS brand and version number
- Total RAM installed in your computer
- Name and version number of your operating system (MS-DOS or DR DOS)
- Name and version number of your memory manager (such as QEMM or 386MAX)
- Video card brand and model name
- Mouse brand and version number of mouse driver
- Sound card brand and model name
- Contents of AUTOEXEC.BAT and CONFIG.SYS files
- Type `MEM /C` at the DOS prompt and copy down the onscreen listing

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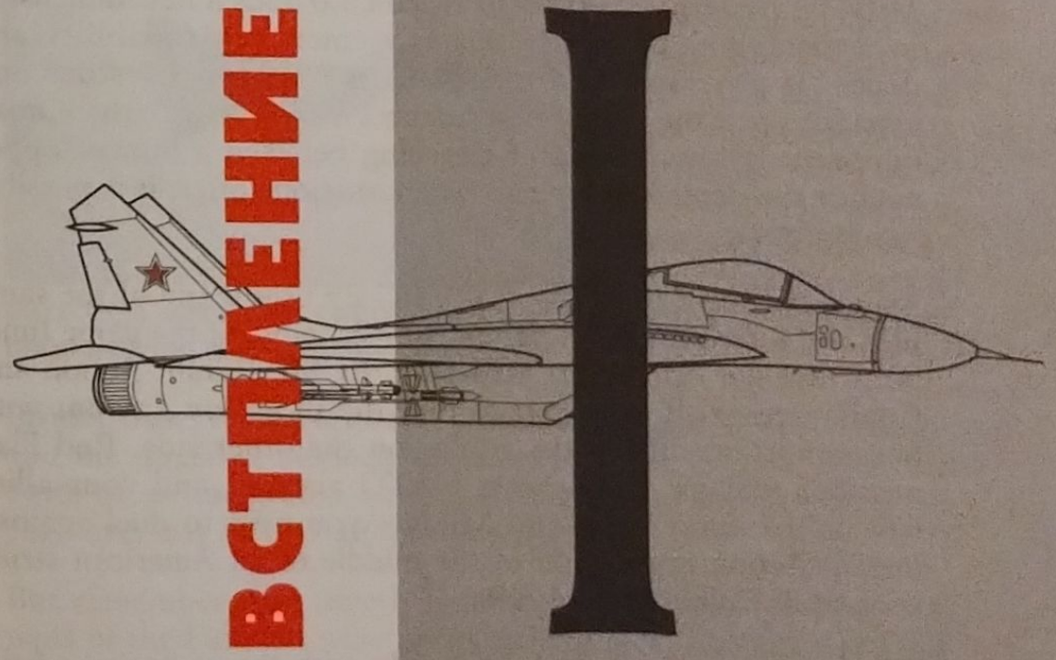
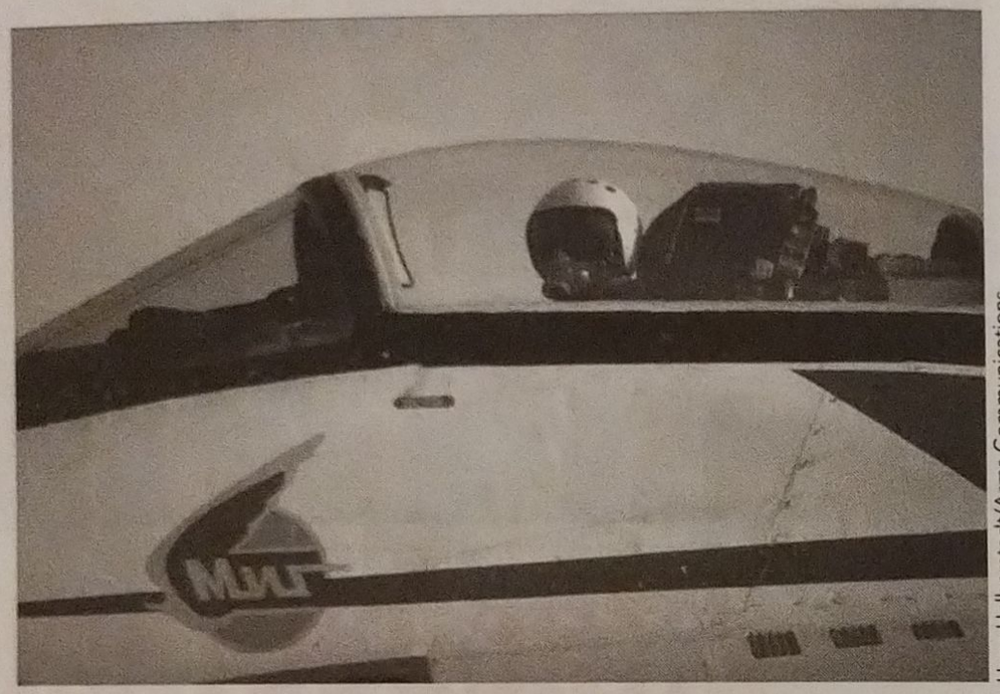
INTRODUCTION



ВСТУПЛЕНИЕ

I

ОДИН



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ABOUT THIS GAME

MiG-29: Deadly Adversary of Falcon 3.0 adds a new dimension to a game which already boasts tremendous flexibility and depth. It provides the flip side to the *Falcon 3.0* story and expands upon the one of best features of *Falcon 3.0*: the human opponent. Now, instead of choosing between a human opponent or the depth of play campaign missions offer, it is possible to have both.

As a supplement to *Falcon 3.0*, *MiG-29* uses many of the same files and interfaces as its parent. In fact, most of the game functions just like *Falcon 3.0*. Red Flag, ACMI, Instant Action and Configuration all work just as they did in *Falcon 3.0*, but with one important difference: you're on the other side. Red Flag missions will pit you against NATO aircraft, and your allies will be the same MiGs and Mirages you used to duel against. Instant Action begins you in the middle of an American strike wing of B-52s, A-10s and A-4s.

This book, therefore, deals primarily with the operation of the MiG-29, the nations and goals you will serve in the reversed campaigns, and the technical references necessary to train you to carry out your mission as a Fulcrum pilot.

Welcome!

Welcome to Frontal Aviation, my friend. This is the branch of the Russian Air Force where the best planes and pilots are assigned, because Frontal Aviation is always the first to enter battle. Here you will learn to fly and fight in the best fighter plane ever made: the MiG-29, which NATO calls the Fulcrum. With time, we have also adopted this name, but some of us call it simply "the 29th." It is the most agile fighter the Mikoyan-Gurevich design bureau (OKB MiG) has ever produced, and you will learn to love it as we do.

But before you begin to work with your new partner, it is proper to make introductions.

The Mikoyan-Gurevich design bureau was named after its founders, Artyom Mikoyan and Mikhail Gurevich. When Gurevich retired in 1964, Mikoyan promised that the bureau's aircraft would continue to bear the name MiG. Mikoyan's successors have kept that promise to this day.

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The 29th

You will notice the smooth lines of fuselage and wings, as well as the raised cockpit which will provide unprecedented visibility to you as a pilot. These graceful curves are part of a carefully engineered airframe capable of feats of maneuvering that will amaze you.

Inside, observe the infrared search-and-track (IRST) systems and the targeting monocle mounted on your helmet. These are parts of an integrated weapons system that will give you the edge over any NATO aircraft in a dogfight.

But remember, the superb handling and deadly weapons systems of the Fulcrum were no miracle of luck. Designers at OKB MiG worked for many years to create the combination of speed, agility and weapons delivery which make the MiG-29 one of the most capable fighters ever.

Design history

During the late 1960s and early 1970s, we began to see several new NATO designs. Among these were the F-14 "Tomcat," F-15 "Eagle," YF-16 (later named F-16 "Fighting Falcon"), YF-17 (later named F/A-18 "Hornet") and the MRCA (later called Tornado). The consensus among our intelligence services, design bureaus, government and military was that existing designs would not be able to compete with these new aircraft.

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Many designs of this period were heavily influenced by events in Vietnam, where maneuverable MiGs did quite well against gunless and comparatively clumsy F-4s. Maneuverability surpassed BVR capabilities on many designers' priority lists.

It was decided that two new types of aircraft were needed. One, a large fighter built for beyond-visual-range (BVR) interception, was given to the Sukhoi design bureau to design. The other, a smaller craft built for dogfighting maneuverability, was to be created by the experts at OKB MiG.

Sukhoi's design for a large and powerful aircraft eventually became the Su-27; Mikoyan-Gurevich's much more agile dogfighter became the formidable MiG-29.

Technical Assignment

In 1972 OKB MiG, in collaboration with the Air Force Commander-in-Chief and the Minister of the Aircraft Industry, issued the Technical Assignment for the MiG-29. This document outlined what the new plane would be. It specified which weapons, engines, avionics and other equipment would be used in the new aircraft. The Technical Assignment also set the goals for combat effectiveness, performance and other operational characteristics.

The Technical Assignment called for a Frontal Aviation fighter with a primary mission of air superiority and a secondary

ground-attack ability. The new aircraft would replace several older models still in use, including the MiG-21, MiG-23, Su-7, Su-15 and Su-17.

It is usual for an aircraft design to go through a prototype phase, during which it is tested and evaluated before it is ordered. Our need for this aircraft was so great, however, that a production order was placed at the same time as the Technical Assignment was issued—before the plane was even designed.

Design concerns

It was crucial to the Mikoyan-Gurevich designers that the new aircraft surpass its NATO opponents. As design work was beginning, the F-14, F-15, F-111 and Tornado had just entered service, and the prototypes for the F-16 and F/A-18 had flown their maiden flights. The new MiG had to be able to outfly the agile new fighters, intercept the new low-flying attack aircraft and outshoot the best NATO weapons systems.

The designers also had to meet the tough Mikoyan-Gurevich criteria of cost effectiveness, mission-capable rate, forward airfield use (where the airstrips are often short and rough), bad-weather operation, reliability and simplicity of use.

Research and development

The chief of OKB MiG, Rostislav Belyakov, and his deputy, Mikhail Waldenberg, took personal control of the MiG-29 project. They worked closely with the aerospace engineers at the Central Aero/Hydrodynamic Research Institute (TsAGI) to create an airframe which could outperform all existing fighter craft, especially at high angles of attack where most planes are in serious danger of stalling.

To achieve the performance required by the Technical Assignment, TsAGI gave OKB MiG many innovations to implement. Some of these included the characteristic twin tail fins, rectangular intakes under each wing root and a raised, bubble-canopied cockpit.

Similarities in appearance between the MiG-29 and Su-27 are largely due to the fact that both bureaus were working with the same information from TsAGI while designing the aircraft.

But nimbleness and visibility alone would not defeat such NATO dogfighting aircraft as the F-16. A definite edge in weapons systems was needed, so the designers took the best that

researchers had to offer. The weapons systems in the Fulcrum include infrared tracking, radar, laser range finding and a helmet-mounted sight in a dogfighting combination that NATO fighters cannot match.

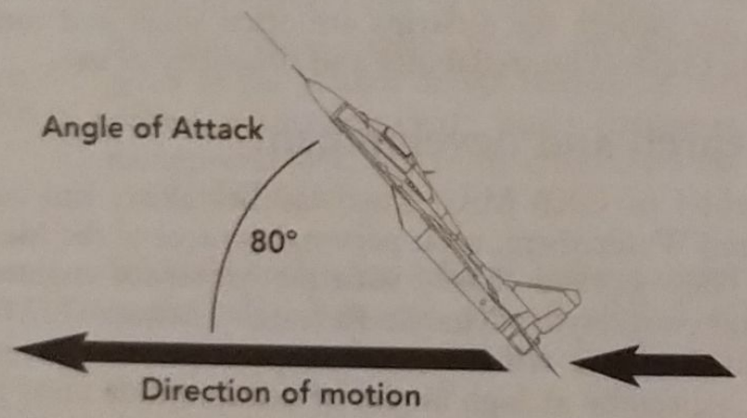
By the time the MiG-29 made its maiden flight in October 1977, Belyakov and Waldenberg had built a fighter with the best visibility, best performance and best weapons systems of any MiG ever made.

Performance

It is important for pilots to understand the advantages they will enjoy over their opponents. The MiG-29 has some unique abilities in both its aerodynamic qualities and its weapons systems.

Aerobatics

The Fulcrum is capable of some unusual maneuvers due to its stability at high angles of attack (AOA). It has been shown in test flights to be capable of remaining in flight at over 80° AOA!



Of course, such high AOA will not be necessary for normal flight operations. While the aircraft is capable of these maneuvers, it requires great pilot skill to perform them. But the option is there.

This is not always as simple to do as it sounds. Chapter 4 discusses proper control techniques for high-AOA situations.

One advantage high AOA gives in a dogfight is the ability to quickly point the nose (and thus the weapons) of the plane directly at your opponent. After weapons release, simply centering the stick should restore the aircraft to 0° AOA and full aileron control. As a result, the MiG-29 can point, shoot, and be back on its way before its opponent can even come close to a firing solution.

Another practical application of this high-AOA capability is the maneuver known as "Pugachev's Cobra." This is a maneuver in which the Fulcrum pilot pulls the nose of his plane so far back that it is actually beyond perpendicular to the direction of travel. This causes the plane to slow drastically and even stop all forward motion.

The loss of airspeed is so rapid that opponents will often overshoot, caught by surprise. The MiG-29 pilot then simply noses forward and opens the throttle to recover completely from what would be an unrecoverable stall in any other aircraft.

Weapons systems

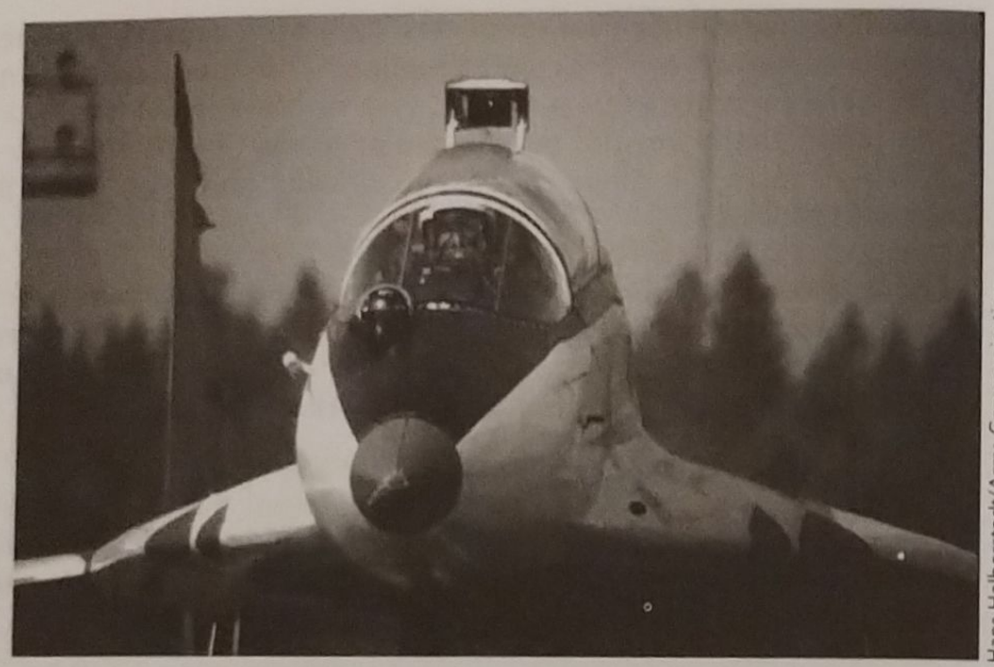
The great maneuverability of this aircraft is just part of its lethality. Integrated and versatile weapons systems give the Fulcrum a powerful bite, both in stealth and in accuracy.

The MiG-29 uses an infrared tracking system (IRST) that can detect and lock on to a target without using radar. This ability to "passively" acquire a target is a great benefit, as the target will have no warning that it has been locked up.

Most targeting systems, especially those used by NATO and its allies, rely on active radar, where a radar signal is bounced off the target to determine its position and motion. Threat warning systems detect these radar signals and warn the target of lock-on. Our IRST system gives no such warning. Since it does not broadcast, there is no way for the target to detect a lock-on.

As a pilot, you may have noticed with some concern that your plane only carries 150 rounds. In fact, the gun is so accurate that Mikhail Waldenberg has said that he would feel comfortable reducing the ammunition store by half! The laser range finder, which is housed with the IRST in a bubble on the nose of the plane, makes all the weapons systems more accurate—especially the gun. When the gun is set to automatic fire, five 30mm shells are fired in a quick burst the instant a target is in the bullet path.

Chapter 3 contains training lessons for all available MiG-29 weaponry.



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USING THIS MANUAL

This flight manual will guide you through the training necessary to prepare you for battle in the cockpit of a MiG-29. Even after you master the plane, this book will continue to be a valuable tactical and technical reference for you. Each section has its purpose.

Chapter 2 - Cockpit

Here you will find complete instruction on the instruments and controls of the MiG-29 cockpit. While this section does go step by step through all the aircraft's systems and explains each in detail, the best way to use it is to skim through, paying closer attention to the parts relating to the controls and weapons. You can refer to the more detailed portions later when you need more information.

Chapter 3 - Weapons training

After you understand how to fly, it is then time to learn to fight. You will be guided through training missions which will teach you how to use each of the weapons systems available to the MiG-29.

Chapter 4 - Tactics

Once you understand the basics of flying and fighting, you are ready to learn the strategies and tactics involved in modern fighter combat. American fighter pilot Pete Bonanni will go over some useful single-ship and two-ship tactics, after which we will discuss other important air-to-air and air-to-ground tactics and techniques.

Chapter 5 - Communications

Before you can test your mettle against real pilots, you will need proper instruction in setting up communications between your simulator and theirs. Chapter 5 contains important information and directions for arranging such communications.

Chapter 6 - Theater briefings

This chapter contains a briefing on each of the possible theaters you may be dispatched to (either as part of Frontal Aviation or as a military adviser).

Chapter 7 - Reference

You will find complete statistics on your aircraft, its weapons and enemy vehicles in this section, along with a complete glossary and bibliography.

But enough of this talk. The cockpit awaits!

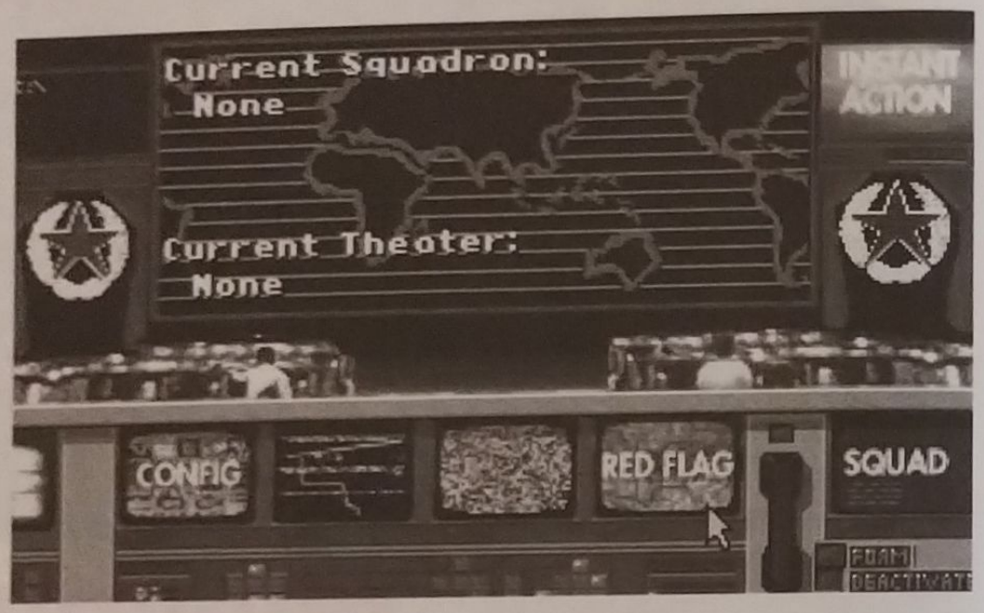
Hans Halberstadt/Arms Communications

COCKPIT

КАБИНА

2

ДВА



THE INSTANT ACTION SIMULATOR

The best place to familiarize yourself with the controls is in the Instant Action simulator. You can enter the Instant Action simulator from the War Room by clicking on the Instant Action monitor over the doorway on the right side of the screen.

You will immediately be placed in the cockpit of a MiG-29 in an extremely hostile environment. Your weapons system will already be locked on to a target, you'll see several aircraft in front of you, and below you are quite a few very active ground units.

Your mission here is a simple one: given unlimited fuel and weapons, prove how good a pilot you are by destroying as many targets, air and ground, as you can before you are destroyed. Instant Action is an excellent place to practice your skills and to become familiar with the aircraft.

CONTROLS

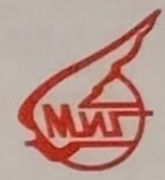
The MiG-29's cockpit and controls were designed to be as familiar as possible to pilots of other planes in the MiG series to minimize retraining. Similarly, the controls and keyboard commands for *MiG-29* share much in common with its parent, *Falcon 3.0*.

KEYBOARD COMMANDS

The following keyboard commands are specific to *MiG-29*.

Bombing pipper airspeed setting

The bombing pipper is set for a very specific bombing profile. It is only accurate if you are at 2,000 ft and diving at a 30° angle. The speed at which it is accurate is adjustable, however. The bombing pipper can be set for airspeeds of 300 to 550 kts, and is initially set for 400 kts. Each time you press [D], the bombing pipper speed is decreased by 50 kts. When it is set for the minimum setting of 300 kts, pressing [D] will jump the setting up to the maximum of 550 kts.



Bombing techniques for the MiG-29 are given in Chapter 3.

Flap system lock-out

The flaps on the MiG-29 are automated and need not be manually actuated (not even for landing). Press [F] if you wish to lock out the automated flap system (which retracts the flaps to the "up" position and locks them there). Pressing [F] again restores the flaps to automatic control.

Automatic cannon

The GSh-301 cannon can be linked to the targeting system and set to automatically fire as soon as a target is in the bullet path. Press [K] to enable or disable the automatic cannon.

Landing drogue chute

The Fulcrum carries a drogue chute for landing. Using the drogue chute, the MiG-29 can land on a much shorter runway than a plane not so equipped. Pressing [O] (the letter "O")opens the drogue chute, and [Shift] [O] disconnects it from the plane. The drogue chute can only be deployed if the plane is traveling 170 kts or below, under 25 ft above ground level (AGL) and with the landing gear down. If 200 kts is exceeded with the chute open, the release mechanism may fail, making it impossible to release the drogue chute.

Trim tab controls

The trim tabs are controlled by the "coolie hat" on a ThrustMaster FCS and CH FlightStick Pro, by the arrow keys

or by [8], [4], [2] and [6] on the numeric keypad if you selected the joystick as the control device. If the keyboard is the selected control device, use [Ctrl] with [8], [4], [2] and [6] on the numeric keypad.

ThrustMaster FCS and CH FlightStick Pro coolie hat toggle

The trim controls are diagrammed later in this chapter. Trimming is explained in detail in Chapter 4.

The "coolie hat" on the ThrustMaster FCS and CH FlightStick Pro normally controls the trim tabs. If you wish, the coolie hat can be switched to controlling the views instead. Pressing [V] toggles the coolie hat between trim and view control.

With the hat in view control mode, forward activates the Front View, left activates the Left View, right activates the Right View and back activates the Padlock View. If you are already in Front View, pushing the hat forward shifts the view upward (just like pressing [Scroll Lock]). Pressing forward again shifts the view back down to the instrument panel. See "Flight Controls" below for details.

HUD airspeed toggle

The airspeed scale on the MiG-29 initially shows true airspeed rather than indicated airspeed. Pressing [Ctrl] [A] toggles the display between true and indicated airspeed.

SCN and IRST radar modes

The SCN radar mode, activated by pressing [F5], is similar to *Falcon 3.0's* NAM mode. The IRST radar mode, activated by pressing [F6], is similar to STT mode in *Falcon 3.0*. The main difference between the IRST and STT modes is that the IRST can track F-117As and helicopters by heat, although this infrared tracking ability is limited to a range of 10 miles. All new radar modes are explained in detail later in this chapter.

Helmet mounted system

The helmet tracking device, also called the monocle, can be activated for use in the Padlock View. Pressing [F7] activates and deactivates the helmet tracking device. The monocle, when in use, slaves the targeting system to the pilot's helmet. Whatever target the pilot is looking at is automatically tracked by the weapons system whenever possible.

FLIGHT CONTROLS

The plane can be flown using the keyboard, but a joystick is preferable. Throttle controls and rudder pedals are also supported. If you are having difficulty with the handling of your MiG-29, you may want to switch to a simpler flight model (see note below). Chapter 4 contains a discussion of aerodynamics and control techniques for the Difficult and Realistic flight models.

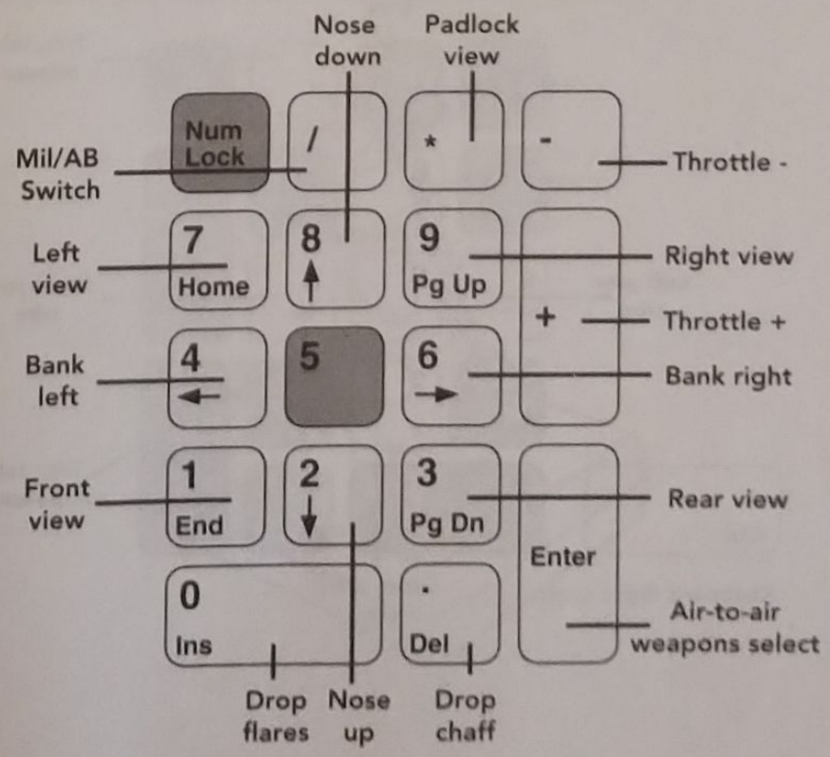
One important thing to note about controlling the MiG-29 is that it is not a fly-by-wire (FBW) aircraft, which means it's going to be a little more work to fly than an F-16 or other similar FBW plane. When using the Realistic flight model, for example, you must manually adjust the trim control (trim controls are discussed in Chapter 4). The trim controls are shown below, along with the other flight controls.

There are four flight models, from Easy to Realistic. Each is more complex and requires greater skill to fly. To change flight models, go to the Difficulty Levels screen by clicking on the Config monitor in the War Room. See the *Falcon 3.0* manual for details.

Keyboard flight controls

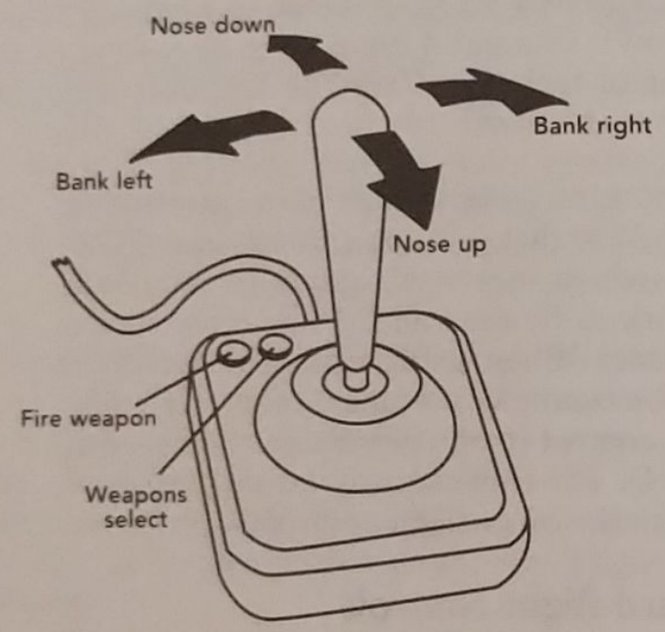
If you are flying with the keyboard, all essential controls have been concentrated on the numeric keypad.

You can also use [Spacebar] as the weapons trigger. Refer to your keyboard chart for other important keys.

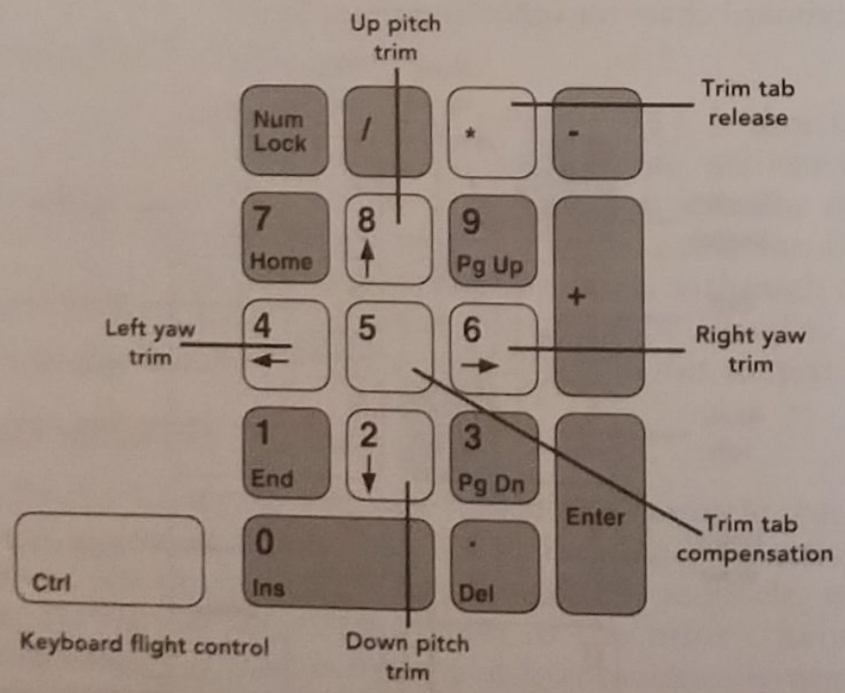


JOYSTICK

On a standard joystick, the stick controls pitch and roll while the first button (button A) acts as a weapons trigger. A second button (button B), if there is one, acts as a weapons select button.

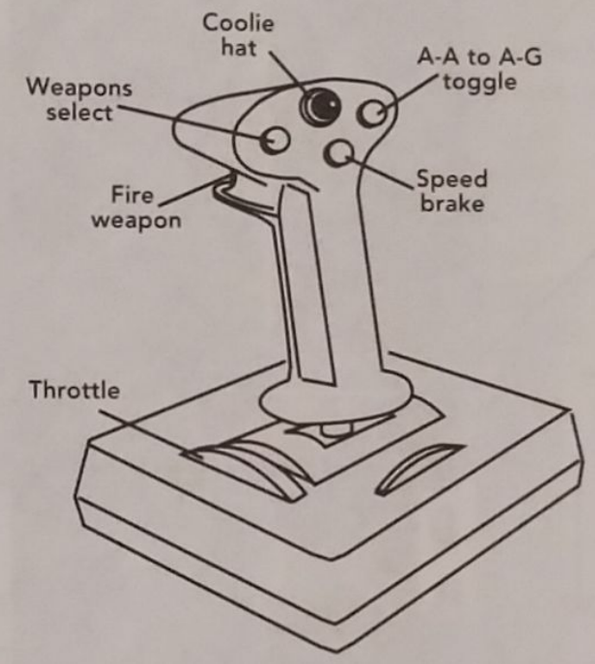


If you have selected the joystick as your control device on the System Setup screen or from the CONTROL menu in the simulation, the trim controls are as follows:



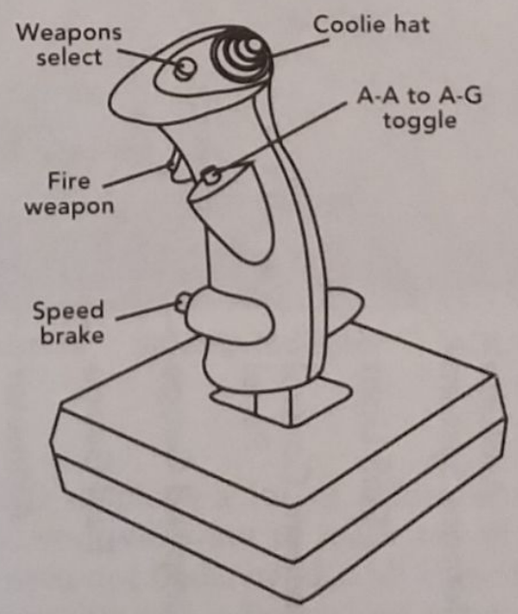
CH FlightStick Pro

MiG-29 and *Falcon 3.0* (if you have installed *MiG-29*) support CH Products' FlightStick Pro. On the System Setup screen, you can now choose the FlightStick Pro as your control device. If you are using a FlightStick Pro, the stick still controls pitch and roll, but it adds many other functions as well:



ThrustMaster FCS

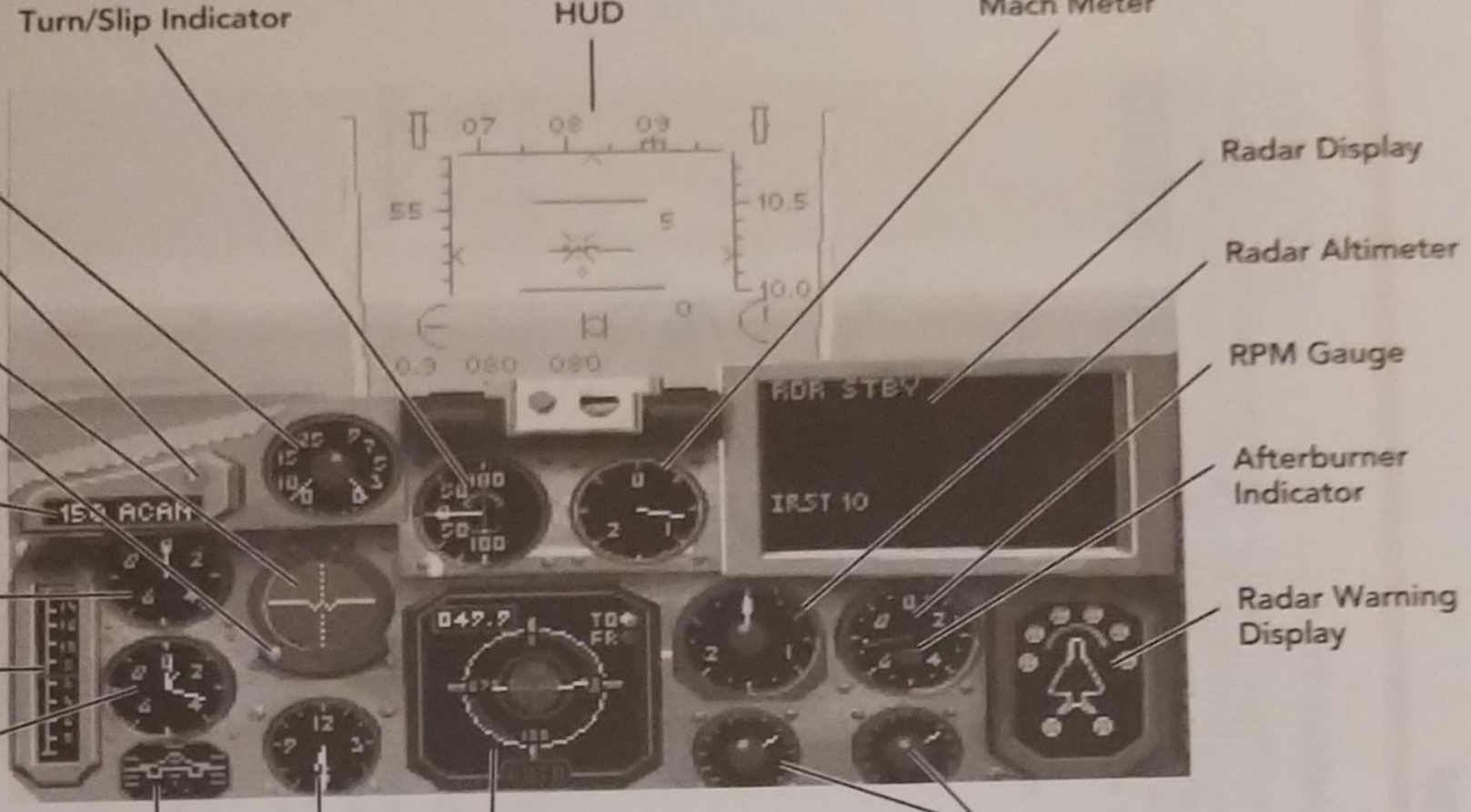
The ThrustMaster Flight Control System (FCS) controls pitch and roll, and performs several additonal functions:



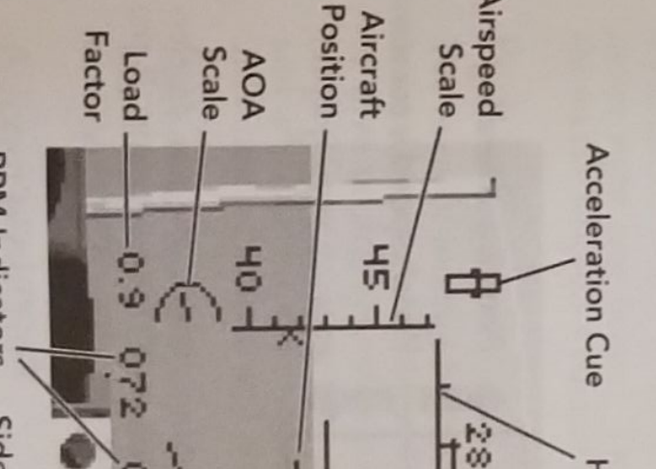
COCKPIT

Press P to pause, returning to the action by pressing P again. Now get out the keyboard chart and have it handy while we have a look at your cockpit.

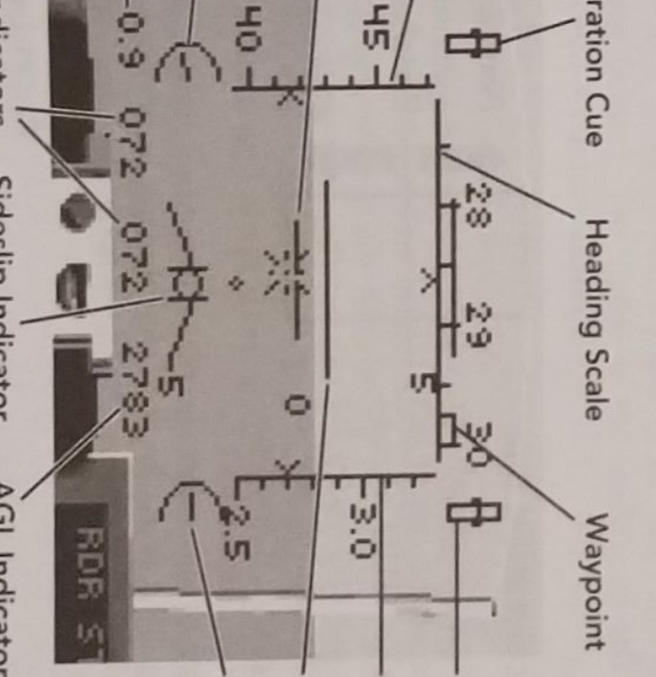
The cockpit of the MiG-29 can be divided into four major sections: head up display (HUD), forward instrument panel, left view and right view. Right now the HUD and forward instrument panel should be visible.



Hans Halberstadt/Arms Communications



You can also pause the simulation by pressing Esc, which brings up the menu bar. Just press Esc again or select Return to Game from the FILE menu to unpause.



HUD

The MiG-29 HUD displays a lot of information in a small space. Eventually, you will want to make use of all this information, but you need not understand it all immediately. As you fly, pay attention to just one indicator and watch how it reacts

to your maneuvers. When you feel comfortable with how that indicator works, pause the simulation while you read about the next one. Then resume play and observe how it behaves, and so on until you've explored the entire HUD.

Heading scale

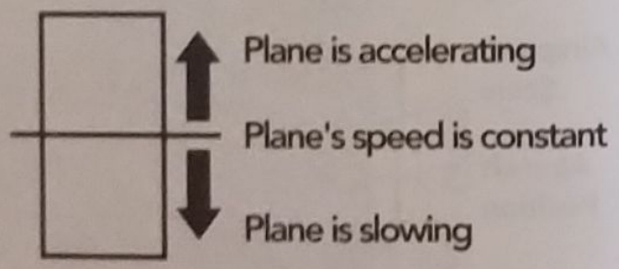
The horizontal scale at the top of the HUD is the heading scale, marked in tens of degrees. The center mark shows your current heading. As you turn, you should see the scale slide left or right to reflect the change in your heading.

Waypoint heading

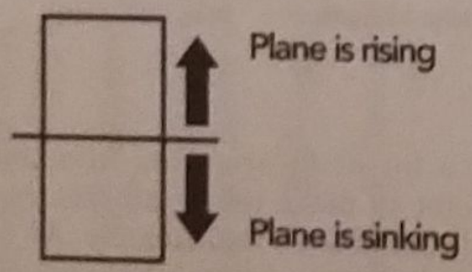
The heading to the current waypoint is marked by a box on the heading scale at the top of the HUD. If the box is centered, the plane is headed directly towards its waypoint. Otherwise, the box shows which way to turn to get back on course. As you turn toward the box, eventually it will slide to the center of the heading scale.

In Instant Action, you have no mission plan to follow, so there are no waypoints. As a result, you will not see a waypoint heading marker in Instant Action.

Acceleration cue



A small vertical bar scale at the upper left of the HUD shows the acceleration or deceleration rate of the plane. If the horizontal line is at the center of the scale, the plane is neither accelerating nor decelerating. If it is at the top, the plane is accelerating greatly. If the line is at the bottom, the plane is decelerating rapidly.



Ascent cue

In the upper right corner of the HUD is a small vertical bar scale, similar to the acceleration cue, showing the relative acceleration of the plane in its vertical

axis. If the horizontal line is centered, the plane is neither rising nor falling relative to its velocity vector. If the line is at the top of the scale, the plane is rising at a rapid rate relative to its velocity vector. If the line is at the bottom, the plane is descending quickly relative to its velocity vector.

Airspeed scale

A scale on the left-hand side shows the plane's current airspeed in tens of knots. This scale initially shows true airspeed. With the airspeed scale set for indicated airspeed (press Ctrl A to switch between indicated and true airspeed), keep the center mark on this scale between 30 and 50 (300 and 500 kts) for maximum maneuverability.

If your speed gets too low, your plane may stall. The best way out of a stall is to open the throttle and bring the nose down. See "AOA Scale" and "Stall Light" below for more details.

Altitude scale

On the right side of the HUD is the altitude scale, showing altitude above sea level in thousands of feet. Note that this is not the altitude above ground level (AGL)—that is displayed by the AGL indicator in the lower right (see below). The mark at the middle of the scale shows the MiG-29's current altitude.

Pitch ladder

The rows of moving horizontal lines in the middle of the HUD make up the pitch ladder. The ladder shows the angle of climb or dive in degrees, with each line being 5° apart. Chevrons (angled lines pointing upward) are used to show a diving angle and are marked with negative numbers (-5, etc.) The straight lines indicate a climb and are marked with positive numbers.

The heavier line at 0° shows the position of the horizon relative to your plane. As you bank the plane, the pitch ladder will tilt to reflect the position of the horizon.

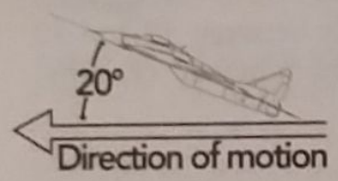
Aircraft position

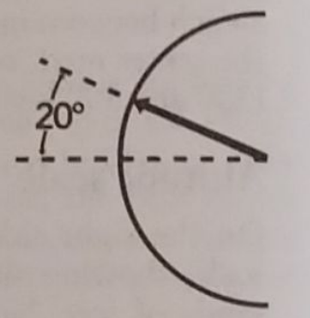
The stationary horizontal lines at the center of the HUD show the position of your plane relative to the horizon lines on the pitch ladder. This is mainly for visual reference and for targeting. When a target is between the horizontal lines, the plane is pointing right at it.

AOA scale

A dial-and-needle scale in the lower left of the HUD shows the plane's current angle of attack (AOA). As the plane's AOA increases, the needle points upward. As the plane's AOA decreases, the needle dips. At 0° AOA, the needle is level at the center of the dial. The needle's position exactly corresponds to the plane's AOA. If the needle is 20° above horizontal, the plane is at 20° AOA.

The surest warning of a possible stall is high AOA. If your AOA gets above 40°, you may stall soon. The MiG-29 has excellent stall handling, however, so recovery is not difficult. See "Stall Light" below for more details.





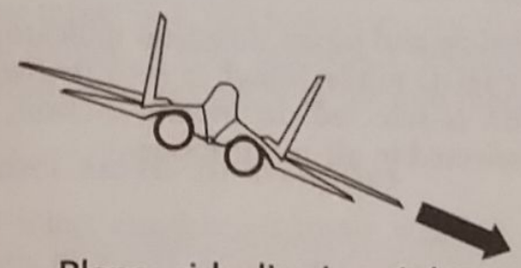
AOA Scale

Vertical velocity scale

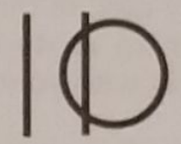
A dial-and-needle scale in the lower right of the HUD indicates the plane's current vertical velocity. If the needle is centered, the plane's vertical velocity is 0 kts. If the needle is above center, the plane is climbing. If the needle points below center, the plane is falling. The farther the needle is from horizontal, the faster the plane is climbing or diving.

Sideslip indicator

Just below the pitch ladder and above the RPM indicators at the bottom center of the HUD is the sideslip indicator, which consists of a circle and two vertical bars. As the plane sideslips to the left or right, the circle (normally centered between the two bars) slides right or left as well. The faster the plane sideslips, the farther off center the circle will be.



Plane sideslipping right



Sideslip indicator

Load factor

The number displayed in the lower left of the HUD is the plane's current load factor. This is the number of Gs the plane is pulling at the moment. Blackout usually occurs around 9 Gs. Redout usually sets in at -3 Gs.

Blackout is a condition caused by excessive G-forces. The pilot passes out as G-forces pull blood down and away from the brain. Redout is the opposite, where negative G-forces increase blood flow to the head until the pilot cannot see and passes out. Chapter 4 discusses the effects of G-forces on the pilot in detail.

RPM indicators

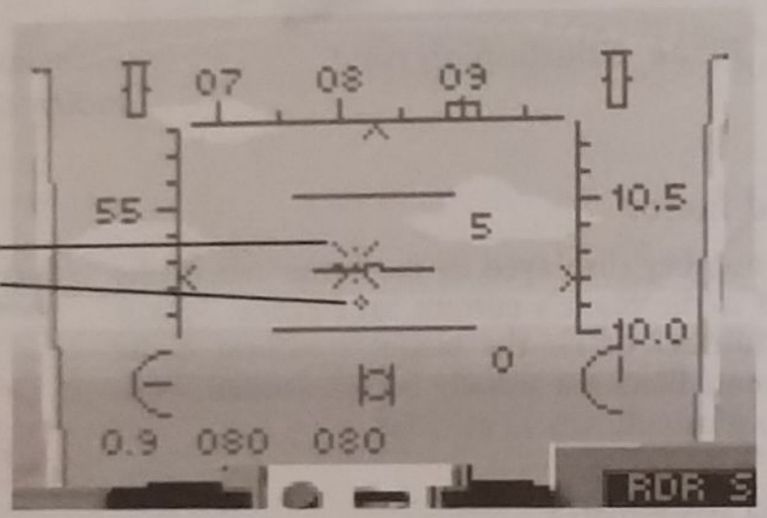
The two numbers in the bottom center of the HUD show the current RPM of each engine as a percentage of maximum power. If the numbers read "066 066," both engines are running at about two-thirds power.

Altitude (AGL) indicator

When you get below 5,000 feet from the ground, a number will appear in the lower right corner of the HUD showing the plane's current altitude above ground level (AGL) in feet (as opposed to the altitude above sea level, which is shown on the altitude scale). This is actually more important than the altitude scale, since it shows how close you are to crashing into the dirt.

THE HUD IN AIR-TO-AIR COMBAT

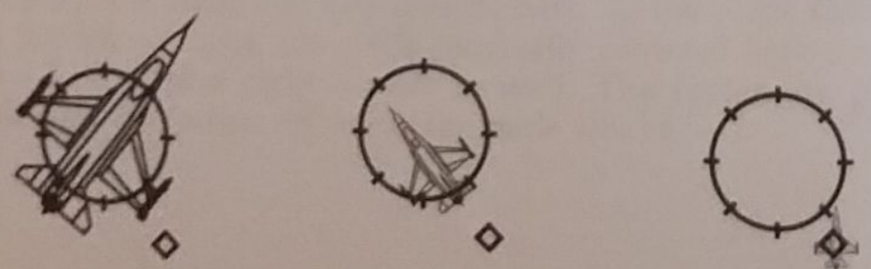
The gun pipper, missile reticle and target direction indicator are only used in the air-to-air HUD mode, which is activated whenever an air-to-air weapon is selected. In Instant Action, you begin with your cannon selected in air-to-air mode.



Gun pipper

The gun pipper looks like a star followed by a tiny diamond. The markers around the outside are scaled for firing solutions at 1,000 and 1,500 feet. This means that if your target is within the inside edge of the pipper's marks and is 1,500 feet away, you have an excellent chance of hitting if you fire. If your target is 1,000 feet away and within the outside edges of the pipper, your chances of hitting are also very good. The trailing diamond shows the bullet path to 2,000 feet.

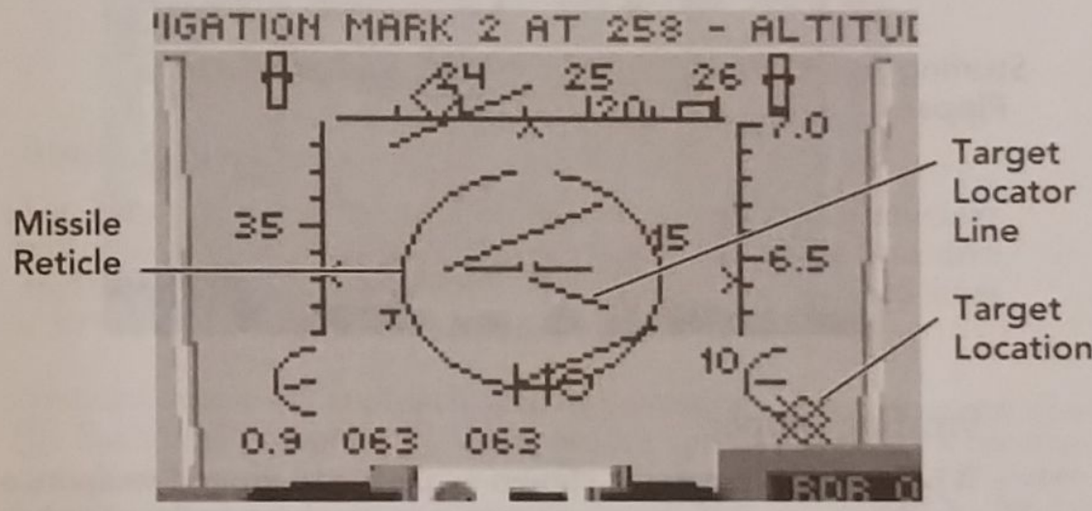
The MiG-29's cannon can be set to automatically fire a burst of five rounds when the locked target is in an optimal firing position. Press [K] to turn the automatic cannon on and off. See "Armament Indicator" below.



If your target is 2,000 feet away, place it under the diamond. If your target is 1,500 feet away, try to place it in the center of the gun pipper. If your target is 1,000 feet away, maneuver to get it within the outer edges of the gun pipper.

Other air-to-air weapons

To select another air-to-air weapon (such as any of the three kinds of air-to-air missile), press [Enter] or button B on your joystick. Your HUD should now look something like this:



Missile reticle

The circle at the center of the HUD is the missile reticle. When your target gets inside this circle, your missile can lock on. When your missile is locked on, the word ">>LOCK<<" will flash across the middle of your HUD. Only fire missiles when they are locked on; you waste them otherwise.

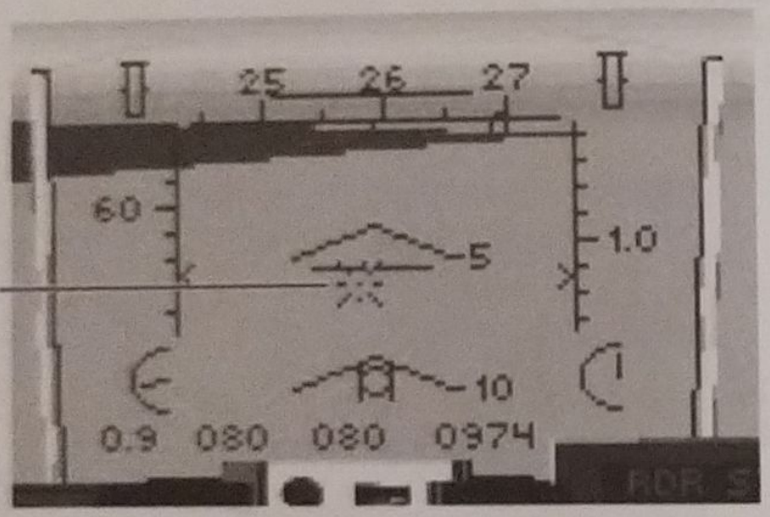
Target direction indicator

Whenever the locked target leaves the HUD, a line will extend from the center of the HUD and point toward the target. This is the target direction indicator line, and it should help you when you're chasing down a target.

The three kinds of air-to-air missile operate very differently. For beginners, the easiest to use is the R-27R. Press [Enter] or button B on your joystick until you see "R27R" displayed on the armament control panel (see below). Chapter 3 has training missions for each type of missile.

AIR-TO-GROUND HUD INFORMATION

The MiG-29 HUD has a different mode for each of four different air-to-ground weapons: the 30mm cannon, bombs, unguided rockets and the Kh-23 air-to-ground missile. Press Backspace to cycle through the various air-to-ground weapons.



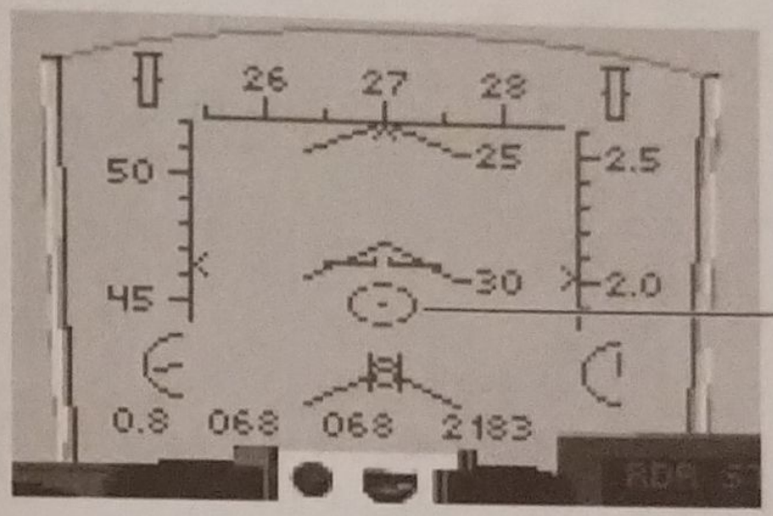
Strafing pipper

Whenever the cannon is chosen as the air-to-ground weapon, a pipper appears, showing approximately where bullets fired at that instant will strike. The pipper moves to adjust for the plane's motion. Place the pipper over your target and fire short bursts.

Rocket pipper

The aiming reticle for unguided rockets looks just like the gun pipper, except it does not move. It is a stationary aim point at the center of the HUD used primarily for visual reference. You will have to judge speed and lead your targets accordingly. Remember that the unguided rockets do not fly in a perfectly straight line; they fall off just as bullets do. Be sure to allow for this effect.

If you are having trouble hitting your targets with the rockets (or any other weapon), try adjusting the weapon effectiveness. To change weapon effectiveness, select the Config monitor in the War Room and go to the Difficulty Levels screen. See the Falcon 3.0 manual for details.



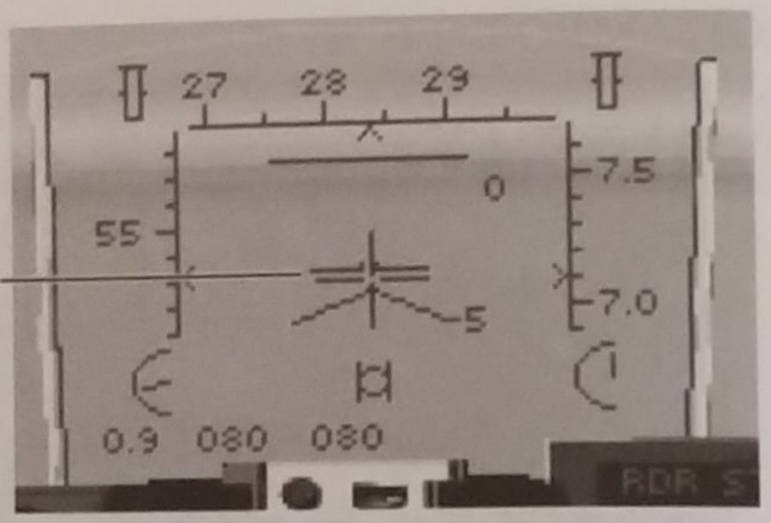
Bomb pipper

The bomb pipper shows where a bomb released at that moment will strike—if you are at the proper speed, altitude and dive angle. The pipper shows where a bomb will hit if released from a 30° dive at 400 kts at an altitude of 2,000 ft.

You can adjust the approach speed by pressing D. Each time you press D, the approach speed is reduced by 50 kts; if you press D when the approach speed is at the minimum of 300 knots, it is set to 550 knots (the maximum value). The pipper will move to reflect the new approach speed.

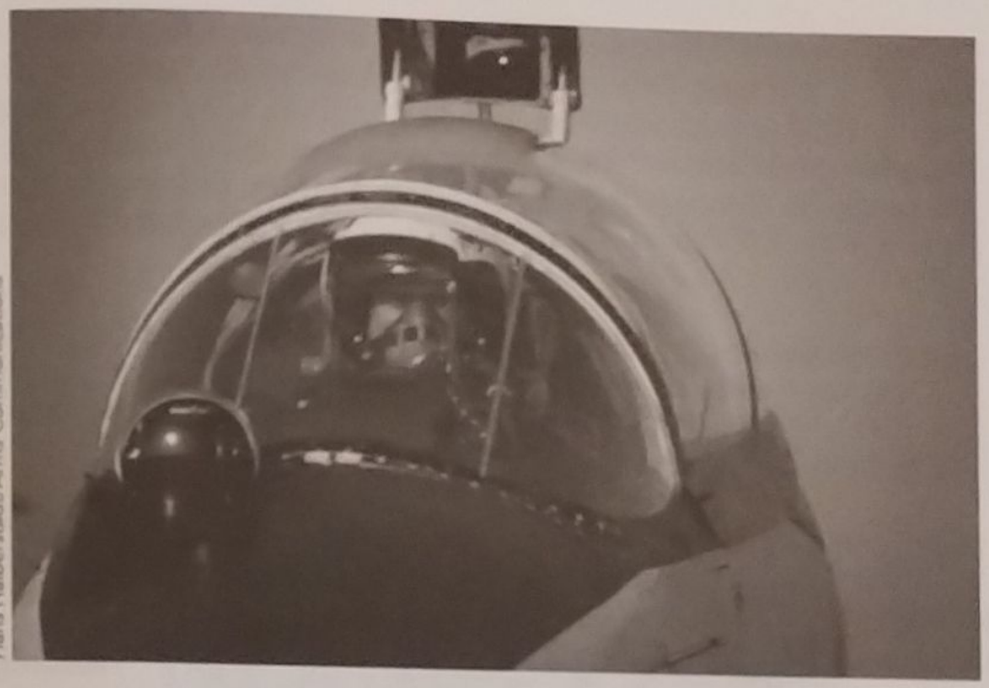
No matter what speed you set the bombing pipper for, it is only accurate at 2,000 ft and a 30° dive angle. These parameters are fixed and cannot be changed.

Once you are within the parameters set for the pipper, place the pipper on the target and release the ordnance. If, for example, you expect to approach your target at 350 kts, press D once. If you match your parameters on your approach, your bombs will hit (if you release while the pipper is on the target).



Missile cross hairs

The missile cross hairs (a large plus sign in the center of the HUD) come up whenever the Kh-23 is selected. This air-to-ground missile requires that the target be illuminated all the way until impact. Put the target in the cross hairs, lock on, fire the missile and keep the target in the cross hairs until the missile strikes. There is no need to keep the target centered. It only has to stay locked, so just keep it within your radar's scan area. A good rule of thumb is to keep the target in the HUD until impact.



Hans Halberstadt/Arms Communications

RADAR

The radar screen on the upper right of the instrument panel displays combined targeting information from the radar, infrared search-and-track system (IRST) and laser range finder.

In the Difficulty Levels screen (accessible from the War Room by clicking on the Config monitor), you can choose from three radar models: SAD (situational awareness display), SWT (scan-while-track) and HFR (high fidelity radar).

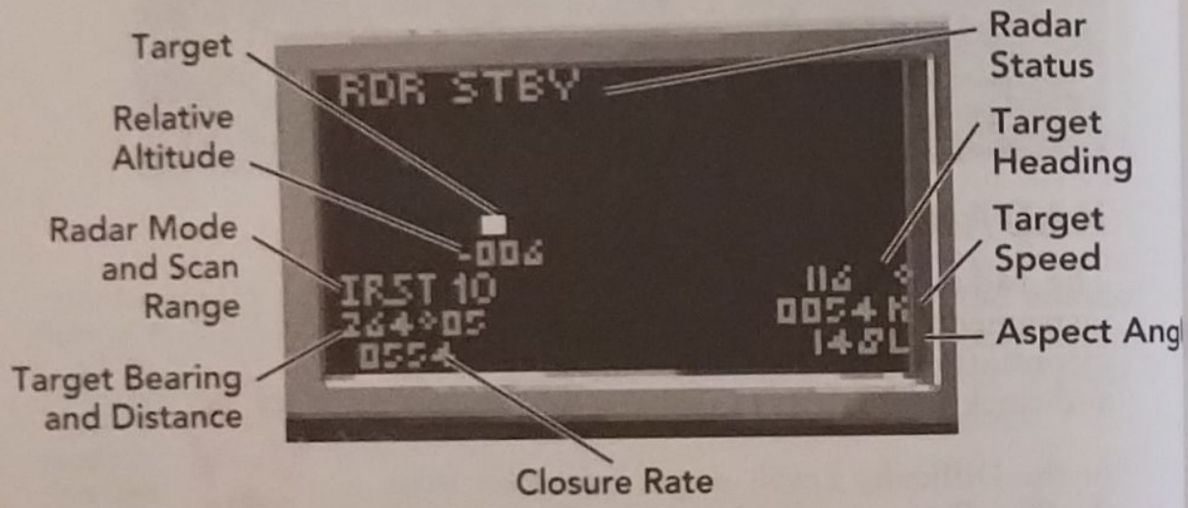
SAD is a simplified radar model which shows all enemies at all times, just as it does in *Falcon 3.0*. It also includes the IRST dogfighting mode (see below). When using the SAD radar model, press F5 to switch to SAD mode. Press F6 to switch back to IRST mode.

The SWT radar model operates like the SWT mode described below, with two main differences: it detects all targets in front of you and you can cycle through the available targets by pressing T. It also includes the IRST dogfighting mode (see below). Press F5 to switch to SWT radar mode; press F6 to switch back to IRST mode.

The IRST is an infrared search-and-track system on the MiG-29. It tracks targets by heat and is, therefore, undetectable by threat warning systems (which can only sense radar emissions). If you turn off your radar (press R) and stay in IRST mode, your opponents will never be aware that you have a lock-on.

If you select HFR as your radar model, there are four different radar modes available when in the simulation: scan (SCN), track (TRK), scan-while-track (SWT) and the IRST dogfighting mode (see below). Press [F5] to switch to SCN radar mode. Once in SCN mode, press [T] to select a target, then press [Z] to change to SWT or TRK modes. Press [F6] to switch back to IRST mode.

In all modes except SAD, your plane can be thought of as being at the very bottom of the radar screen. For example, if a target is displayed as being 16 miles from the bottom of the radar screen, it is 16 miles in front of you.



IRST mode

If you have chosen the HFR model, the radar defaults to the IRST mode. If you are in another mode and want to switch back to IRST mode, press [F5]. In this mode, the plane's targeting systems track only one target at a time. The advantage to this is that you receive constantly updated information, which can be quite useful when you are trying to get the upper hand in a close dogfight.

In IRST mode, the radar is kept on standby for as long as possible to minimize the target's chances of detecting you. The IRST, which is not detectable, is used to track the target. The radar is switched on only when it is necessary to maintain the lock (if the target goes behind a cloud bank, for example).

Since the IRST tracks by heat, it can pick up some targets that radar cannot. For example, the F-117A and all helicopters can be detected and targeted by the IRST. One drawback is that the IRST has only a 10-mile range.

The **radar status** is shown in the upper left. "RDR STBY" means the radar is ready, but is not emitting (the IRST is tracking the target). "RDR OFF" means the radar has been turned off manually and cannot emit (press [R] to switch the radar on or off). When the radar is on and emitting, the notation will read "RDR ON."

In IRST mode, you get quite a lot of information about the locked target. This information is displayed under the target blip and in the bottom two corners of the radar screen. The scan-while-track (SWT) and track (TRK) modes, which we will look at shortly, provide this information in the same manner.

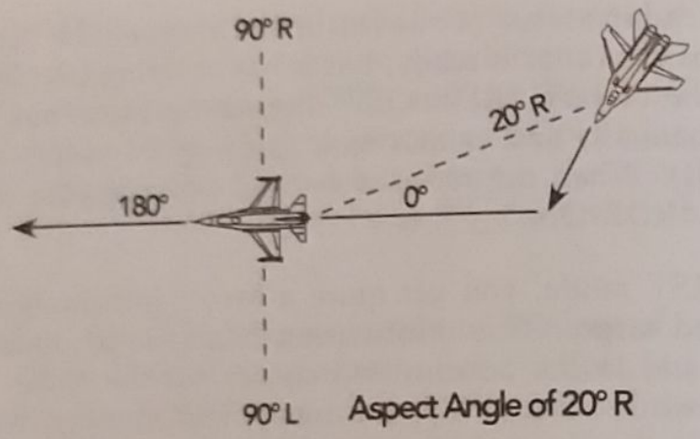
The **target** appears as a square blip, with its **altitude** relative to you printed right below it (in thousands of feet). A negative number means the target is below you, while a positive number means the target is above you.

In the lower left are three notations. The first line is the **radar mode** and **scan range.** The scan range is the distance from the bottom of the screen to the top in miles. For example, if it says "IRST 10," this means that the radar is in IRST mode and that the distance from the bottom of the radar screen to the top is 10 miles. If the blip was halfway down the screen, your target would be five miles away.

The second line shows the **target bearing and distance.** If it read "132° 06," the target would be at a bearing of 132° and six miles away. When the target gets within a mile, the distance is shown in tenths of miles instead. The last number at the bottom left is the **closure rate** in knots. This is how fast you are approaching the target. A negative number means the target is getting farther away. A reading of "0195" means you are approaching your target at a rate of 195 kts. If it reads "-0250," your target is separating from you at a rate of 250 kts.

When a heading is given in degrees, it refers to a compass direction. 0° is north, 90° is east, 180° is south and 270° is west.

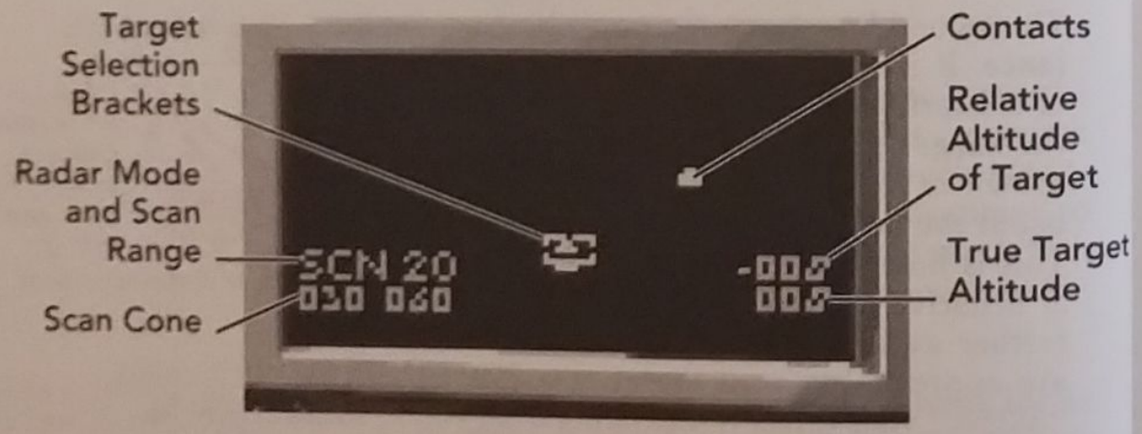
On the lower right are three other indicators. The top line is the **target heading** and the second is the **target speed.** If it says "270°" and "350K," for instance, the target is going 350 kts due west.



Aspect Angle of 20° R

The last indicator on the lower right shows the **aspect angle**. This gives you an idea of your positional relationship to the target. The aspect angle ranges from 0° to 180° and is followed by an "R" or "L." The degree reading shows your relationship to the target's flight path, where 180 is directly in front of the target and 0 is directly behind it. The letter tells you which side you are approaching the target from (right or left).

For example, if the aspect angle is 45° R, your position is diagonally behind the target and to its right. If the aspect angle is 135° L, you are in front of the target and to its left. The idea is to stay behind your opponent, so keep the aspect angle as low as possible.



SCN mode

To switch to scan (SCN) mode, press [F5]. In scan mode, your radar sweeps a wide area instead of focusing on one target as it does in IRST mode. As a result, it can detect many targets, but SCN mode does not track individual contacts as well as the IRST mode.

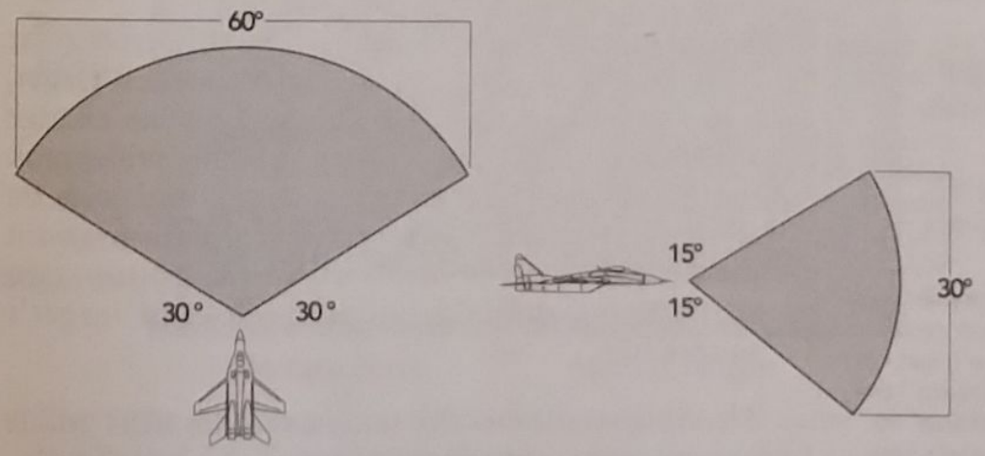
Because SCN mode also depends on active radar broadcasts to detect targets, enemies will detect you on their threat warning indicators and other radar detection gear if you switch to SCN mode.

SCN mode is best used intermittently to check for contacts at long range. A good technique is to set the radar to SCN (press [F5]), turn the radar on (press [R]), note the position and altitude of targets, and then turn the radar off (press [R] again).

All radar **contacts** appear as roughly triangular blips on the display. To get information about a particular target, press [T] until the **target selection brackets** appear on the target you want. In the lower right, you will see two numbers. The top one is the target's **altitude** relative to you. The bottom one is its **true altitude**. These altitudes are given in thousands of feet. If the top number (relative altitude) says "-002" and the lower number (true altitude) says "004," the target is 2,000 feet below you and 4,000 feet above the ground.

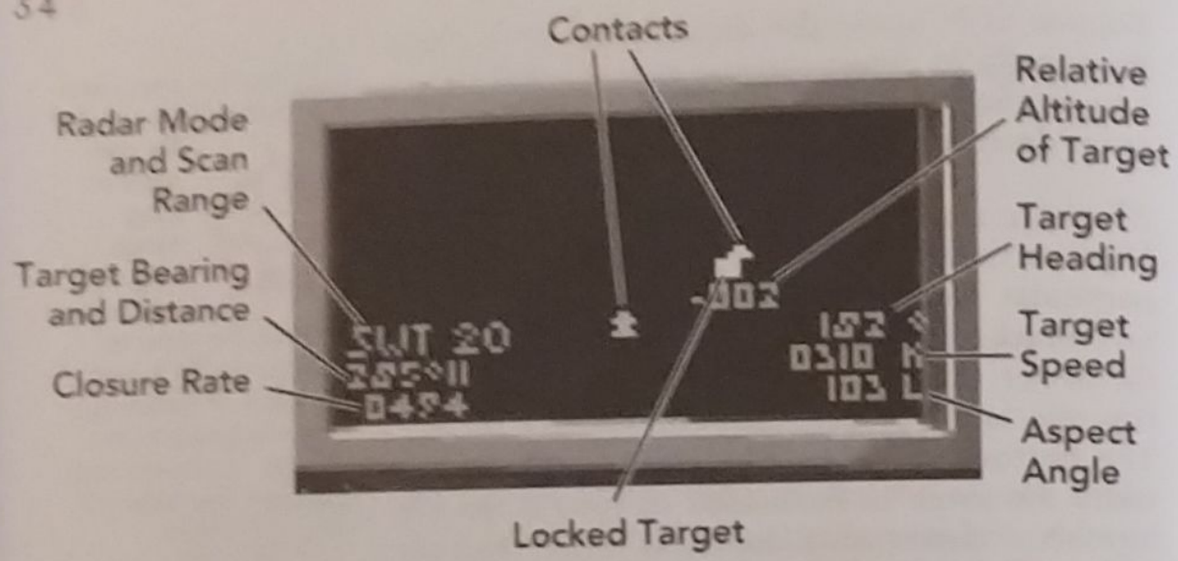
In the lower left, the **radar mode** (SCN) and **scan range** are shown. The scan range can be adjusted to 10, 20, 40 or 80 miles by pressing [F8]. As in other radar modes, the scan range is the distance from the top of the screen to the bottom.

Beneath the mode and scan range is the horizontal and vertical dimensions of the radar **scan cone**. The radar broadcasts in a cone extending outward from the nose of your aircraft. This cone extends 15° each direction vertically (a 30° arc) and 30° each direction horizontally (a 60° arc).



SWT mode

When you have selected a target in SCN mode and you want to begin tracking it more closely, press [Z]. This switches you into scan-while-track mode (SWT). In SWT mode, the radar pays



special attention to the specific target you have selected while still keeping track of the other targets in the area.

As with SCN mode, SWT depends on active radar broadcasts to detect targets. If you use SWT mode, enemies will detect you on their threat warning indicators and other radar detection gear.

One difference between SWT and SCN is that the radar scales its scan range automatically based on where the target is. If the locked target moves away, the radar will increase the scan range to accommodate the target. Likewise, if you close on the target, the radar will automatically change to a shorter scan range when it becomes possible.

Radar **contacts** still appear as roughly triangular blips on the display, except for the **locked target**, which is now shown as a square. You can change the locked target by pressing [X]; this will break the radar lock on the current target and lock on to the next available target. Information about the locked target is displayed in the bottom two corners and directly under the locked target's square blip.

In SWT and TRK modes, the lock on the target can be broken, which returns you to SCN mode. There are many ways to break radar lock, but the most common is to "beam" the radar. "Beaming" is when the target flies perpendicular to the radar so that its relative motion drops to zero, which causes the target to "disappear" for a moment. See "Avoiding Radar" in Chapter 4 for details.

This information is the same as in the IRST mode above: beneath the locked target is its **relative altitude** in thousands of feet. On the lower left, the **radar mode** (SWT) and **scan range** is shown, followed by the locked target's **bearing and distance** in miles. At the bottom is the target's **closure rate**

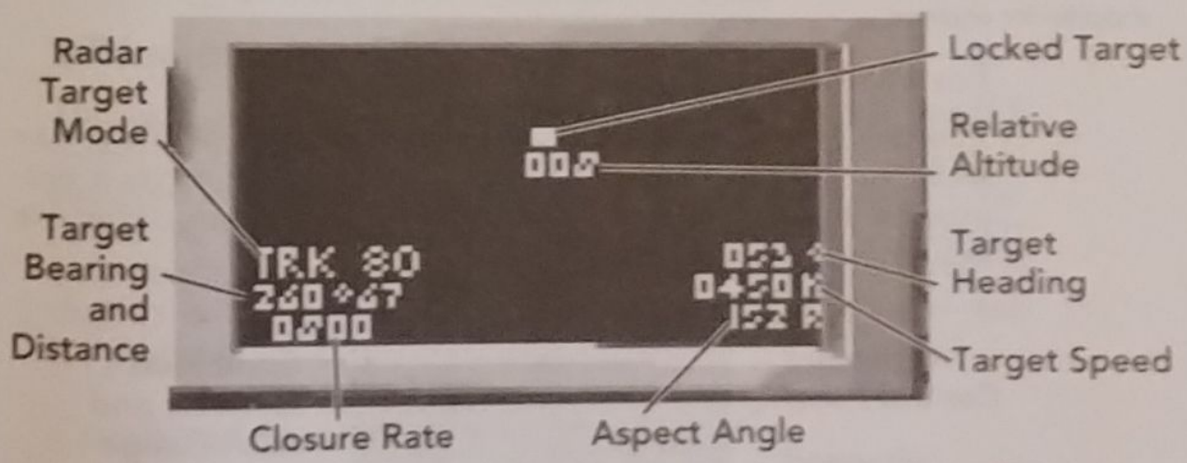
in knots. On the lower right, the locked target's **heading** is shown, followed by its **speed** (in knots) and the **aspect angle**. See the section on the IRST mode for full details.

TRK mode

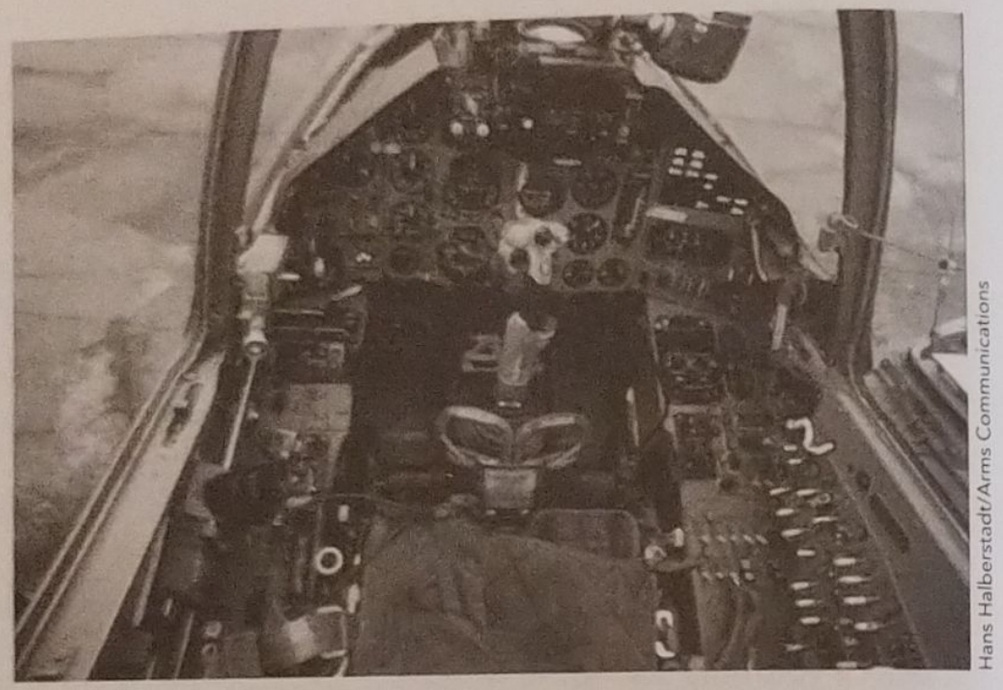
To pay even closer attention to your target, you can switch to tracking (TRK) mode. Select the target in SWT mode and press [Z]. TRK mode presents the same information about the locked target that the IRST and SWT modes do, but updates it much more frequently. This is because the radar is now no longer scanning the entire area; it is locked on that one target. All other radar contacts drop off the screen. The **locked target** appears as a square blip.

As with SCN and SWT mode, TRK depends on active radar broadcasts to detect targets. If you use TRK mode, enemies will detect you on their threat warning indicators and other radar detection gear.

In TRK mode, as in SWT mode, the radar scales its scan range automatically based on where the target is. If the locked target moves away, the radar increases the scan range to accommodate the target. If you close on the target, the radar will change to a shorter scan range as soon as possible.



The information about the locked target is the same as in the IRST and SWT modes above. Beneath the locked target is its **relative altitude** in thousands of feet. On the lower left, the **radar mode** (TRK) and **scan range** is shown, then the locked target's **bearing and distance** in miles, then the target's **closure rate** in knots. On the lower right, the locked target's **heading** is shown, followed by its **speed** (in knots) and the **aspect angle**. See the section on the IRST mode for full details.



Hans Halberstadt/Arms Communications

You can look out the rear of the canopy by pressing 5. There are no instruments in this view, but you can look over your shoulder for enemies.

INSTRUMENTATION

Virtually all of the flight instruments are on the forward control panel. The left side of the cockpit (visible by pressing 4) contains the in-flight recording system control panel, while the Right View (visible by pressing 6) contains the caution light panel. Return to the Front View by pressing 3.

Forward control panel

This is where all the critical gauges are collected. While it may seem like a sea of dials and numbers, every gauge serves a purpose. With time, these instruments will become an instinctive link between you and your MiG-29.

Armament Control Panel
Airspeed Gauge
Fuel Gauge
Altimeter
Configuration Display
Combined G Meter and AOA Gauge
IFF Indicator
Attitude Reference System
Stall Light
Clock

Armament control panel

In the upper left corner of the forward control panel is the armament control panel, which shows what the currently selected weapon is. The possible readouts are:

- CAN — GSh-301 30mm cannon (air-to-air mode)
- ACAN — GSh-301 30mm cannon (air-to-air mode) set to automatic fire
- R60 — R-60 rear-aspect infrared-homing missile
- R73 — R-73 all-aspect infrared-homing missile
- R23R — R-23R radar-guided missile
- KH23 — Kh-23 air-to-surface missile
- STRF — GSh-301 30mm cannon (air-to-ground mode)
- UB57 — UB-16-57 or UB-32-57 rocket pod
- F250 — FAB-250 bomb
- F500 — FAB-500 bomb
- FBHD — FAB-250 bomb (high-drag version)
- RBK5 — RBK-500 cluster bomb

Press Backspace to cycle through the air-to-ground armaments or Enter to select air-to-air weapons.

IFF indicator

The light just above the armament control panel is the IFF (identify friend or foe) indicator. Every friendly plane is equipped with an IFF transponder. Pressing N will send a signal to the currently locked target. If the target is friendly, its transponder will return a signal and this light will come on. If there is no return signal, the IFF indicator will not light—and it's probably the enemy.

Combined G meter and AOA gauge

To the right of the armament control panel is the combined G meter and AOA gauge. The G meter, which shows the G-forces you and your plane are under, is on the right side of the dial. The AOA gauge, which shows the angle of attack your plane is flying at, is on the left.

The longer needle on the right shows the number of Gs the plane is currently pulling. The shorter needle on the right indicates the maximum Gs pulled since the last time the G meter was reset. Pressing Alt G resets the G meter by setting the shorter needle to the current G level.

The needle on the left indicates the current AOA in degrees. The AOA gauge tops out at 30°.

Fuel gauge

The fuel gauge is the vertical gauge on the far left of the control panel. It shows the total amount of fuel remaining in all tanks in thousands of pounds. So if the needle is on 4, it means 4,000 pounds of fuel remain.

Keep an eye on your fuel. While you have an unlimited amount in Instant Action, you can quickly find yourself in trouble in a campaign if you use the afterburner too much or don't bring an external fuel tank on a long mission.

Airspeed indicator

The airspeed indicator is just below the armament control panel and to the right of the fuel gauge. It shows the plane's current speed in hundreds of knots. This gauge shows indicated airspeed, so named because it is literally "the airspeed shown on the indicator." True airspeed is indicated airspeed corrected for the air pressure and air density at sea level and the local airflow around the plane. Indicated airspeed is more important for judging when you may be about to stall, since it relates directly to the airflow over the wings (which determines lift and control surface authority).

Altimeter

Below the airspeed indicator is the altimeter, a gauge which shows the plane's altitude in thousands of feet above sea level. Note the difference between this and altitude above ground level (AGL); if you are flying over a mountain range, the difference between this altitude and your AGL altitude could be thousands of feet! On the right side of the control panel is the radar altimeter, which displays your AGL altitude (see below).

Attitude reference system

To the right of the airspeed indicator and below the combined G and AOA meter is the attitude reference system. This device shows the plane's orientation with respect to the horizon. The line across the middle of the gauge represents the position of the

MiG-29's wings. As the plane turns, the artificial horizon (the division between the two colors in the gauge) tilts to match the actual horizon. When you dive or climb, the artificial horizon reacts appropriately.

The attitude reference system is of most use when you cannot see the real horizon (due to bad weather or the orientation of the plane).

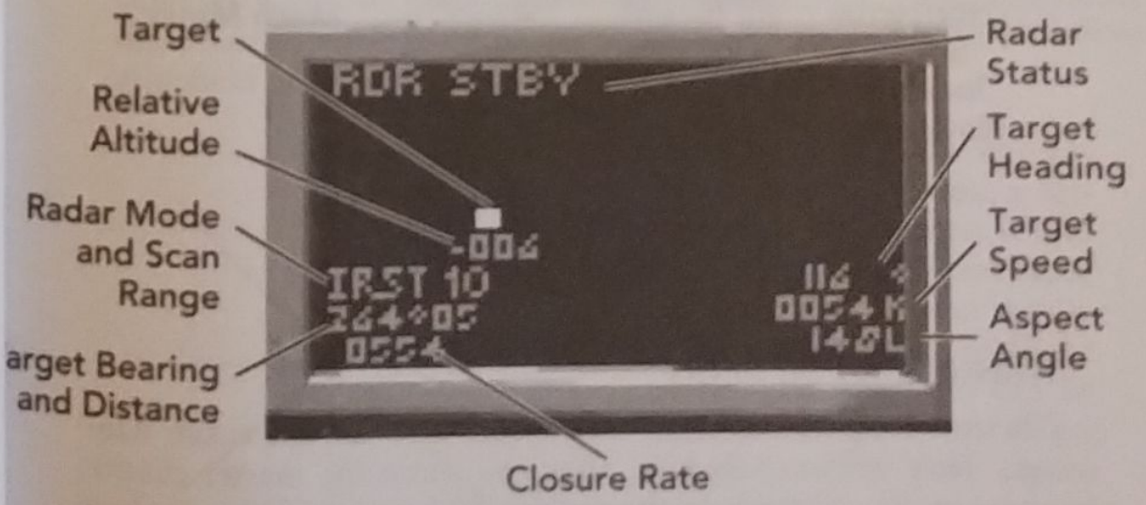
Stall light

The stall light, right below the attitude reference system, comes on when you are entering a stall. Stalls occur when there is not enough air flowing over the wings and other control surfaces to allow the plane to remain in flight. As AOA increases, the direction of airflow changes until the wings and other control surfaces cannot generate enough lift to maintain control. In a stall, the controls cease to have much effect, and the plane begins to fall.

There may be times when a stall appears useful, since a stalled plane moves too slowly to be picked up on radar. But in reality, the momentary benefits gained from such a maneuver are far outweighed by the vulnerability of a stalled plane in an air combat situation.

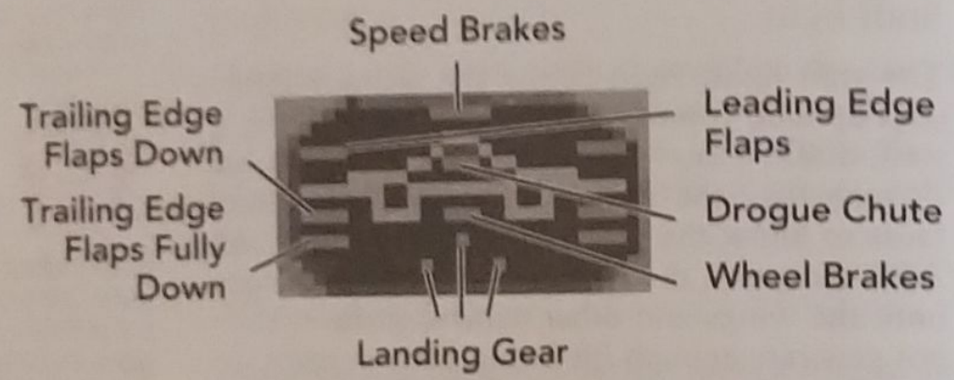
Fortunately, once the plane begins falling, enough air begins flowing over the airframe to allow control to return. Also, opening the throttle or lighting the afterburner can increase the plane's speed and reduce AOA, which is essential for recovering from a stall.

The MiG-29 retains control at much higher AOA than most other planes, but even it is in danger of entering a stall if the AOA exceeds 40° for very long.



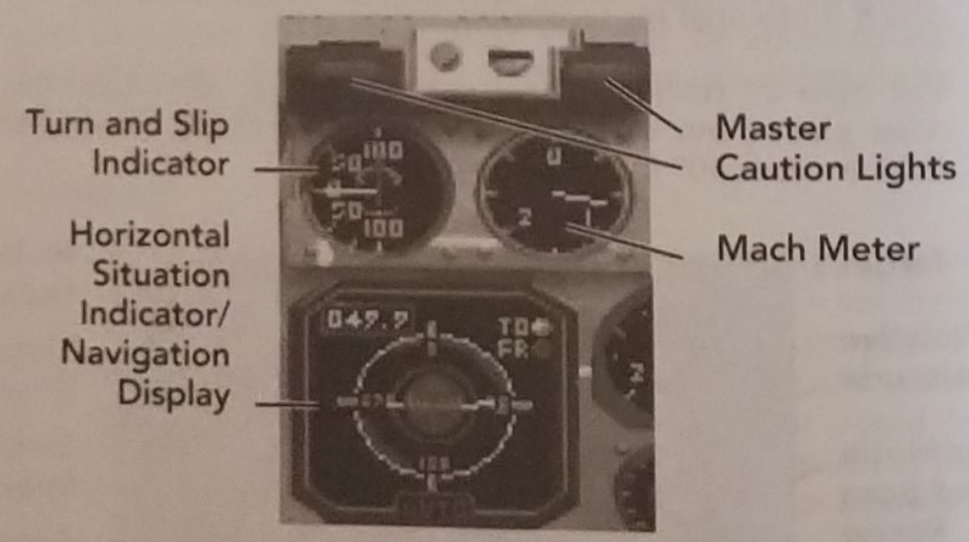
Configuration display

The configuration display is below the altimeter on the left side of the forward instrument panel. It shows the positions of the gear and flaps. As the flaps on the MiG-29 are largely automated, you can use this display to keep track of the flap position. It also shows if your speed brakes are on and if your landing gear is deployed.



Clock

The clock, just to the right of the configuration display, keeps mission time. For example, if your mission begins at 0600 hours, the clock will show 6:00 on the runway at mission start.



Master caution lights

These two lights will flash when your MiG-29 has taken damage. They will stop flashing after you check the master caution panel in the right side of the cockpit by pressing [6] (press [3] to return to the forward view).

Turn and slip indicator

In the top center of the forward instrument panel is the turn and slip indicator, which includes the rate of turn indicator. The yellow needle on this gauge shows the left or right turn rate in tens of degrees per minute. The red needle shows the number of minutes it will take to complete a 360° turn.

Toward the bottom of this gauge is a small backup slip indicator. The yellow ball will slide left or right as appropriate to show sideslip.

Mach meter

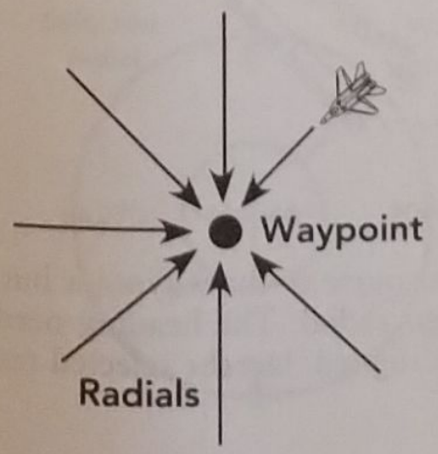
The Mach meter, just to the right of the turn and slip indicator, shows the plane's present speed in Mach units. If the needle is halfway between the 1 and the 2, you are traveling Mach 1.5.

Horizontal situation indicator and navigation display

The horizontal situation indicator and navigation display (HSI) is just below the turn and slip indicator and Mach meter in the center of the forward control panel.

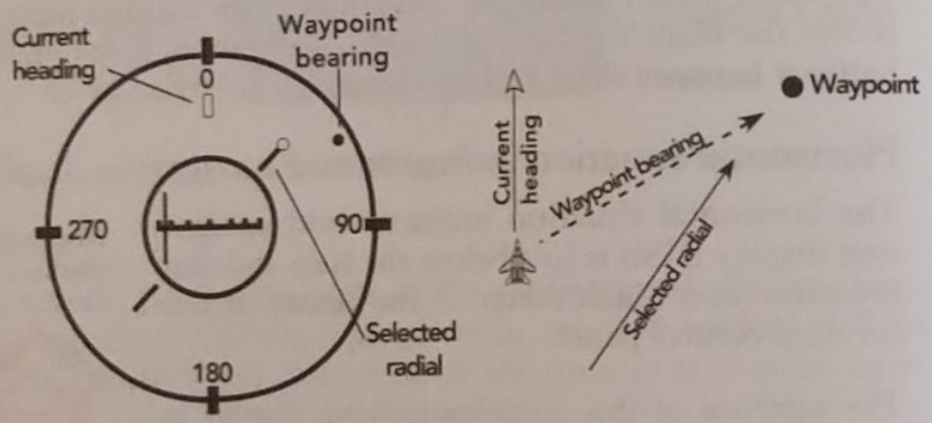
The purpose of this complex-looking display is to aid you in following your flight plan. When a mission is laid out, there are waypoints to which you are expected to go. How you approach those waypoints is also part of your flight plan. For example, the flight path of a well-planned mission will avoid concentrations of enemy air defenses whenever possible. If you deviate from this planned route, you could be stepping into a minefield of SAMs and AAA.

When setting up your missions in the campaign, take advantage of this navigation display. Since you can, with this instrument, approach waypoints along the exact path laid out in the mission plan, it is worth it to lay out flight paths which avoid enemy air defenses.

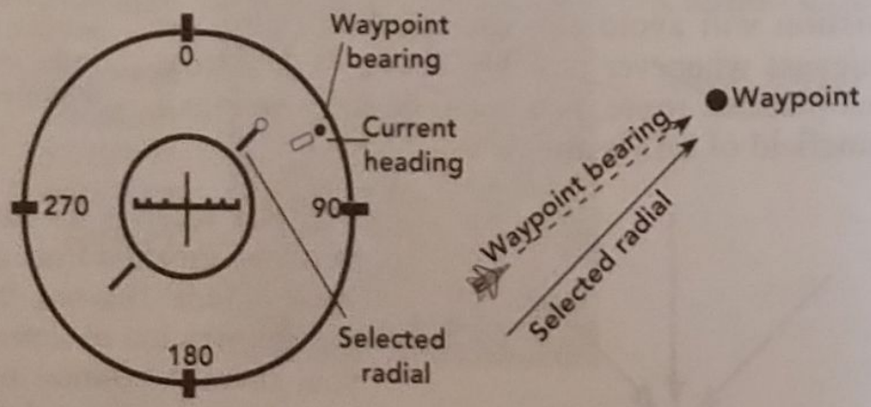


The HSI receives its navigational information from a network of radio beacons. It uses this information to determine your aircraft's position in relation to your planned flight path. Each waypoint can be approached from any direction (or radial). Your chosen flight path is your selected radial to that waypoint.

The HSI shows the distance to the current waypoint in miles in the upper left-hand corner. A pair of lights in the upper right show whether you are approaching ("TO") or leaving ("FR") the current waypoint. If neither light is lit, you are not yet within receiving range of a navigational beacon. In addition, two vertical red bars (one on either edge of the dial) will light to warn you if you are not receiving navigational information. A white needle indicates your current heading. The small red mark on the compass ring shows the bearing to the current waypoint. The two yellow lines on the center ring show your selected radial. The one with the white tip shows the way you should go.



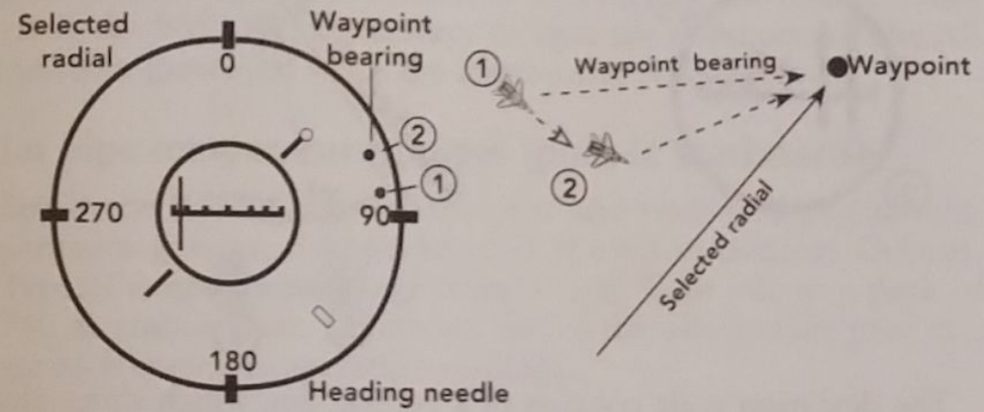
To get to your waypoint, line up the waypoint bearing marker with the current heading needle. Refer to the diagram below:



In this diagram, the MiG-29 is on course to the waypoint but is not approaching along the selected radial. The heading needle and waypoint bearing marker are aligned, but the selected radial lines are not.

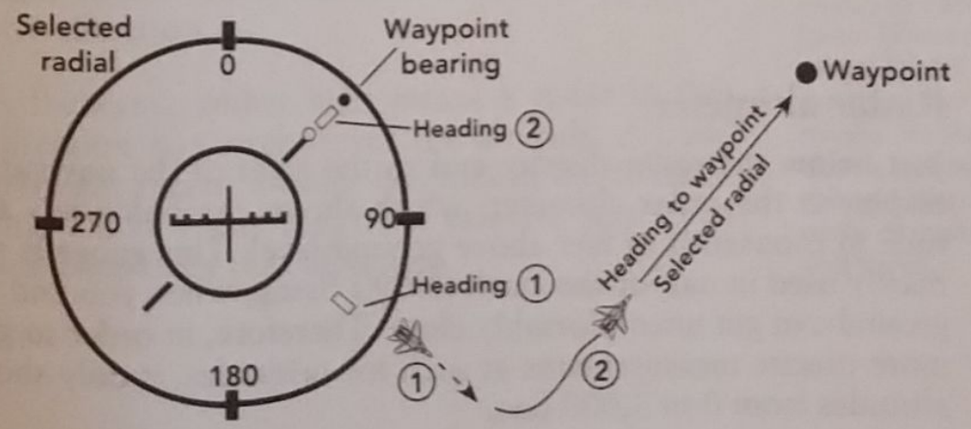
To get on the proper approach path, try to line up the heading needle, the selected radial lines and the waypoint bearing marker. The best way to do this is as follows.

First, maneuver so that the heading needle and waypoint bearing marker are on the same side of the selected radial lines. Then turn until the waypoint bearing marker is between the heading needle and selected radial. Watch which way the waypoint bearing marker slides. You want it to slide closer to the selected radial. Refer to the following diagram:



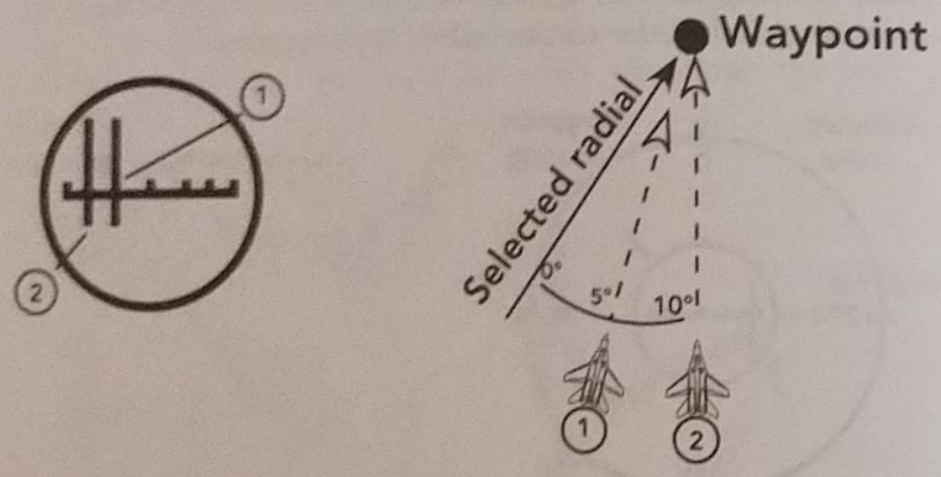
Notice that the waypoint bearing marker is between the heading needle and the selected radial. As the Fulcrum moves from position 1 to position 2, it gets closer to the selected radial. To reflect the change in position, the waypoint bearing marker slides from position 1 to position 2, getting closer to the selected radial lines.

Once the waypoint bearing marker and the selected radial lines are aligned, turn quickly to bring your heading needle in line as well. Refer to the following diagram:

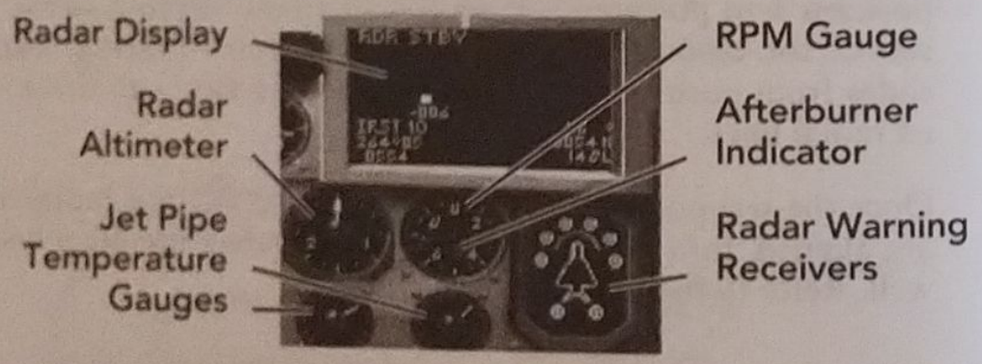


When the MiG-29 is in position 1, the waypoint bearing marker and the selected radial lines are aligned, but the current heading needle is off. By turning, the Fulcrum reaches position 2, where all three items are lined up.

When all three are lined up, you are on the proper approach path to your waypoint. Use the deviation scale in the center of the display for fine-tuning your approach.



The deviation scale consists of a vertical line which slides along a scale to the left or right to show how far off the proper approach path you are. Each mark on the scale represents 5°.



Radar altimeter

Just below the radar display and to the right of the navigation display is the radar altimeter, which shows the Fulcrum's altitude in thousands of feet above ground level. This gauge is primarily used in nap-of-the-earth (NOE) flying, when you and the ground can get uncomfortably close. Therefore, in order to give more precise measurements at such low altitudes, it only shows altitudes from 0 to 3,000 feet.

RPM gauge

To the right of the radar altimeter is the RPM gauge. A small marker on the rim of the gauge reflects the current RPM setting. The needle marks the actual percentage of maximum thrust being applied. As you open or close the throttle, the small marker will move immediately to the setting you have selected. The needle will gradually come around to the selected point as the engines spool up or down.

In the middle of the RPM gauge is the afterburner indicator. It glows amber if the afterburner is selected but the engine is not yet at the 90% RPM necessary to light the afterburner. The AB indicator glows red when the afterburner is engaged.

Jet pipe temperature gauges (port & starboard)

Below the RPM gauge are the port and starboard engine temperature gauges. Temperature is shown in degrees Celsius. Typical temperatures range from 450 at flight idle to a peak of 750 at engine start. Continued use of the afterburner may create high engine temperature readings.

If the engines are inoperative, these gauges may show very low temperatures. If there is an engine fire, the gauges may read very high—and it may be time to eject.

SRO-15 radar warning receiver display

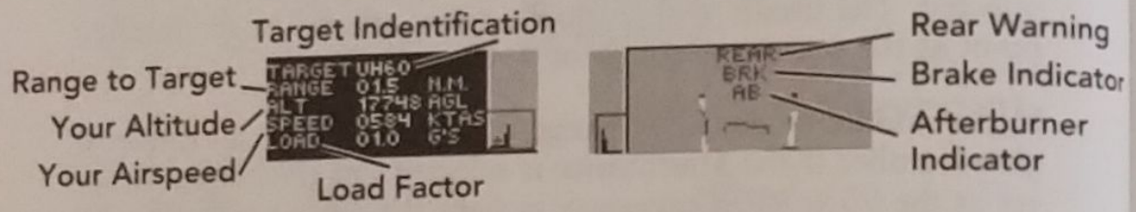
The SRO-15 radar warning receiver display is in the bottom right corner of the forward instrument panel. It consists of an airplane silhouette surrounded by eight warning lights. Each light corresponds to the position of a threat relative to your aircraft. If a warning light comes on, it indicates a radar is active in that direction.

A flickering amber light means a radar in that direction has swept over the aircraft. A solid red light means a radar has locked on. A flashing white light means a missile has been launched from that direction.

If you get a lock warning, maneuver quickly to break the lock. Once you get a launch warning, switch to Padlock View (press 8) and then press T until you are watching the incoming missile. As the range closes, get ready to roll and bank hard while dumping chaff and flares.

PADLOCK VIEW

Pressing [8] will switch to the Padlock View, which is used for keeping an eye on your opponent at all times. If you activate your helmet mounted system (by pressing [F7]), you can keep your weapons on your opponent as well.



New padlock view information

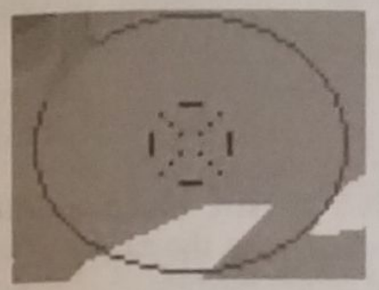
The Padlock View in *MiG-29* (and in *Falcon 3.0*, once *MiG-29* is installed) shows some new information.

In the upper left corner window is the target identification, followed by the range to target in nautical miles (NM). The range now shows tenths of miles. Next is your altitude above ground level (AGL) and airspeed in knots. The airspeed shown is either indicated (KIAS) or true (KTAS), switchable by pressing [Ctrl] [A]. Last is the current load factor (in Gs).

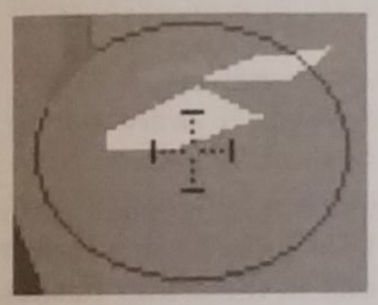
In the upper right corner window are three indicators. As before, the warning "REAR" appears if the target is behind you. Beneath this, the indication "BRK" appears if your speed brakes are on. Finally, the indication "AB" comes on if your afterburner is activated. If the "AB" is green, the afterburner is selected but not engaged since the engine is not yet up to the 90% RPM necessary to light the afterburner. When the "AB" is shown in red, the afterburner is fully engaged.

Helmet mounted system (HMS)

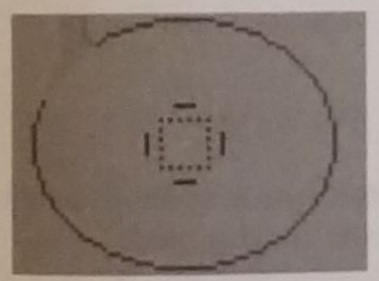
When the helmet mounted system (HMS) is activated, your fire control computer will track whatever you are looking at. In the center of the monocle, you will see one of eight different symbols depending on the situation. The symbols and their meanings follow.



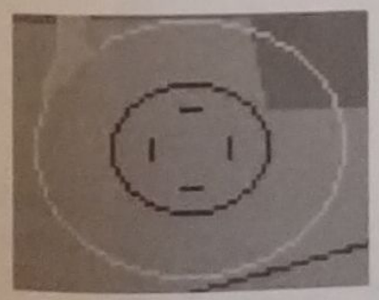
The target you are looking at is beyond the weapons system's tracking abilities (too far to one side or too far above you, for example).



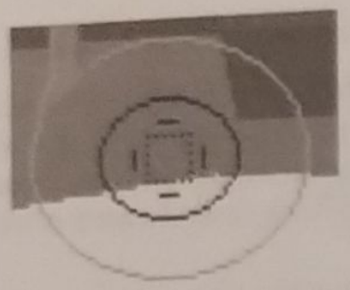
The IRST is scanning for a target in the area you are looking at. The cross flashes on and off.



The IRST is tracking the target.

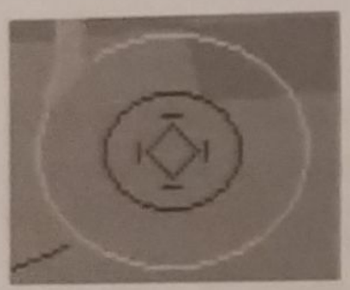


The radar is scanning for a target in the area you are looking at. This occurs if the IRST has been disabled or cannot track the target for some reason. The circle flashes on and off.

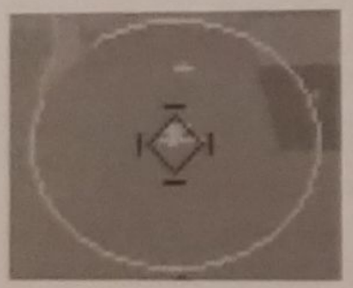


If the circle is flashing, it means a radar-guided missile (such as the R-23R) is trying to lock onto the target you are looking at.

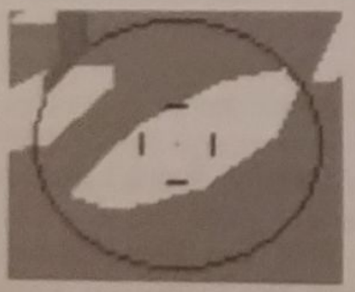
If the circle is steady, it means the radar is locked onto the target (which only happens if the IRST is disabled or cannot track that target for some reason).



A radar-guided missile (such as the R-23R) has locked onto the target. The diamond in the center flashes.



An infrared-guided missile (such as the R-60 or R-73) has locked onto the target. The diamond in the center flashes.



The helmet mounted system or the weapons systems are disabled.

LEFT SIDE PANEL

Press [4] to look at the left side of the cockpit.

The in-flight camera panel is on the far left side of this view. It shows how much of your available in-flight recording time remains and whether or not the camera is presently recording. Press [Ctrl] [R] to turn the in-flight camera on or off. If you run out of recording time on the tape, it loops around and starts over, recording on top of the beginning portion.

The in-flight camera in *MiG-29: Deadly Adversary of Falcon 3.0* operates exactly like the ACMI in *Falcon 3.0*. Refer to that manual for details.

RIGHT SIDE PANEL

Press [6] to look at the right side of the cockpit.

Chaff/flare indicators

These show how many charges each of chaff and flares remain on board. Use chaff to foil radar-guided missiles and flares to spoof infrared-guided missiles.

ECM indicator

When the on-board ECM (electronic countermeasures) and jamming package is activated, this light will come on. Press [E] to switch ECM on and off.

Master caution panel

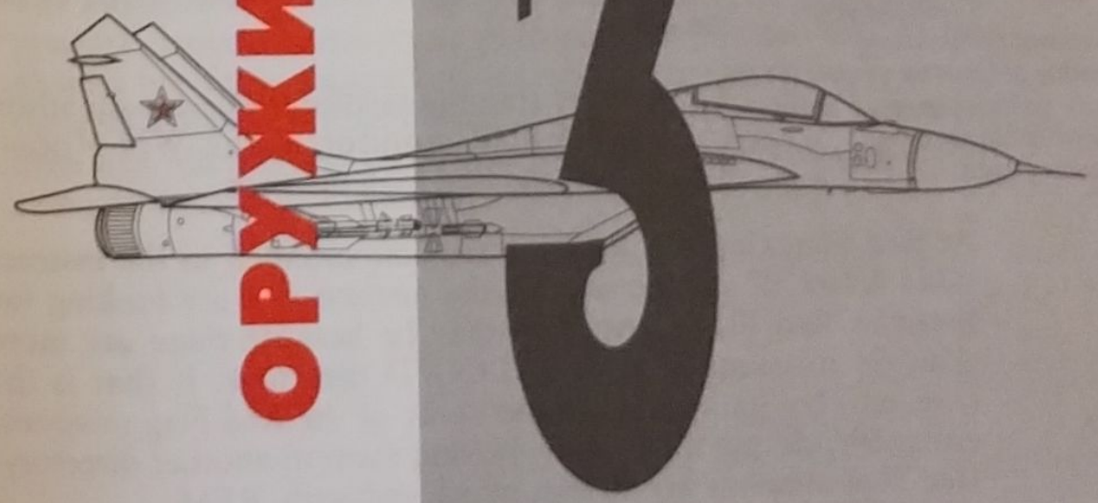
The lights on this panel show what systems, if any, are damaged. Each light corresponds to an aircraft system:

HUD	The head-up display is inoperative.
RDR	The radar is damaged.
CAN	The GSh-301 cannon will not work.
ENG	The engine is out.
OXY	Oxygen supplies are lost. Blackout will occur at altitudes above 10,000 feet.
ECM	Electronic countermeasures are nonfunctional.
BKS	The speed brakes are locked.
FLP	The flaps are frozen in place. Since the leading edge flaps have a great deal to do with the MiG-29's performance at high AOA, you may notice a greater tendency to stall if the flaps are frozen.
LGR	The landing gear is damaged.
ARM	The weapons arming system is inoperative. No weapons can be fired.
BRN	The afterburner system is not functioning. Afterburning is no longer possible.
JP4	The fuel storage area has been damaged. Land as soon as possible.
ORD	The external ordnance is damaged. The cannon is the only usable weapon.
NAV	Navigational aids are out. This includes the navigational display.

If you take too much damage, it is time to take advantage of the excellent safety features of the K-36D ejection seat (press [Ctrl] [E] three times).

Robert Prothero/Arms Communications

WEAPONS TRAINING



ОРУЖИЕ

3

ТРИ

WEAPONS TRAINING

The R-60 and R-73 dog-fighting missiles are specifically designed to be fired "off bore-sight." This means you need not be pointing directly at your target in order to be able to launch a successsful missile attack.

Extensive practice with the weapons systems is essential for pilots of the MiG-29. The R-27R's range, the off-boresight abilities of the R-60 and R-73, the accuracy of the GSh-301 cannon, the striking power of the Kh-23 missile and UB-57 rocket pod—all are wasted in the hands of an amateur.

To fly these training missions, select a squadron from the Squad monitor in the War Room. Then select the Red Flag monitor.

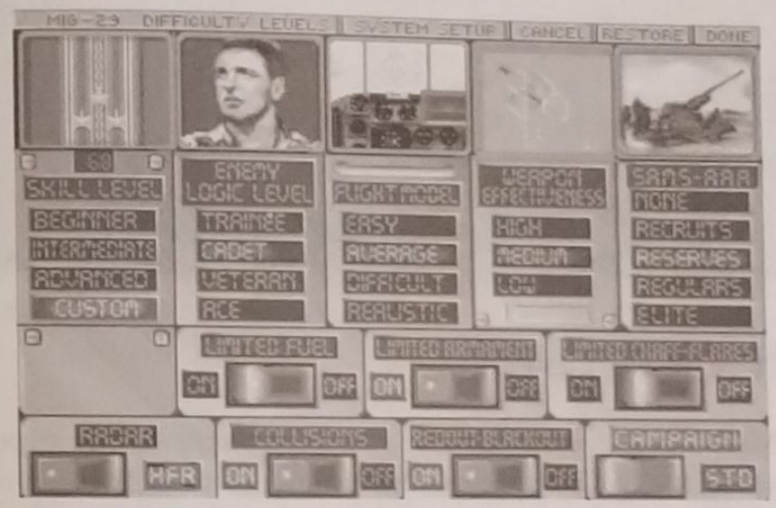
At Red Flag Control, load the mission specified in the instructions below. If you do not see the mission you are looking for listed in Red Flag Control, it may be because there are more than 64 missions in your FALCON3 directory. If that is the case, exit the game and remove some of the Red Flag missions, either by deleting them or by moving them to another directory. Red Flag missions are all files which end with .RFM.

Click on the bottom right-hand button of every screen until you are airborne.

The instructions in this chapter assume you are flying at Intermediate level with radar set to HFR. From the War Room, select the Config monitor. At the Difficulty Levels screen which

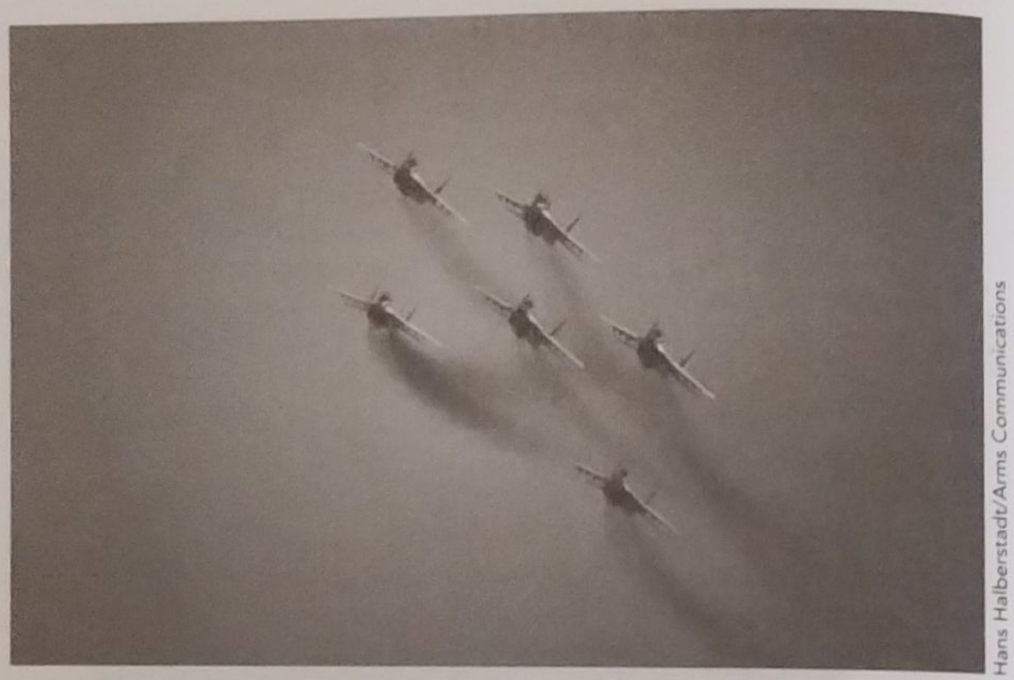
should appear, select Intermediate from the list in the upper left of the screen. Then click on the switch below Radar until it says "HFR." Select Done from the bar at the top of the screen.

If the training missions are too difficult, select Beginner instead of Intermediate on the Difficulty Levels screen. Some of the instructions will not make as much sense since they assume Intermediate difficulty, but the missions themselves will be much easier to complete.



Difficulty Levels Screen

The missions should be flown over and over again until you feel you are satisfied with your performance. At the end of this chapter are some suggestions for ways to get even more out of these missions. Different weapons, different enemy behavior, even just altering the technique you use to complete each mission can teach you a whole new set of skills.



Hans Halberstadt/Arms Communications

AIR COMBAT

The MiG-29 was built for one primary purpose: air superiority. It is specifically designed to hunt down and destroy other aircraft. In the hands of an expert pilot, the weapons and targeting systems of the Fulcrum are excellent for this task. With time, study and practice, you will be able to get the most out of your Fulcrum.

Target practice with the Gsh-301 cannon

The GSh-301 cannon, coupled with the MiG-29's laser range finder and targeting system, is arguably the most accurate aircraft gun available. It can even be set to automatically fire when a target enters the bullet path.

Chapter 2 contains descriptions and diagrams of the flight controls. It also details the functions of all the cockpit displays.

This mission is a simple guns-only attack on a group of unarmed transport planes. The C-130s are large, slow and can take a fair amount of punishment before they go down. The first few times you try this mission, use the automatic cannon (activated by pressing [K]). It will probably be about all you can do to keep the target in your sights until you master the flight controls.

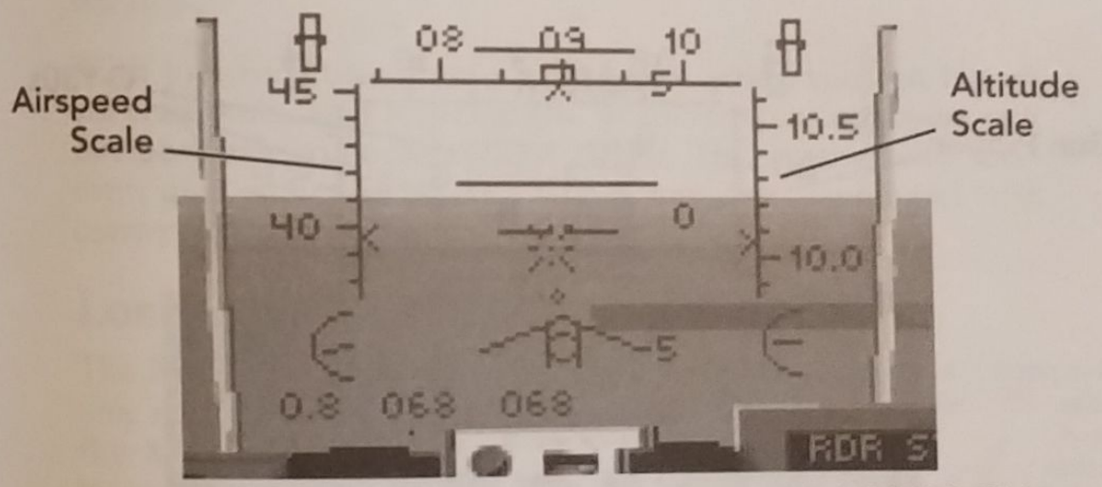
After you are more comfortable, try this mission with the automatic cannon turned off. See how many you can shoot down with one load of ammunition. Conservation of ammunition is a valuable skill to learn. After all, in a real combat situation, you definitely do not want to be out of bullets.

Mission preparation

- Read the sections in Chapter 2 that deal with the air-to-air HUD and the IRST tracking mode on the radar display.
- At Red Flag Control, load the mission titled "MIG29GUN." Once it is loaded, click on the bottom right-hand button of every screen until you are airborne.

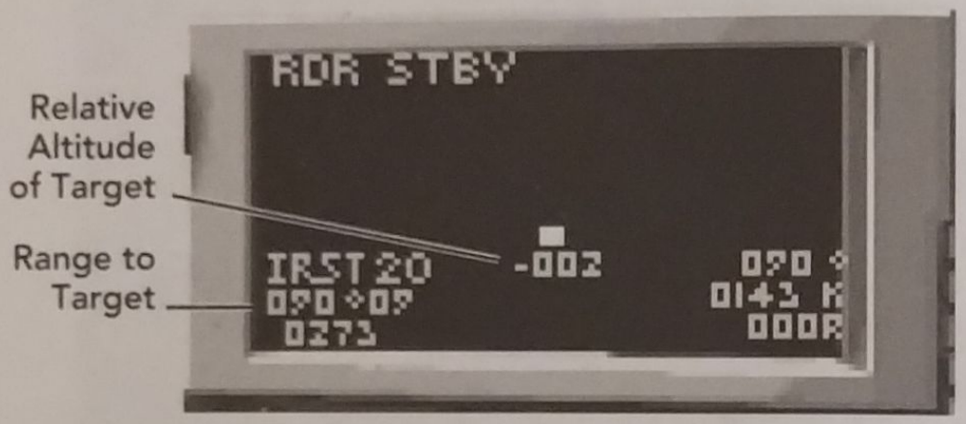
Spotting the target

You should begin the mission at 400 kts at an altitude of 10,000 ft. Check the airspeed and altitude scales on your HUD to be certain.

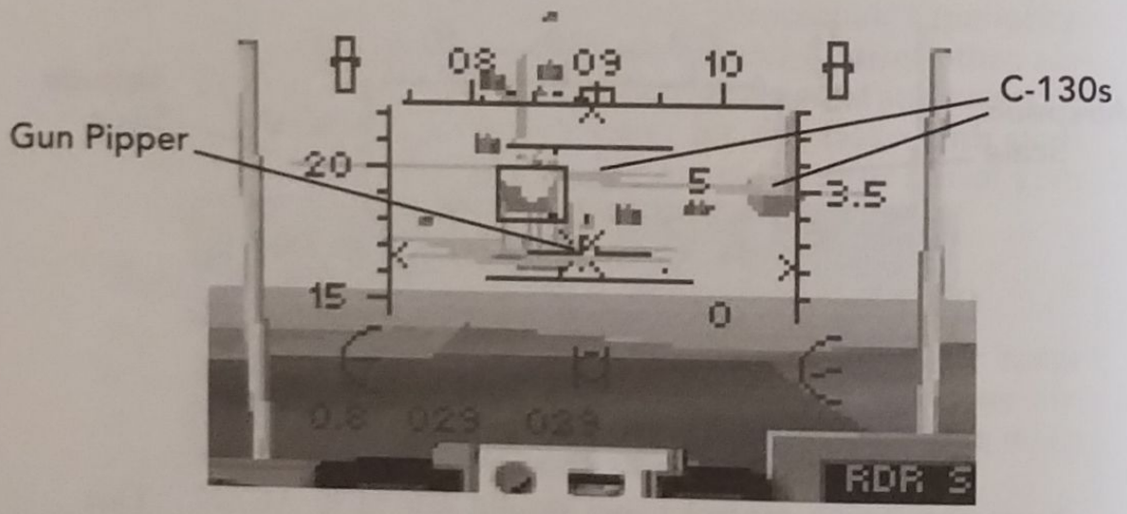


Your armament control panel should say "150 ACAN." This means that the cannon is the active weapon, it is set to automatically fire when a locked target is within gun kill parameters, and you have a full load of 150 rounds remaining.

The target flight is about 20 miles straight ahead of you. Just keep cruising until your IRST picks them up. Your radar screen should look something like this:



Maneuver until the targets are directly in front of you and at the same altitude. Watch as the range closes; the cannon has a range of under a mile.



Closing in

When the range shows one mile, pay close attention to the gun pipper. Keep the target under the small diamond at first, then gradually slide the reticle over the target and keep it there.

You should see and hear the cannon fire repeatedly. The automatic cannon does not know when a target is disabled; as long as the target is in the pipper, it will keep on firing until it is empty (or until the target is torn to pieces so small the tracking system cannot detect them).

While five rounds from the GSh-301 cannon is enough to down most planes, don't expect large aircraft such as the C-130 or B-52 to disintegrate from cannon fire.

To control the amount of ammo expended, switch the automatic cannon off by pressing [K] after it has fired about 10 or 15 rounds into a target. For large targets, you may want to allow 15 or 20 rounds—or more—to be sure of a kill. Watch the target to see if it is burning or spinning out of control. The surest sign of a kill is when the crew bails out; look for the parachutes.

If the target is still not disabled, switch the automatic cannon back on and let it fire a couple more bursts. When a target is disabled or destroyed, switch targets by pressing [X] to break the weapons lock. Line up and commence another firing pass like the first.

With judicious use of your ammunition store, you can kill all six transports. In a real dogfight, of course, your opponent will not be nearly so docile as these targets. This mission will, however, give you good practice with aiming the cannon and with conserving ammunition.

Long-range interception with the R-27R

The R-27R is a radar-guided missile with a reliable range of about 17 miles. When it is aboard the MiG-29, it can lock onto a target using information from the IRST. This means that the first warning your opponent may get is the missile's homing signal. To make best use of its range, however, you will have to use your radar to detect your opponent. The IRST can only reliably lock onto a target under 10 miles away—less than two-thirds of the R-27R's range.

If you are flying in Padlock View (discussed in Chapter 2) with the helmet mounted system (HMS) active, the R-27R can lock on to targets which are outside the HUD.

The purpose of this training mission is to familiarize you with using the radar and R-27R missile to intercept opponents beyond visual range (BVR). You will be up against a captured F-5 which has been modified as a target drone. It will fly a relatively straight course, using countermeasures and maneuvering

only when it detects a threat. You will approach the target, using your radar sparingly to establish where your opponent is. When you get within 20 miles, you will establish a radar lock and close rapidly to within firing parameters.

Your MiG-29 is loaded with two R-27R all-aspect active radar homing missiles for this mission. Unfortunately, with 17 miles' warning, even a C-130 has a fair chance of evading one missile using chaff and hard turns. So, at 17 miles out, launch the first missile; then, while the target maneuvers wildly to evade it, fire the second from closer range.

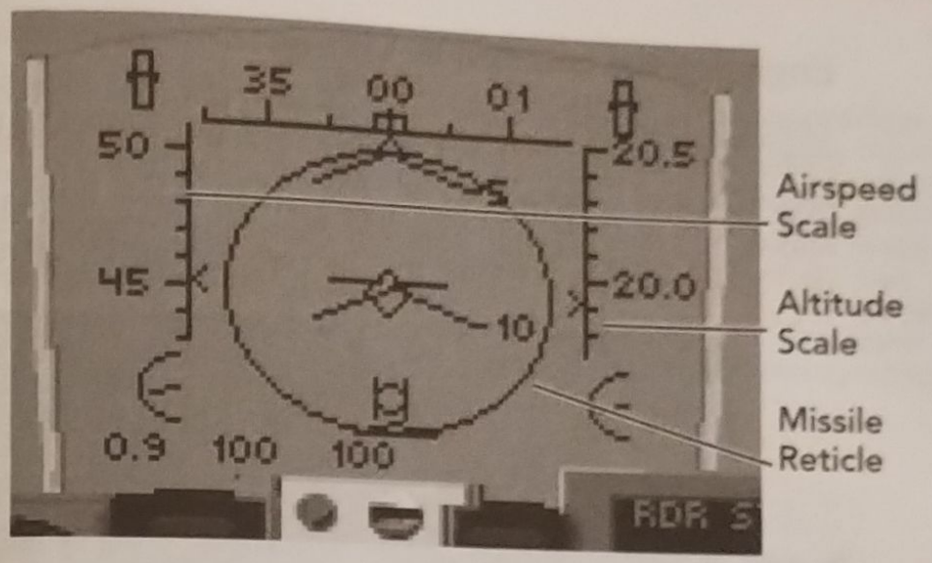
This tactic is effective even against live opponents. Distracted by the first missile, they often cannot see the second until it is too late. A pilot will pay close attention to an incoming missile, and it is this concentration which can allow a second missile to escape notice and get in for the kill.

Mission preparation

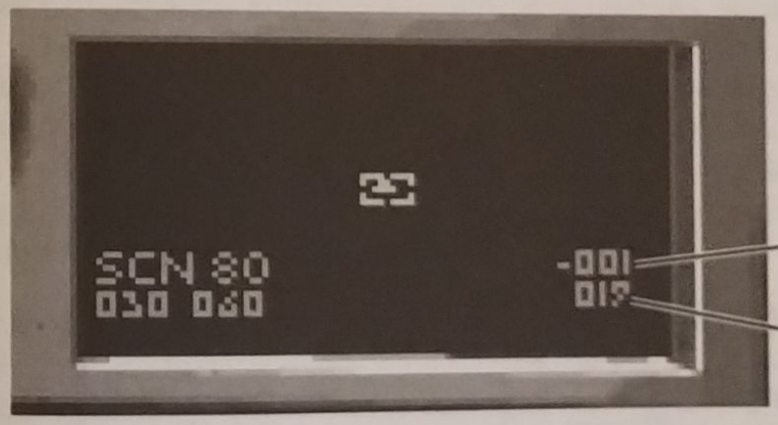
- Carefully read the sections on the air-to-air HUD and the IRST, SCN and TRK radar modes in Chapter 2. Pay special attention to the missile and radar information.
- At Red Flag Control, load the mission titled "MIG29BVR." Once it is loaded, click on the bottom right-hand button of every screen until you are airborne.

Initial contact

You should be in flight at an altitude of 20,000 ft at a speed of 450 kts. Check the altitude scale and the airspeed scale on your HUD to verify this. Then select the R-27R as your active weapon by pressing [Enter] (or button B on your joystick) until you see the missile reticle.



Press [5] to switch your radar to SCN mode. Make sure your radar is set to an 80-mile scan range (use [F8]) to give the earliest warning of enemy aircraft. In a real engagement, you will have no idea where your opponent may be coming from. Almost immediately you should have a radar contact directly in front of you.



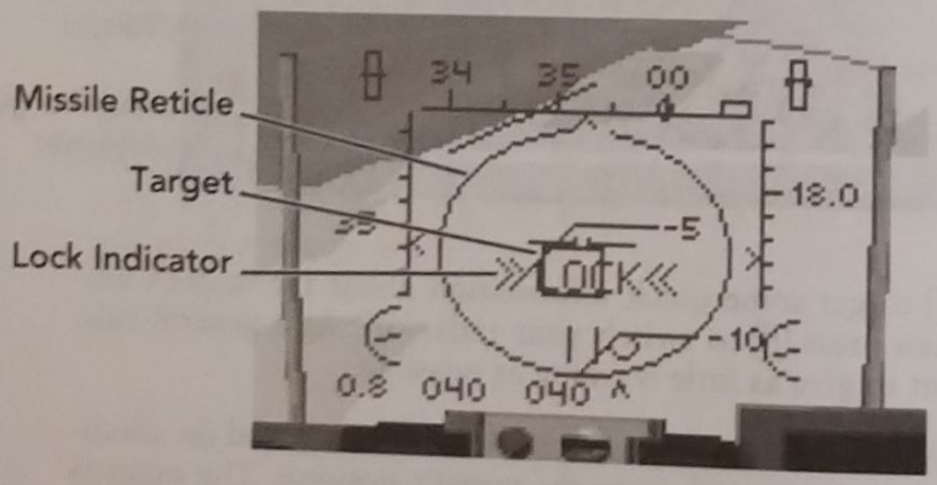
Press [T] to get some quick information about the target's altitude, then press [R] to switch your radar off. As a general rule, you want to give as little warning as possible.

Every five or ten seconds, turn the radar back on and do another sweep to get an update on the target's position. The range is especially important. Remember that the scan range at the lower left is the distance from the bottom of the screen to the top. For example, if your scan range is 40 and the target is halfway down the screen, it's 20 miles away. When the range gets down to 20 miles, it's time to go into action.

Engaging the enemy

Turn on your radar (if it's not already on) and watch for the contact to appear again. Press [T] to acquire the target, then press [Z] twice to lock it up in TRK mode. Now that your target knows you are coming, you had best get there quickly. Open the throttle to its maximum and hit the afterburner (press [/]).

In the lower left, you should see the target's range gradually decrease. Maneuver so that the target is centered in your HUD, then glance back at the radar. When you get down to 17 miles away, look back to the HUD. It should be flashing the word ">>LOCK<<" across the middle of the missile reticle (assuming the target is still within the missile reticle).



Press [Spacebar] or button A on your joystick to launch the first R-27R. Wait a few seconds to close on the target and let it begin evading that first shot. Keep the target in the missile reticle on the HUD; you need to be locked on in order to fire the second missile. Now fire again.

You can press [Q] to see how the target plane is doing against the missiles, or press [Q] twice to see what your missiles are doing. If you don't score a kill, don't be too concerned. While beyond visual range (BVR) intercepts can eliminate a foe before the fight even begins, they also suffer from a fairly low success rate. It is difficult to score a kill when your opponent has so much warning.

If the target evades both missiles, close in and try to down it with your gun for some dogfighting practice.

An ambush using the R-73

The R-73 is a very advanced infrared-homing missile. It is designed specifically for dogfights and, therefore, has a range of only about four miles. Since the R-73 does not require radar to track its target, however, there is little warning if it is used with the IRST.

The purpose of this mission is to teach you how to "sneak up" on multiple opponents using the IRST and then launch a deadly missile attack from any angle. The target is a pair of F-4s flown by aggressor squadron pilots who will do their best to react like real NATO pilots. You spot the targets using a brief radar scan, then come in low and from behind to avoid being detected. Using your IRST to detect your opponents, you then pitch the nose of your plane upward and launch two R-73 missiles at each opponent. You then have two missiles left to finish off any survivors in a dogfight.

In a combat situation, especially when you are outnumbered, surprise is everything. If you are in an area with few surface-to-air defenses, one of the best places to hide from your opponents is below them. Using passive detection of enemies will also aid you.

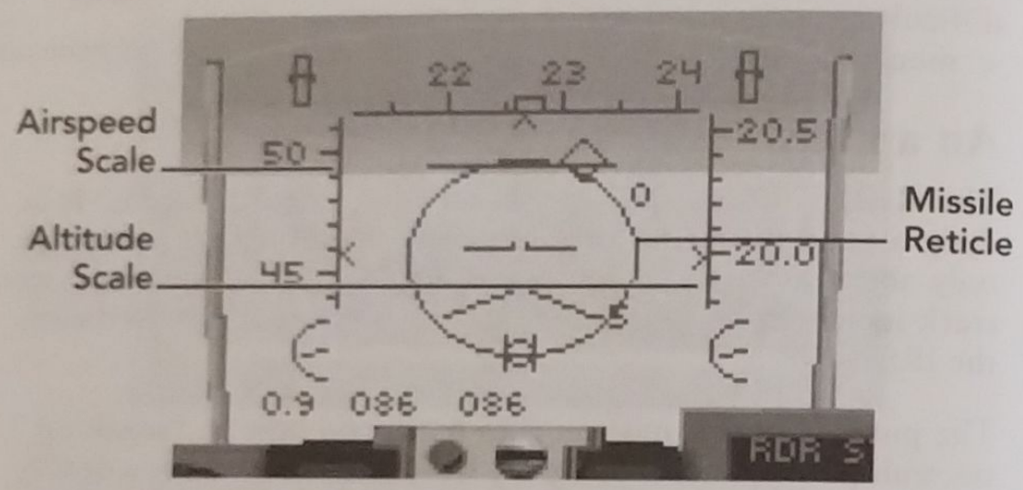
"Passive detection" refers to detection techniques which do not radiate any traceable signals. The IRST uses a passive infrared detection method to track targets.

Mission preparation

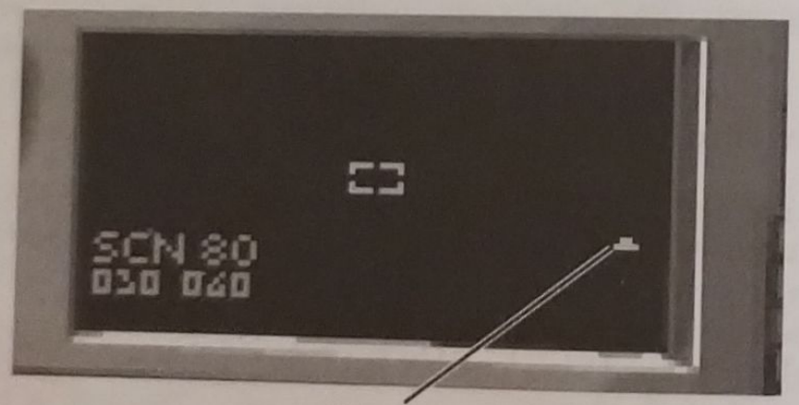
- Fly the R-27R mission until you are comfortable with your ability to intercept a target with missiles.
- Reread the sections of Chapter 2 which deal with the air-to-air HUD and the IRST and SCN radar modes.
- At Red Flag Control, load the mission titled "MIG29R73." Once it is loaded, click on the bottom right-hand button of every screen until you are airborne.

Spotting the enemy

You should be in flight at an altitude of 20,000 ft at a speed of 450 kts. Check the altitude scale and the airspeed scale on your HUD to verify this. Then select the R-73 as your active weapon by pressing Enter (or button B on your joystick) until you see the missile reticle.



Press F5 to switch your radar to SCN mode with an 80-mile scan range. Right away, you should have a radar contact about 40 miles ahead and slightly to your right.

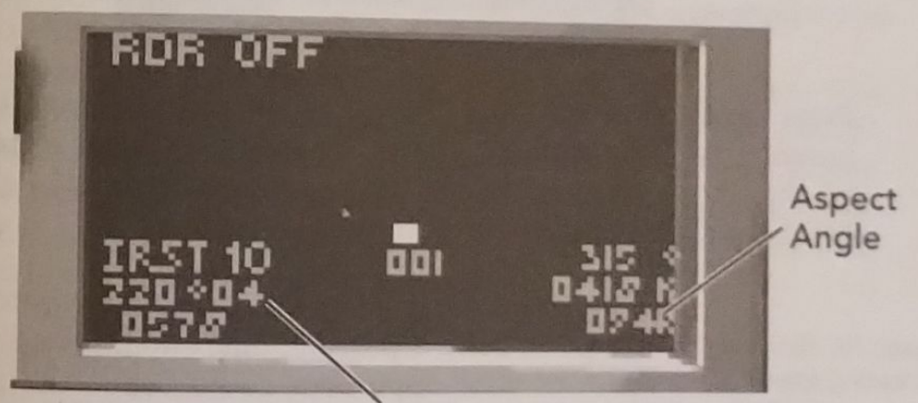


Press T to get some quick information about the target's altitude, then press R to switch your radar off. Unlike the target drone in the first mission, these two planes mean business. If they detect you, you're in for a two-versus-one dogfight.

Use your radar as sparingly as possible! Keep it turned off except for brief scans. The three things to keep track of are range, altitude and whether the F-4s have broken formation yet. As long as their blips stay close together, you can bet they haven't detected you yet. Tight formation is good for transit from one waypoint to another, but it's useless in a dogfight.

Head for the deck. Nose down and dive until your radar shows the targets as being 5,000–6,000 ft above you (press T and check the top number in the lower right), then level out. When the range gets down under 20 miles, switch your radar to IRST mode and wait until you pick them up on your scope. If you think they might have spotted you, switch to SCN mode for a moment to see if they are still in formation. Once you have them on IRST, it's time to assess the situation.

One way to tell if enemies are on to you is if any of the lights on your radar warning receiver (in the lower right of your cockpit) flashes yellow. This means you're being scanned by active radar. If the light goes red, they've locked on. If it flashes white, they've launched a missile!

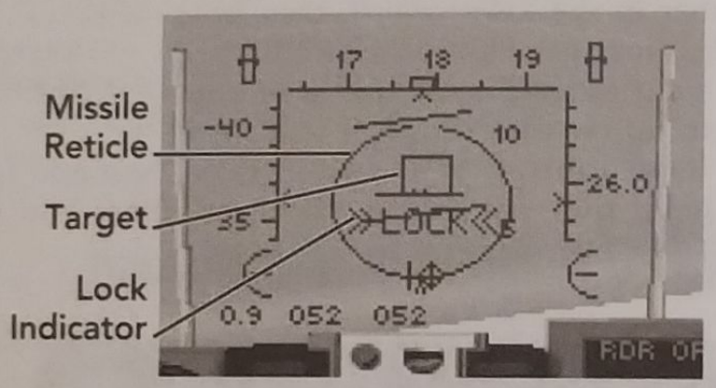


Press X to break the lock on one target and pick up the other one. Look at the aspect angle to determine if you are behind them or not. If it's a low number, such as 015R or 002L, you're in good shape; 000 is directly behind your opponent. If you can maneuver to reduce this aspect angle, do so.

Moving in for the kill

Watch the range as you close on them. Remember that the R-73 won't lock until four miles away, so be ready. Try to keep your closure rate positive but low, around 100 kts or less. Use the throttle to control your speed.

When you get to a range of five miles, pull back on the stick (or press [2] on the numeric keypad if you're using the keyboard for flight control) to pitch upward. If you've stayed 6,000 ft below the targets, this should be about a 10° climb. Wait for the lock indicator to flash across your HUD.



Press [Spacebar] or button A on your joystick to launch the first missile. Fire a second one at the same target almost immediately, then press [X] to break the lock and switch to the other target. Launch two missiles at that one as well.

The reason for climbing and increasing speed is to simultaneously increase energy and get closer to corner velocity (the speed at which you are most maneuverable).

You could sit back at this point and watch what happens, but it's possible that one or possibly even both of your opponents could evade the missiles. Now is the moment to get yourself into position to finish off whoever survived. Open the throttle all the way, engage the afterburner (press [/]) and climb until you get to corner velocity (your best turning speed, which is 400–450 kts). Kill the afterburner (press [/]) and do your best to get any survivors back into that missile reticle on the HUD. When the lock indicator flashes, fire immediately.

If this mission was carried out properly and you got a little lucky, both targets should be destroyed. But this was a 1v2 scenario, which is a difficult fight even for an experienced pilot. If you didn't get them both, just go back and try it again. If you did succeed, do it again anyway to make sure it wasn't just luck. Experience is more dependable than luck.

Feint and attack with the R-60

The R-60 is an infrared-guided missile with a range of about three miles. It is an older missile designed for much the same purpose as the R-73. Unfortunately it is not as reliable as the R-73 when it comes to all-aspect target tracking. The R-73 can home on the heat generated from the friction of air flowing over a target's wing, while the R-60 is not quite so sensitive. The R-60, however, is perfectly reliable when fired at the rear of a target, and is both cheap and plentiful.

This mission will train you in the use of the R-60 in a combat situation against multiple opponents. You will also learn the value of missiles as distractions. No sane pilot will ignore a missile headed towards his plane; thus, a missile fired as a distraction can be used to break up defensive formations (plus there is always the chance that the missile may hit).

Given the close range at which the R-60 operates, it is not uncommon to use it to distract an opponent and then close in on that same opponent with guns to finish the job.

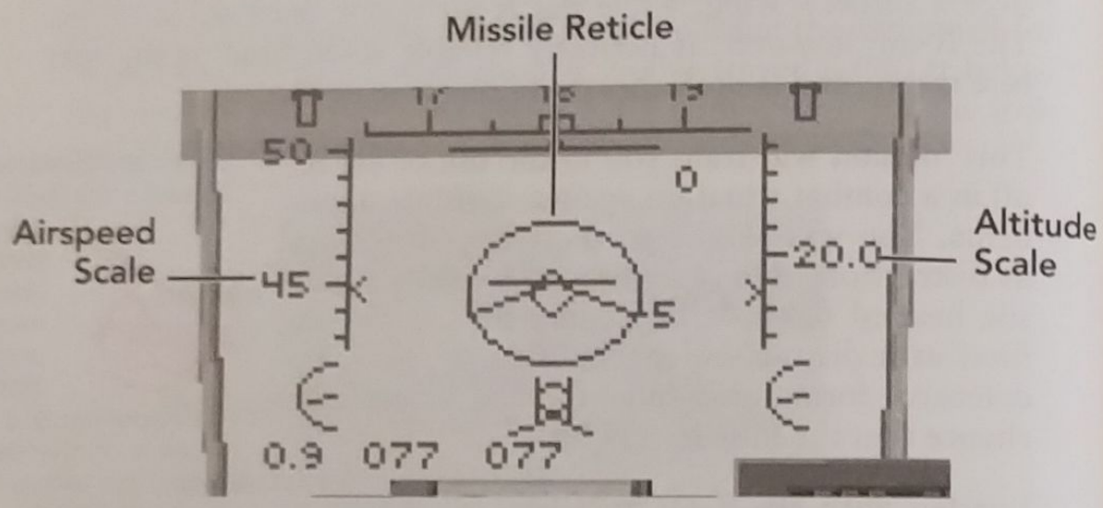
Loaded with six R-60 missiles, you begin in position behind a B-52 escorted by two F-16s. After closing to visual range and identifying the bomber, you fire one missile at each escort, causing them to break off and go into defensive maneuvers. In the few moments you have before they recover, you send your remaining four missiles towards the bomber, your real target. You can then engage the F-16s with your cannon or attempt to escape.

Mission preparation

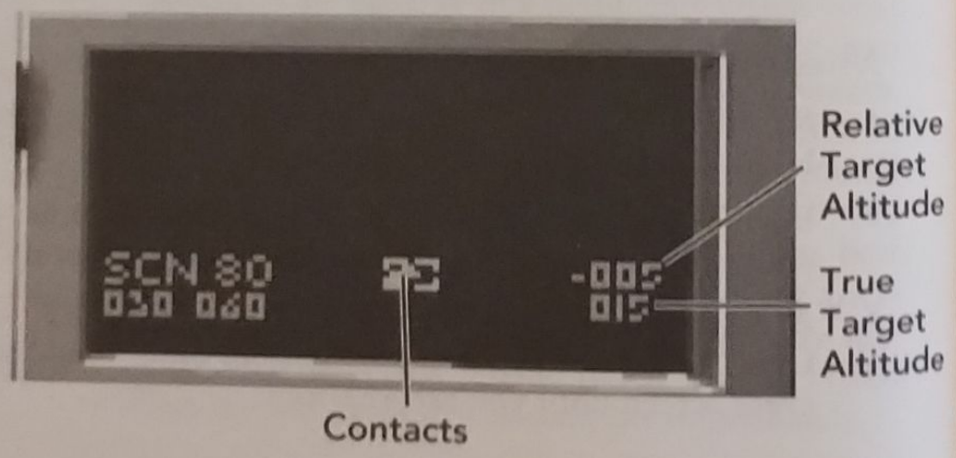
- Reread the parts of Chapter 2 which cover the air-to-air HUD and the SCN and IRST radar modes.
- Fly the R-73 training mission until you are confident in your ability to ambush and engage multiple aircraft.
- At Red Flag Control, load the mission titled "MIG29R60." Once it is loaded, click on the bottom right-hand button of every screen until you are airborne.

Identifying the targets

You should be in flight at an altitude of 20,000 ft at a speed of 450 kts. Check the altitude scale and the airspeed scale on your HUD to verify this. Then select the R-60 as your active weapon by pressing [Enter] (or button B on your joystick) until you see the missile reticle.

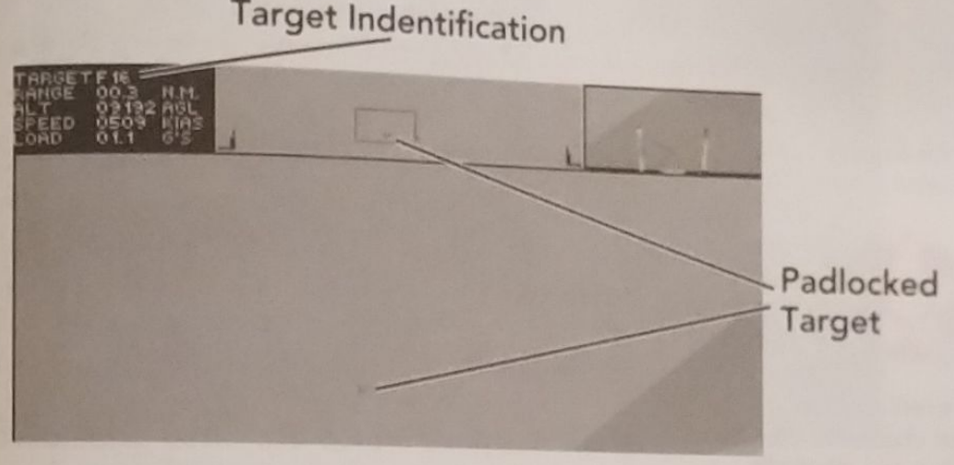


Press [F5] to switch your radar to SCN mode. You should detect a cluster of three contacts almost dead ahead. Keep track of their formation and distance by flipping the radar on and off every now and again (press [R]). Use [T] to cycle through the contacts and find out their altitude.



Close to within 10 miles, then switch to IRST mode. Press [X] to cycle through the three targets. Watch the targeting box which appears over each target on the HUD as it is selected. By watching where the box hops to, you can tell that two of the targets are in tight formation; the third is separated, flying at a slightly lower altitude.

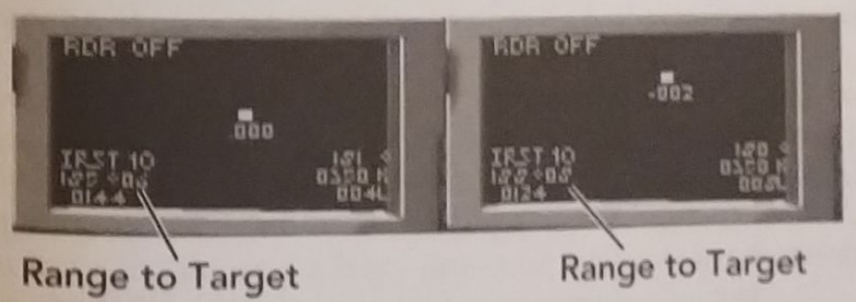
The two high-altitude contacts are probably the escort. Get them in the missile reticle first. Watch the distance close up. At about three or four miles, you can probably visually identify them in Padlock View. Press [8] to go to the Padlock View, then cycle through the visible targets by pressing [T].



There is no doubt now; the two high flyers are the escort. Time to go to work.

The attack

Very soon now, you should have a lock on one or the other of the escorting F-16s. But don't fire yet; take a quick look at what's ahead of you. Press [X] to cycle through the targets. Note the difference in distance (shown on the radar screen) between the F-16s and the B-52.



It looks as though they're about two miles apart, doesn't it? That means you should probably wait until you're about one or two miles from the escort before you fire. If you fire now, while they're three miles out, you'll have to cover the two-mile gap to the bomber before the Falcons recover. It's better to wait.

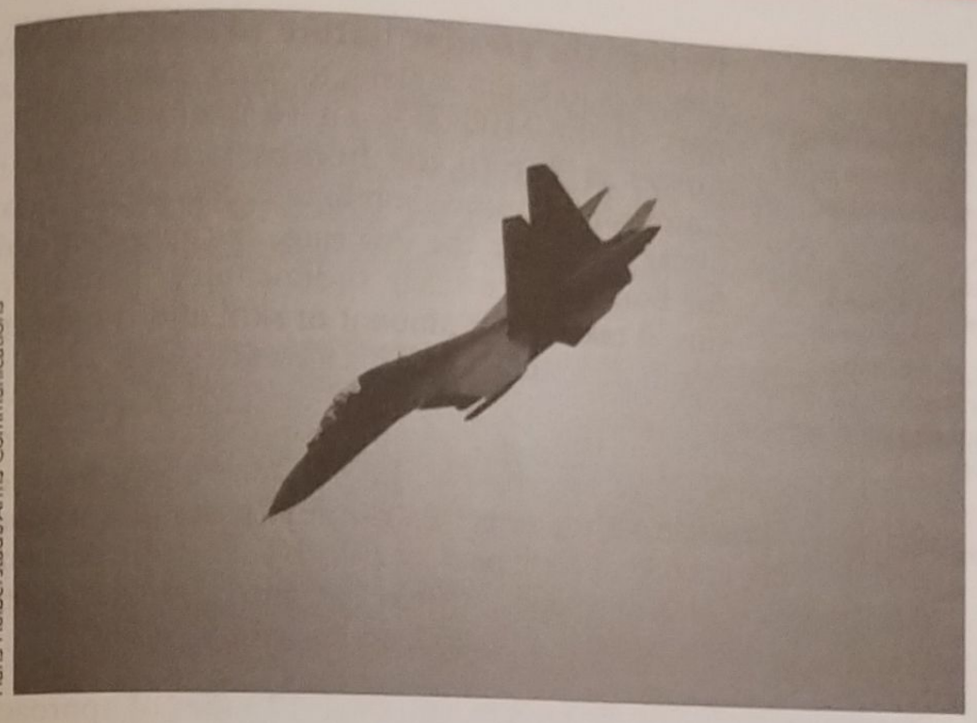
Once the range closes to one or two miles, launch by pressing Spacebar or button A. Then press X to change targets to the second F-16 and launch as soon as you see the lock indicator. Then press X again, lock onto the bomber, and release the last four at about one-second intervals.

The R-60 doesn't carry a very large warhead, so it will take more than one—maybe more than two—to bring down a B-52.

Since this is a practice mission, engaging the two F-16s would be good dogfighting practice. But in a real mission, your best choice when faced with a 1v2 dogfight against equal opponents is usually to avoid it.

Now you have three choices. You can bug out at high speed, hoping the Falcons will be too busy with your missiles to chase you right away; you can follow the B-52 in with your cannon to make sure; or you can duke it out with the surviving escorts. No matter what happens now, though, the R-60s have more than done their job.

Even weapons which are often considered obsolete or a handicap have tactical uses; it just takes a creative and skilled pilot to discover and implement them.



Hans Halberstadt/Arms Communications

AIR-TO-GROUND

While the MiG-29's central mission is air-to-air combat, it was designed as a true multirole aircraft. The Fulcrum's suite of air-to-surface weapons, while limited, is quite effective if used properly.

Bombing

Unguided general-purpose bombs are the cheapest and most plentiful munition available. In addition, they have capabilities that other weapons cannot match. Since they are free-fall weapons, there is no need for engines or fuel systems. As unguided weapons, they do not require guidance packages. Virtually the entire weapon is dedicated to warhead. This gives bombs capabilities that no other weapon can match. For example, the only MiG-29 ordnance powerful enough to take out hardened targets (such as reinforced buildings) is the big FAB-500. And if you have to take out a large group of infantry or other soft targets, the RBK-500 cluster bomb is extremely effective.

Experienced pilots can often tell just by looking when to release their bombs. If used properly, however, the bombing pipper is much more accurate than such guesswork.

Perhaps the greatest feature of bombs is how easy it is to equip a plane to carry them. A great part of the MiG-29's air-to-ground striking power is due to the bombs it can deliver. Unfortunately, since bombing is considered a secondary duty for the Fulcrum, the targeting aids for bombing are fairly rudimentary. Because of this, it takes a fair amount of skill and training to bomb accurately with this aircraft.

This mission is a simple run against a row of practice targets. Fly this mission over and over again until you can hit the target on the first pass. In a combat environment, even that first pass is risky. Taking second and third passes over a defended target is extremely unwise unless you are absolutely certain of little or no resistance.

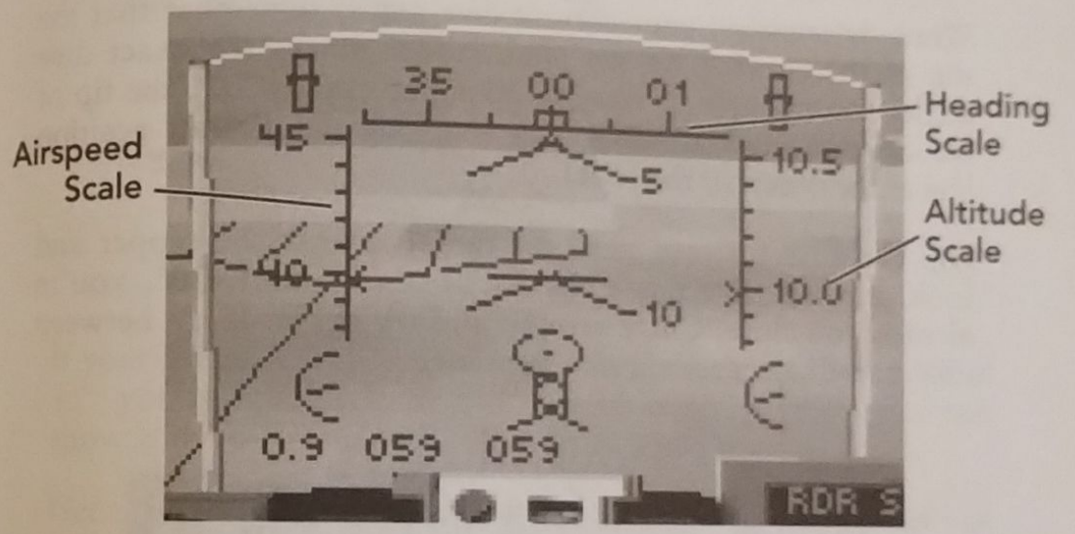
Starting due south of the targets, you have 12 miles of approach to set up your bombing run. The target is right at your waypoint, so line up on it and bring your speed to 400 kts. Your altitude should be 2,500 ft AGL. On final approach, dip the nose to a 30° dive angle, placing the pipper on the target right as you reach 2,000 ft AGL. Upon release, climb out to avoid the blast and then bank and see how you did.

Mission preparation

- Reread the parts of Chapter 2 that deal with the air-to-ground HUD, paying particular attention to the bombing information.
- Work on your control until you can rapidly get to a specified altitude, speed and dive angle.
- At Red Flag Control, load the mission titled "MIG29BMB." Once it is loaded, click on the bottom right-hand button of every screen until you are airborne.
- There are parts of this mission which happen very quickly, so be ready to pause while you read the instructions. The events in the final part, "Bombs away," happen in a matter of seconds.

The objective

At the start of this mission, you should be cruising due north (heading zero) at 10,000 ft and 400 kts. Check your airspeed scale, heading scale and altitude scale to be sure. Make sure your FAB-500 bombs are the active weapons by pressing Backspace until the armament control panel says "F500."

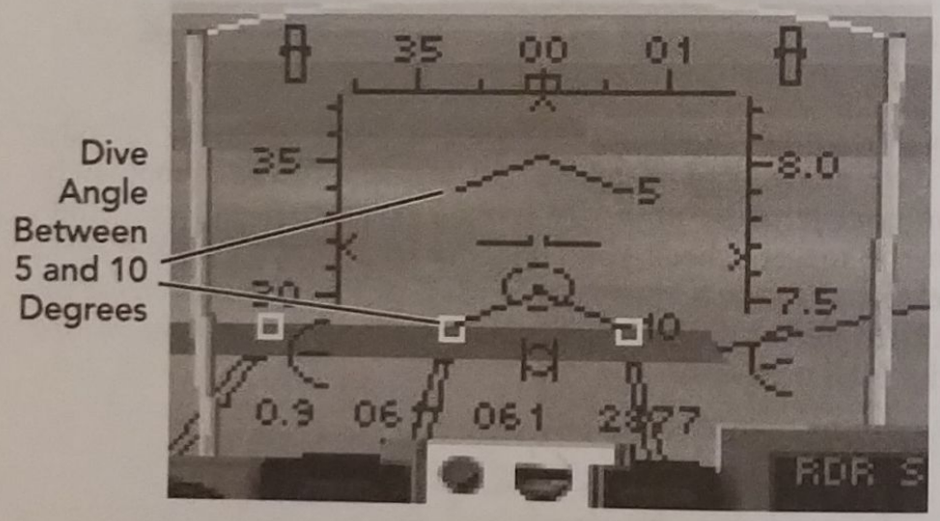


Since the pipper is calibrated to be accurate at 2,000 ft and a 30° dive angle, you should begin losing altitude immediately. Your goal is to be at 2,500 ft AGL when you are two miles from the target (check your distance to waypoint on the HSI). Level off. You can verify your AGL altitude by checking the radar altimeter on the instrument panel and the AGL indicator on the HUD.

Adjust the throttle until your airspeed is a fairly steady 400 kts. As you get closer to the target, you should look around until you spot it. At around two miles away, you should be able to see the bombing targets. Adjust your approach so that you are lined up with the target you intend to bomb. Now it's time to set up the third component of the bombing formula: the dive angle.

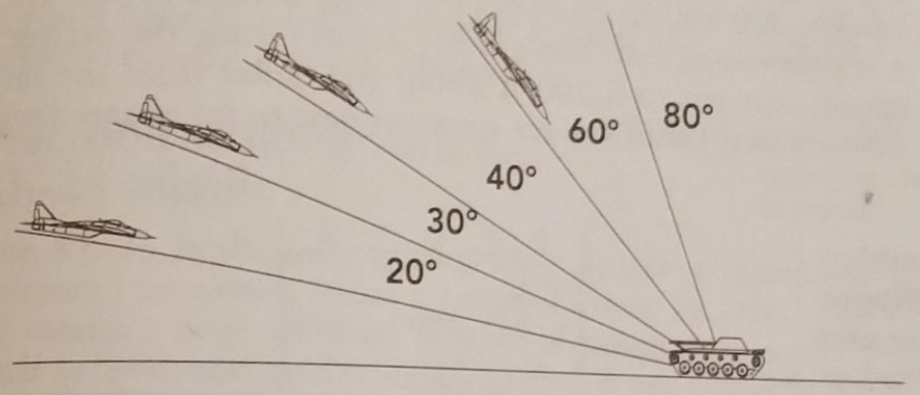
When determining your current dive angle, remember that the top of the chevron on the pitch ladder marks the exact dive angle. For example, if you are diving at exactly 10°, the tip of the chevron marked -10 will be touching the aircraft position line at the center of the HUD.

Dip the nose forward until the target is near the pipper and look at what dive angle that is. If it's beyond -30°, you're already too close. Circle around and try again. If it's between -30° and 0°, you're in a fine approach.



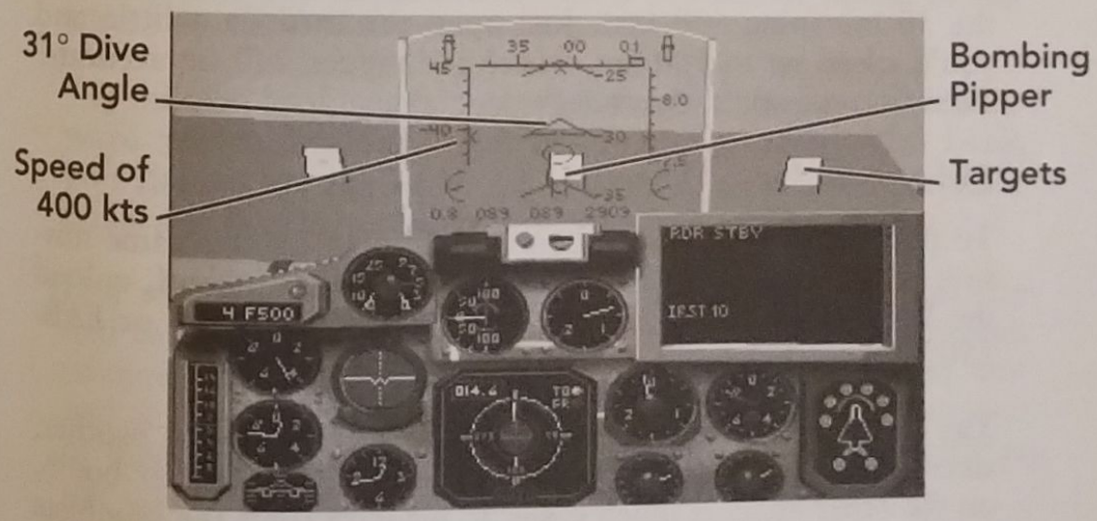
Bombs away

As you get closer to the target, the dive angle necessary to place the pipper on the target becomes steeper and steeper.



If your initial test showed that the target was at a dive angle of 15°, you should level out and fly a bit closer. If your initial test showed the dive angle was close to 25°, you're ready to go.

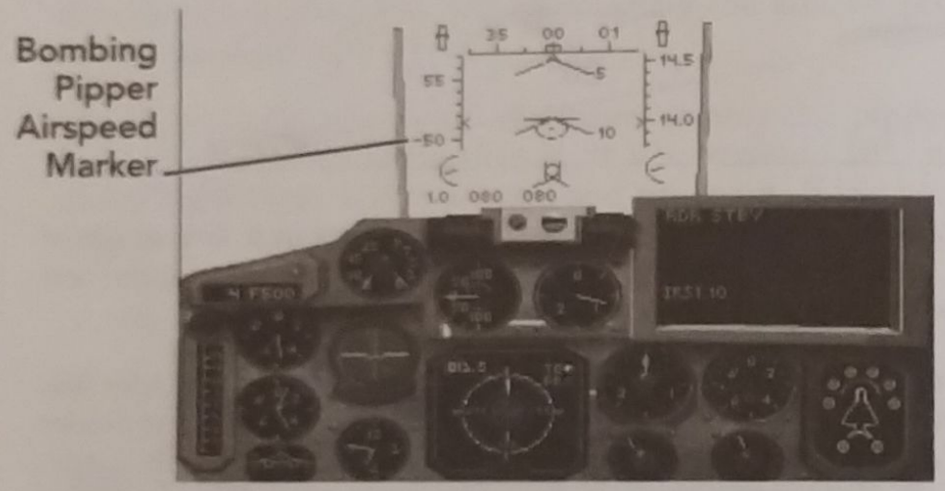
Get close enough to the target that you can place the pipper just below the target at a 30° dive angle. As you get closer, the pipper will eventually rest right on the target. If your approach is perfect, the pipper will cross the target right as your altitude crosses 2,000 ft. Press the trigger or Spacebar to release the bombs.



Immediately roll to one side and pull back hard to turn and climb rapidly. With bombs as large as the FAB-500, you should be as far away from them as possible when they hit in order to reduce the chances of damaging your plane in the blast. To check your work, circle around and take a look in Satellite View (press 1).

Changing the pipper setting

In a real combat situation, you are unlikely to have the luxury of setting up a perfect 2,000-ft, 400-kt, 30° approach. This is why the bombing pipper is adjustable. Every time you press [D], the airspeed the pipper is set for is decreased by 50 kts until it reaches 300 kts, at which point it wraps around to 550 kts. On a real bombing run, don't worry about getting the perfect airspeed; just get to 2,500 ft about two miles from the target. Then press [D] until you see a marker next to the airspeed scale.



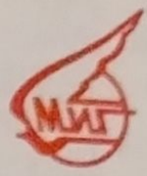
This marker shows what speed the pipper is set for. For example, if you're going 480 kts, press [D] until you see the mark by the 50 (meaning 500 kts). Just open the throttle a little and you'll speed up to 500 kts, where the pipper will be accurate. Then you can set up the dive angle.

Other bombs

To practice with the other bombs, just load up this same mission. Then, at the Armament screen just before takeoff, unload the FAB-500s and put on some FAB-250s, RBK-500s or FAB-250 HDs.

The FAB-250 is a bit harder to hit with than its big brother, since it has a smaller blast radius. The RBK-500 cluster bomb, on the other hand, is much easier to hit with since the bomblets spread out over a large area.

If you want to practice with high-drag bombs (such as the FAB-250 HD), the important thing to note is that you can release at a much lower altitude than with the normal bombs. A lower altitude means a much reduced chance of detection. But since the pipper is calibrated to 2,000 ft, you'll have to learn to eyeball it (which, fortunately, is easier at lower altitudes).



While high-drag bombs allow lower safe release altitudes, the pipper is still only accurate at 2,000 ft. To bomb at a lower altitude, practice until you know by experience when to release.

Precision striking with the Kh-23 missile

The Kh-23 is the only guided air-to-surface missile in the Fulcrum's inventory. It has a range of about 10 miles. The Kh-23 carries a large warhead capable of penetrating hard targets such as tanks and smaller buildings, but is not powerful enough to damage an ultrahardened target such as a bunker.

It is used for situations where the target is small, mobile or in a civilian area. One of the best uses for the Kh-23 is to disable enemy surface-to-air installations, but such missions are highly risky and probably best left to dedicated strike aircraft. In this mission, you will be attacking an armored column.

The tanks are relatively small and mobile; hitting them with a bomb would be quite difficult, and their armor would shrug off cannon fire and rockets. It is clearly a situation calling for the Kh-23 missile.

The mission profile is fairly straightforward: You make a southward approach along the river. On the western (right) bank, you should see the armored column. Lock on to one of the tanks and fire, then change targets to another one and fire.

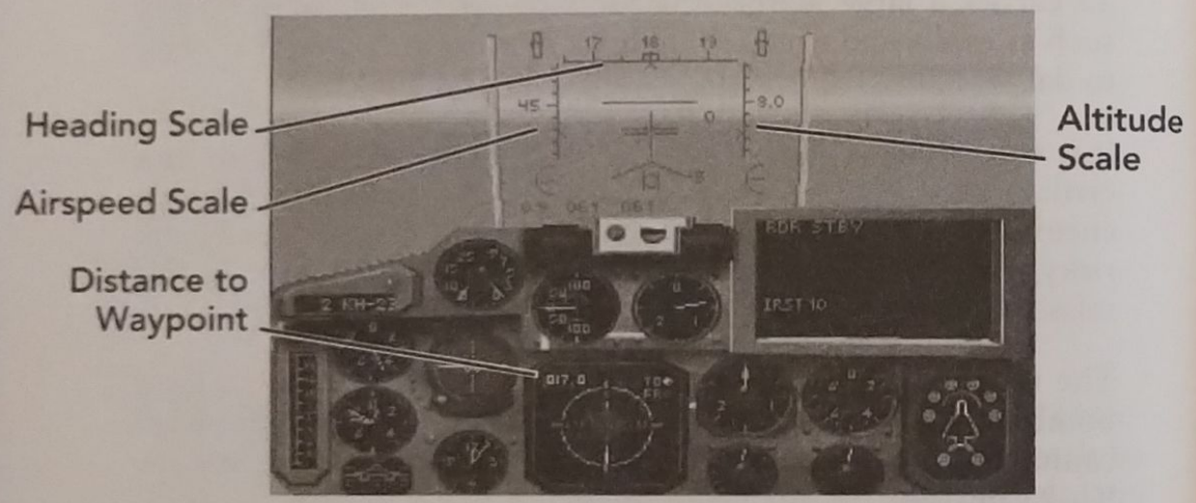
In a real wartime situation, if an armored column of this size had to be stopped, you would probably be sent with enough wingmen to wipe out the entire group. But if this enemy unit had already engaged friendly forces, the destruction of even two of their number could be enough to turn the tide of battle in our favor.

Mission preparation

- Read the sections in Chapter 2 which deal with the HUD, paying particular attention to the air-to-ground targeting information.
- At Red Flag Control, load the mission titled "MIG29K23." Once it is loaded, click on the bottom right-hand button of every screen until you are airborne.

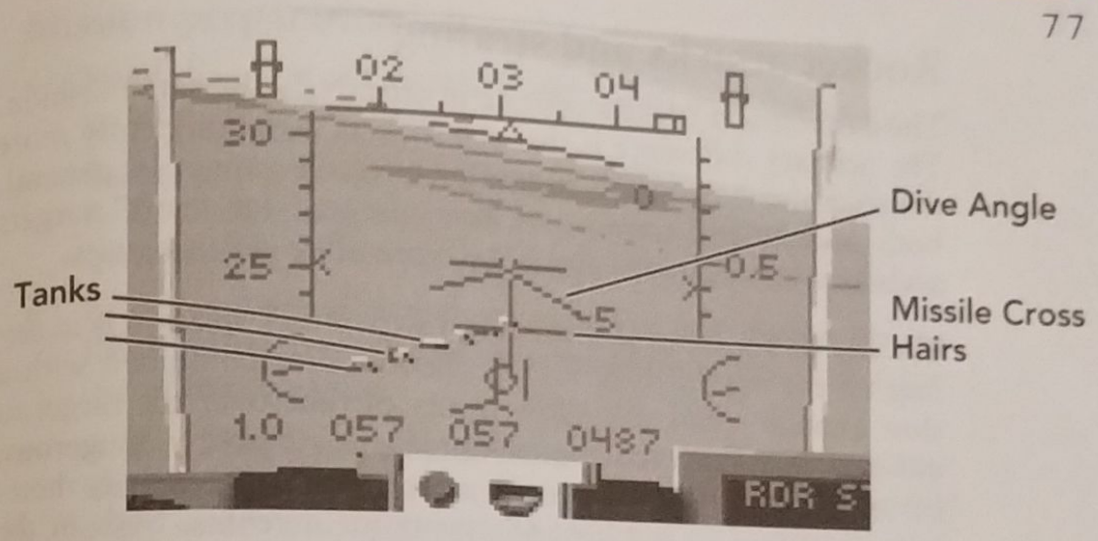
Closing on the objective

At mission start, you should be 8,000 ft up, traveling at 350 kts. Your first waypoint is the last reported position of the enemy column, due south of you by the river. Check your heading scale, altitude scale and distance to waypoint.



In order to spot your targets, you will have to lose some altitude. Dive to about 2,000 ft AGL (check your radar altimeter and AGL indicator). Select the Kh-23 as your active weapon by pressing Backspace until the armament control panel says "KH23."

As you close to within five miles of your waypoint, start looking around for your targets. At about three or four miles from your waypoint, you should see a row of gray-colored vehicles—the enemy.



Level out and approach in a shallow dive, no more than 30°. Place the missile cross hairs on a target. The missile will automatically lock on. You can tell the missile is locked on if the cross hairs "stick" to the target—that is, the cross hairs stay on the target even after you stop pointing at it.

Fire and forget

Press the trigger (Spacebar or button A on your joystick) to fire the first missile. Then, while that missile is in flight, press T to change targets for the remaining missile. After all, one of these missiles is more than powerful enough to destroy a tank. When the second missile is locked, launch it immediately.

Tougher targets

Once you have completed this mission successfully a few times, you should be ready for the second stage. Instead of striking the tanks, go past them to your second waypoint. Two air defense units, a missile launcher and a gun battery, are posted here to provide cover for the advancing armor unit. If you place each of your missiles accurately, you can dispose of both anti-aircraft units.

When given a choice between a SAM site or a AAA battery, always hit the SAM site first.

In a campaign, good use of the Kh-23 missile can clear a corridor in enemy air defenses for your strike aircraft to get through. But it takes good piloting to avoid getting shot down in return.

Rocket attacks and strafing

The cannon and rockets operate in effectively the same fashion. The primary difference is that the rockets do significantly more damage and have a much greater effective range. In general, both rockets and cannon are best reserved for "soft" targets such as infantry, trucks and some types of air defense units.

The key to a successful strafing run with either weapon is in setting up the approach profile. A shallow dive combined with a slow enough speed will allow plenty of time over the target to aim and fire. This type of attack profile is inherently dangerous; flying low and slow makes you an excellent target. It is therefore important to suppress all enemy air defenses, both in the air and on the ground, before strafing.

This mission comes in two parts. The first is a simple practice run against a column of trucks. The second is an attack on a miniature enemy encampment, with anti-aircraft defenses and an F-4 flying combat air patrol (CAP) over the area.

In the practice field, simply strafe the targets as you see fit. In the second area, you should first take out the enemy aircraft and then destroy the SAM launcher. Finally, you can set up your strafing run against the rest of the encampment.

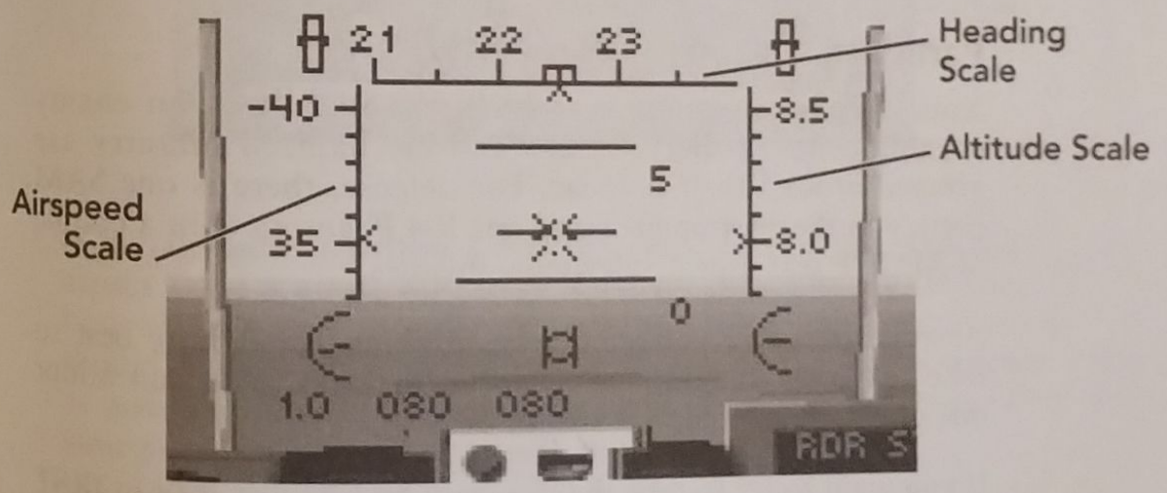
Mission preparation

- Read the parts of Chapter 2 which deal with the air-to-ground HUD.
- Before you go on to the second half of this mission, fly the R-73 mission and air-to-air cannon mission until you are quite comfortable with the basic techniques of air combat.
- Before you go on to the second half of this mission, fly the Kh-23 mission until you are satisfied with your skill in the use of that weapon.
- At Red Flag Control, load the mission titled "MIG29STF." Once it is loaded, click on the bottom right-hand button of every screen until you are airborne.

Milk run

You begin at an altitude of 8,000 ft at a speed of 350 kts, headed southwest toward a river. Just on the other side of this river is a row of enemy supply trucks which has stopped temporarily.

Drop down to 2,000 ft above ground level (AGL). When you



get within five miles of your waypoint, you should see a row of vehicles. Those are your targets. Go into a shallow dive towards them with either cannon or UB-57 rocket pods active. Be sure the cannon is in strafing mode (press Backspace) if that is the weapon you plan to use.

The truck column is parked right on your first waypoint, so check the distance to waypoint indicator on the HSI to find out the distance to your target.

Since the rockets have an effective range of over a mile, you could start firing them as soon as you are lined up on the targets. Unfortunately, it is quite difficult to aim the rockets at this range, so it would be better to wait until you are close enough to be sure of a hit.

When you strafe, fire several short bursts from the gun or brief salvos from the rocket launchers. This uses less ammunition than a long, continuous stream while still covering the same area. If you plan to go on to the second half of this mission, try to conserve ammunition.

You may need to make several passes over the column to finish them all off. Since there are no threats in this area, you have the time to make those extra passes. In the second half of the mission, however, you will not have that luxury (at least not at first).

Practice the first half until you feel you are accurate enough to go on. In the second half of this mission, there is a good chance that you may have to strafe a SAM battery. The need for accuracy in such a situation is obvious.

Part 2: strafing in the real world

Your second waypoint is right in the middle of an enemy encampment. Artillery, trucks and mechanized infantry are arrayed around their bivouac. For defense, there is one SAM battery in the encampment and one F-4 Phantom with a couple of AIM-9P rear-aspect Sidewinder missiles.

As you approach, you should be thinking about how best to take out enemy aircraft on patrol in the area. You can adopt one of two general tactics: stealth or BVR.

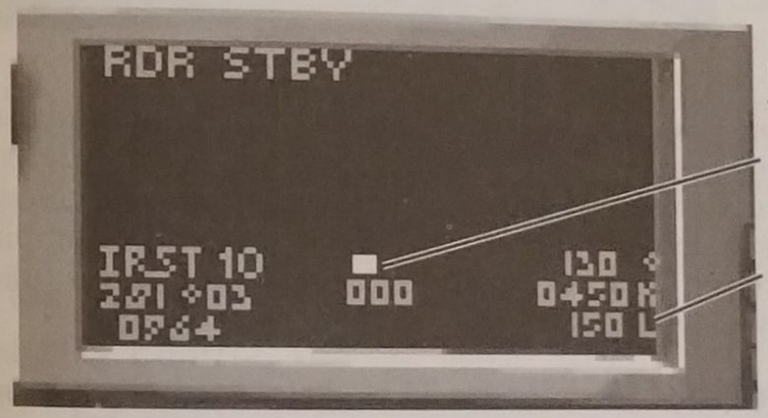
If you want to try to take the Phantom by surprise, stay in IRST radar mode with the radar turned off (press R) to be certain of not giving yourself away with radar broadcasts. Hopefully, as long as you can stay high enough (or low enough) to be outside his radar cone, you should be able to lock him up without his knowledge. Then you can give him a couple of R-73s, switch to automatic cannon and chase him down.

To pull off a complete BVR intercept, you'd have to be carrying some R-23Rs or other long-range missiles. But you can use active radar to lock him up early and perhaps spook him into coming after you, which would keep you away from that SAM battery while you're dogfighting.

A better solution might be a combination of the two, where you use the radar to scan briefly at long range and then switch to IRST for the interception. This is the tactic used in the earlier air-to-air missions.

Keep an eye on the information in your radar display, especially the target altitude and aspect angle.

To get the target's attention with active radar, switch to SCN mode. Once you pick up the target, press T to place the target brackets on it. Press Z to go to TRK mode. This locks the target up in radar, which will definitely get the other pilot's attention.

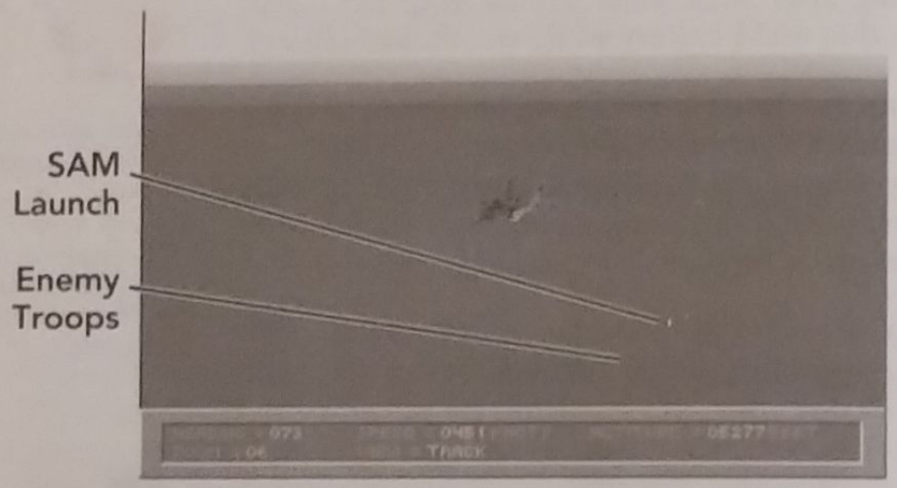


If his altitude is within a few thousand feet of you and the aspect angle is above 60°, there is a good chance he can see you both visually and on radar. In order to surprise the target, you should stay far above or far below him and behind him as much as possible. Ideally, the first knowledge the enemy will have of your presence is a pair of missile launches.

Once the enemy aircraft is dispatched, it's time to remove that SAM battery that has no doubt been busily trying to kill you throughout the dogfight. Switch to the Kh-23 (press Backspace until the armament control panel says "KH23").

Now you must use your eyes; that's the only way to tell which is which down there. Trucks are green, artillery pieces are dark brown, and all the armored vehicles are gray—including the Roland SAM launcher you're hunting for. You could take two random guesses with your missiles, but if you're wrong, this mission might end right here.

You can also take a riskier approach, if you think you're good enough at dodging SAMs. Fly over the target area, keeping the encampment in view as long as possible. Press [7] to go to Track View, then pan around your aircraft using [F3] and [F4].



When you hear the launch warning from your radar warning receiver, notice which general direction the missile is coming from. That should narrow your search quite a bit. Now all you have to do is evade that SAM! Bank hard while dumping flares and chaff. You can switch to Padlock View (press [8]) to keep an eye on the missile.

Once you've identified the SAM site, you can lock it up with a Kh-23 and destroy it. But if you are out of Kh-23s, you'll have to strafe. Use rockets, not the cannon, and don't miss. While it will be difficult for the Roland to hit you head-on, your twin tailpipes are perfect targets.

While the cannon will do just fine against infantry and trucks, you will need to use rockets to destroy the armored vehicles.

Once the SAM site is taken care of, you can begin strafing runs through the enemy forces. In a real wartime situation, you will have considerably more help. No sane commander would send one aircraft into a known hostile area. But, on the other hand, you will not run into many enemy bases guarded only by one SAM and one under-armed fighter.

ADVANCED TRAINING TECHNIQUES

These missions provide a basic training in the weapons systems of the MiG-29. They are designed to be easy enough that you need not be an advanced pilot to complete them. With a few small changes, they can be made significantly more difficult. Here are some suggestions for how to make these missions a little more challenging.

The Difficulty Levels screen

One of the fastest and simplest ways to increase the difficulty of the training missions is to increase the difficulty you fly them at. From the War Room, select the Config monitor. On the Difficulty Levels screen which then comes up, select Advanced. You will suddenly find these missions quite a bit more challenging.

Other items on the Difficulty Levels screen which can make a major impact are Enemy Logic Level, SAMs-AAA and Flight Model. After boosting these difficulty levels, you may soon find yourself the target instead of the hunter.

Red Flag Control

At Red Flag Control, you can make many adjustments to these missions or even create your own. The simplest ways to make the missions more like a real combat situation are to add wingmen, add more air defenses, add enemy aircraft, change the enemy's orders or alter the weapons available to the enemy.

A complete explanation of how to use the Red Flag mission builder can be found in the *Falcon 3.0* manual.

Adding wingmen

From Red Flag Control, click on Nellis Command. On the Nellis Command screen, click on Friendly Aircraft. This will bring up the Friendly Aircraft screen. On this screen, you can add wingmen to help you by clicking Add Aircraft.

Once you're satisfied with the size of your flight, click on the bottom button until you get to the Armament screen. You can set up the armament of each of your wingmen (or just choose Copy Leader to arm them exactly as you are).

Wingmen and wing man commands are explained thoroughly in the manuals for *Falcon 3.0* and *Operation: Fighting Tiger*.

Working with wingmen is an important aspect of air combat operations. In addition, if you are having difficulty with a mission, adding some helpers will improve your success rate somewhat.

Adding air defenses

These missions are quite light on enemy air defenses. In the air-to-ground missions especially, this is not realistic. Virtually every enemy combat element has some type of air defenses.

To add air defenses, click on Nellis Command. Then choose Coalition Forces. The Coalition Forces screen will show you what types of ground units are in each enemy troop concentration. Just add some SAMs, AAA or infantry (who carry shoulder-launched missiles) to the formation, and your job will suddenly get a bit more interesting. When you're done here, just keep choosing the last button until you're airborne.

Adding enemy aircraft

In a real engagement, the enemy will likely have much better air cover than you've seen in these training scenarios. To beef up the enemy's air cover, select Nellis Command and then click on Coalition Aircraft.

On this screen, there are two ways to toughen up the enemy. Click on Add Aircraft to increase the number of hostiles you will face. In the small screen in the upper right corner is a list of available enemy aircraft. Choosing a tougher opponent will convert one of the enemy planes. If your opponent is a single F-4, as in the second half of the strafing mission, you can change it to an F-15 or F-16 for a more challenging engagement.

Changing the enemy's orders

There are a number of different actions the enemy can be ordered to undertake. By changing their orders, you can turn passive targets into aggressive opponents. From Red Flag Control, select Nellis Command. Click on Coalition Aircraft. Finally, choose Coalition Waypoints. Here is where you can change their orders.

Look at what is written in the space next to the button marked Action. If it says CAP or INTERCEPT, the enemy is actively looking for you and will engage. If it says NOTHING, the enemy will blithely cruise around while you shoot at them.

To change their orders, click on the Action button until the orders say what you want. For the purposes of these misssions, the only actions that have any relevance are CAP, INTRCEPT and NOTHING. If you create your own missions, you may want to arrange for other action types.

Other waypoint actions include such choices as BOMB, CAS (close air support of ground troops) and S AND D (search and destroy—attack any targets encountered).

Once you're done with the first enemy waypoint, click the Waypoint # button to go to the next one and modify that waypoint as you see fit. When you're finished changing the enemy orders, just pick the last button on every screen to get to the mission.

Improving enemy weaponry

The weapons available to the enemy have been limited in those missions involving combat aircraft. While this is good for teaching basic techniques, the enemy is unlikely to be so poorly equipped in reality. Altering the enemy's arms can change missions dramatically as foes that previously had to close to guns range can now have you dodging missiles.

From Red Flag Control, click on Nellis Command. Select Coalition Aircraft, then Coalition Waypoints and then Coalition Setup. On the Coalition Setup screen, you'll see a list of armaments with circles next to them. A filled-in circle means the weapon is available to the enemy. An open circle means the enemy will not be armed with that weapon. Just select those weapons you want available to the enemy by clicking on the circles. Click on the last button on every screen to get to the mission.

Sergey Skrynnikov/AviaData/Arms Communications

TACTICS

ТАКТИКА

4

ЧЕТЫРЕ

Sergey Skrynnikov/AviaData/Arms Communications

AERODYNAMICS AND NON-FBW AIRCRAFT

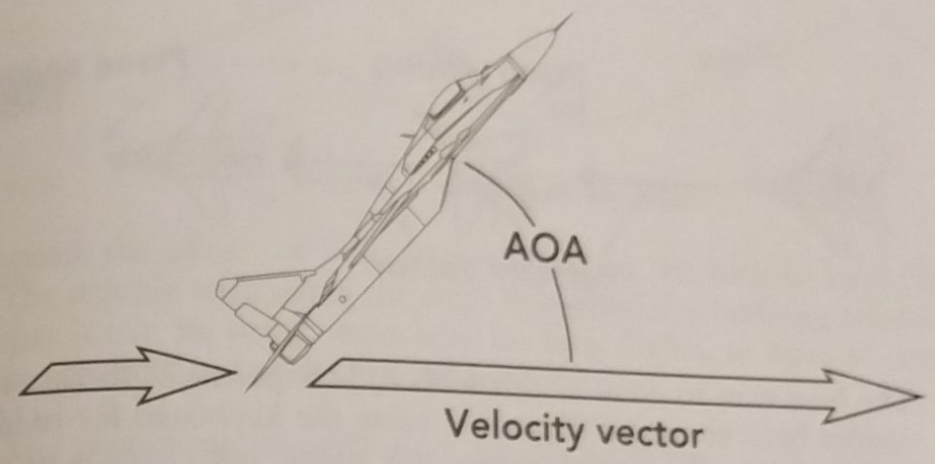
The MiG-29 Fulcrum has a hydro-mechanical flight control system (FCS), as opposed to the fly-by wire (FBW) FCS used in the F-16. The lack of an FBW system has two effects. It allows the pilot to push the outermost edges of the plane's capabilities since there is no computer to override the pilot's control inputs. FBW systems generally have built-in G limiters and other safeguards to keep the pilot from maneuvering the aircraft beyond its controllable flight envelope.

Non-FBW aircraft such as the MiG-29 also require the pilot to do more work in general. FBW systems handle many of the small details of flight, which are now left to the pilot.

AOA, pulling Gs and the flight stick

One area in which there is a significant difference is in pulling Gs to bring your weapons to bear on a target. When you pull Gs, you also create a significant angle of attack (AOA) for as long as you pull back on the stick. As the AOA increases, your nose rises more and more above your velocity vector.

When you center the stick, the plane's tendency is to snap back into line with its velocity vector (depending on its speed and trim; see below).



When you center the stick in an F-16, the FCS interprets this to mean you want your aircraft to fly at exactly 1 G and gradually eases up on the control surfaces so that the snap-back effect is greatly reduced.

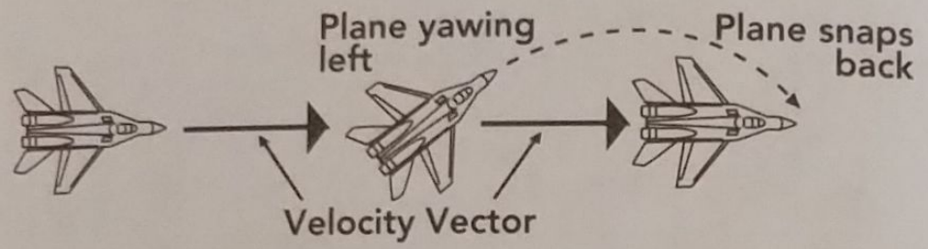
The MiG-29, however, will immediately center the control surfaces (since that is exactly what you told it to do when you centered the stick) and drop back in line with its velocity vector, making it seem as if the nose suddenly dipped downward.

To counter this effect, ease forward on the stick instead of letting it spring back to center, and the plane will stay pointed where it is. The velocity vector will gradually catch up with the nose as the AOA drops, assuming the trim controls are adjusted properly for the current airspeed (see "Trim controls" below).

When flying with the keyboard as your primary flight control, this snap-back effect can make the plane almost unflyable. Unless you have a joystick, we strongly recommend you use either the Easy or Average flight model.

Rudder-induced yaw

When you use the rudder to adjust your attitude, the plane yaws sideways relative to its velocity vector. Just as centering the flight stick in a high-AOA situation causes the nose to suddenly fall back in line with the aircraft's velocity vector, centering the rudder in a high-yaw situation causes the plane to snap sideways to realign with the velocity vector.



The best way to counter this with rudder pedals is to ease the rudder back to center. If you are using the keyboard for rudder control, the only way to reduce this snap-back effect is to tap the rudder key repeatedly instead of just releasing it when you wish to center the rudder.

Trim controls

The trim controls are adjusted automatically in all flight models except Realistic.

Another of the many things an FBW system does for you is adjust the trim. In the MiG-29, trimming is up to the pilot. To understand what the purpose of trim controls are, it is useful to have a basic understanding of certain aerodynamic effects.

All aircraft can remain in level flight at the one particular speed the trim controls are set for. The F-16's FCS automatically adjusts the trim controls to match whatever speed the plane is going. In the MiG-29, the trim controls are manually operated. Every time the aircraft speeds up or slows down, the trim must be adjusted to accommodate the new speed.

If the Fulcrum goes faster than the trim speed, it will gradually nose up. As the plane noses up, it slows down, which brings it back to level flight speed. Once level, it speeds up again and the nose rises, starting the cycle over.

The reverse is true if the plane drops below trim speed: the nose drops, the plane speeds up, the nose returns to level, the plane slows, the nose drops, and so on.

This constant bobbing (called an out-of-trim condition) can be controlled by using the stick. If the plane is going too fast to fly level, the pilot simply pushes forward a bit on the stick and the oscillation stops. If the plane's nose is dipping due to low air-

Aircraft in out-of-trim condition

speed, the pilot just pulls back slightly on the stick to level off. The trouble with using the stick to control this bobbing effect is that it can be tiring, especially for long flights, to have to constantly hold the stick forward (or back).

The trim controls take care of this problem. Applying a little forward trim keeps the plane level if it is going too fast; applying some back trim will level the aircraft if it is going too slow. Once the proper amount of trim has been applied, the pilot can center the stick and the plane will fly straight and level.

As an example, assume you are flying level at 400 kts with the stick centered. You look at your fuel consumption and the distance to your waypoint and decide to increase speed to 500 kts in order to arrive sooner. The nose begins to rise. By pushing forward slightly on the stick, you can fight this tendency and remain in level flight. But you will have to maintain this forward stick pressure for the rest of your flight unless you adjust the trim. Applying some forward trim will allow you to release the stick and still fly level.

The trim controls are [2], [4], [6] and [8] on the numeric keypad if you are using a joystick for flight control. If you have a ThrustMaster FCS or CH FlightStick Pro, the coolie hat can be toggled between view control and trim control by pressing [V]. If you are flying with the keyboard only, the trim keys are [Ctrl] [2], [Ctrl] [4], [Ctrl] [6] and [Ctrl] [8] on the numeric keypad. See Chapter 2 for more information on controls.

Left and right trim are used if your plane is slewing to one side or the other (possibly due to uneven distribution of drag). Simply apply left or right trim as appropriate until the plane straightens out.

Blackout and redout

In the F-16, the FCS will limit the amount of Gs the pilot can pull. In the Fulcrum, the only limit on G loading is how much the pilot can stand and the structural integrity of the airframe.

As Gs increase, blood pools in the lower areas of the body, decreasing the brain's blood supply. For most pilots, at around

9 Gs (or after several seconds at 7–8 Gs) the pilot's brain will actually shut down due to lack of oxygen. This condition is known as blackout.

Blackout can also be caused by the concussive effects of a sudden increase in Gs. If a pilot is already pulling 6 Gs, easing up to 8 or 9 Gs is not likely to induce blackout immediately. But a pilot who yanks on the flight stick, jumping almost instantly from 1 G to 9 Gs or more, is certain to black out from the sudden shock.

Redout is a condition caused by negative Gs. Blood pressure in the upper body increases, eventually causing a loss of consciousness. For most pilots, this occurs at around -3 Gs.

To avoid blackout, do not pull high Gs for extended periods of time. The stress on the pilot builds up unless it is relieved periodically. When you must pull Gs, do so gradually to avoid the concussive effects of sudden G loading.

In *MiG-29* and in *Falcon 3.0* (after you install *MiG-29*), the blackout/redout conditions are dependent on what flight model you have selected on the Difficulty Levels screen. Blackout and redout conditions are defined by pilot G tolerance, maximum blackout duration and degree of control over the plane during blackout. In all cases, increasing your G load gradually instead of suddenly will greatly extend the onset time of blackout and redout.

Your pilot's G tolerance decreases over the course of the mission due to the cumulative strain of repeated G loading. The more Gs you pull, the more strain you put on the pilot. As a pilot's G tolerance decreases, blackout and redout will occur more suddenly and at lower G values.

Pilots flying the Easy flight model (or Simplified flight model in *Falcon 3.0*) have three times the G tolerance of a normal pilot. This means they can pull more Gs for a longer period of time than pilots in the other flight models. The Easy flight model has a maximum blackout time of one second; this means that once you stop pulling Gs (by centering the stick), you will recover consciousness within a second. While you are blacked out in the Easy flight model, you can still control the plane—you just can't see.

In the Average flight model (or Moderate flight model in *Falcon 3.0*), pilots have double the normal G tolerance. As in the Easy flight model, pilots will recover consciousness a maximum of one second after centering the stick. Blacked-out pilots using the Average flight model have no control over the plane.

The Difficult flight model (or Complex flight model in *Falcon 3.0*) gives pilots one and a half times the G tolerance of a normal pilot. The most severe cases of blackout can last as long as 15 seconds after centering the stick, and unconscious pilots cannot control the aircraft at all.

Under the Realistic flight model (or High Fidelity flight model in *Falcon 3.0*), you will be allotted the G tolerance of a normal pilot. In the worst situations, blackout can last as long as 30 seconds after the stick is centered. In the Realistic flight model, as in the real world, you have no control over your Fulcrum while you are unconscious.

Air Combat Tactics

Since the primary mission of the MiG-29 is air superiority, it is important to provide tactical advice to its pilots. A brief technical lecture on BVR (beyond visual range) interception techniques will be followed by discussion of some more general tactical concepts.

The tactical instruction on BVR intercepts is provided by F-16 pilot Pete "Boomer" Bonanni. The 1v1 tactics discussed are valuable in any part of the game, while the two-ship tactics are most useful in the multiplayer campaign or multiplayer allied games, where your wingman is a human pilot.

Pete Bonanni has also done an extensive training aid for air combat tactics when within visual range called the *Fighter Air Combat Trainer (FACT)*. *FACT* includes a workbook, videotape and disk that can be used with *Falcon 3.0* or *MiG-29*.

The sections on passive interception and off-boresight interception are more general in nature. The intent is to give the MiG-29 pilot an understanding of these general principles, which can then be applied in tactical intercepts.

Fulcrum BVR tactical intercepts

This section will focus on one of the most difficult parts of fighter employment: tactical intercepts. Most successful air-to-air engagements start with a radar contact and end with a fireball. There may be a tallyho wedged somewhere in between, but actually seeing the bandit is not always necessary to achieve a kill. The Fulcrum carries the R-23R radar-guided missile that can be launched BVR at an adversary. With this missile and the N-019 radar, the MiG-29 can easily turn a radar contact into a dead bandit without ever seeing the enemy visually. The *MiG-29* simulation (and of course *Falcon 3.0*) allows fighter pilots to work

in two-ship elements in the multiplayer modes, so this tactical discussion will emphasize element intercept tactics. Here goes.

Rules of engagement (ROE) only apply in the Kurile Islands theater. See Chapter 5 for details.

Tactical intercepts have two primary purposes. The first is to drive your jet into weapons parameters, and the second is to arrive in a visual fight with an offensive advantage. If the rules of engagement (ROE) allow you to shoot BVR, then you may kill the bandit before ever getting to a visual fight.

Fighter pilots strive to achieve these basic objectives through two general tactical intercept types: direct confrontation or deception. Direct confrontation intercepts can be performed in any size element. Deception intercepts require at least a two-ship element.

Direct confrontation

As the name implies, direct confrontation involves going right at the bandits by executing a collision, conversion and/or vertical intercept. Another term for a direct confrontation intercept is a "duke it out" intercept.

In most tactical intercepts, the bandits are aware of your presence and are probably flying a tactical intercept on you as well. Therefore, direct confrontation should only be used when your MiG-29 has a definite tactical advantage over your opponent. If you match the Falcon/AMRAAM combination against the Fulcrum/R-23R in the simulation, they are roughly equal; in this case, direct confrontation may get you killed. If you are fighting a Falcon without the AMRAAM, then a "duke it out" game plan is the way to go.

There are two types of direct confrontation intercept: collision and conversion. A collision intercept is used to get to the target as fast as possible. A conversion intercept is used when you plan to engage in a turning fight at the merge. The term "merge" refers to the point in time and space when the fighters transition from a radar intercept to a visual fight.

A collision intercept is generally used when you do not plan to turn at the merge. Fig. 1 shows a collision intercept.

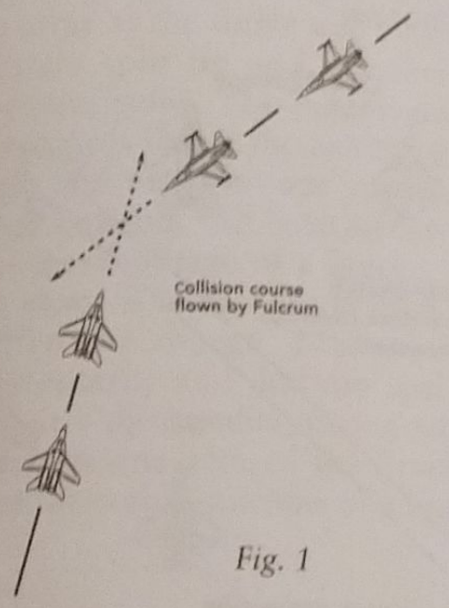


Fig. 1

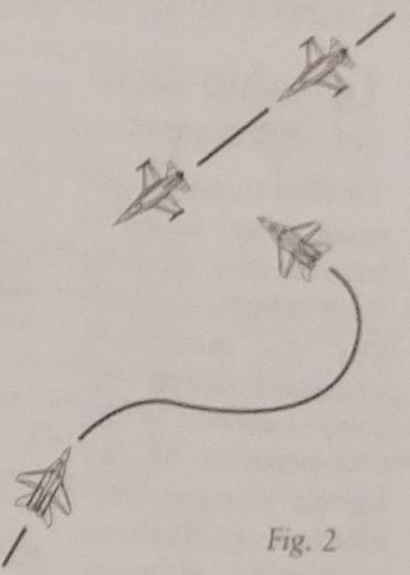


Fig. 2

In a standard (or horizontal) conversion intercept, you offset the bandit to get turning room so you can "convert" to his six o'clock. Fig. 2 shows a conversion intercept.

A vertical intercept is a type of conversion intercept that uses altitude to gain turning room instead of horizontal offset. A vertical intercept is usually used in combination with straight conversion intercept so that a Fulcrum has both horizontal and vertical turning room. Fig. 3 shows a vertical intercept.

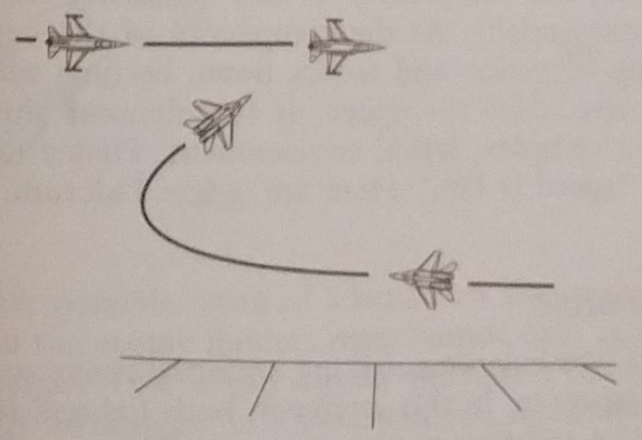


Fig. 3

Formations

Before continuing with our discussion of tactical intercepts, let's discuss element formations. There are a large number of two-ship formations available to the Fulcrum pilot which vary based on the threat. If the element is offensive, then the wingman will

generally fly back from the leader. If the element is defensive, then the wingman will fly line abreast to check six o'clock. Fig. 4 shows various two-ship formations.

Two-ship tactical intercepts

Tactical intercept formations are flown to provide the element with mutual support and to maximize available firepower. Since survival is the top priority of any fighter mission, formations are flown to maximize element survival.

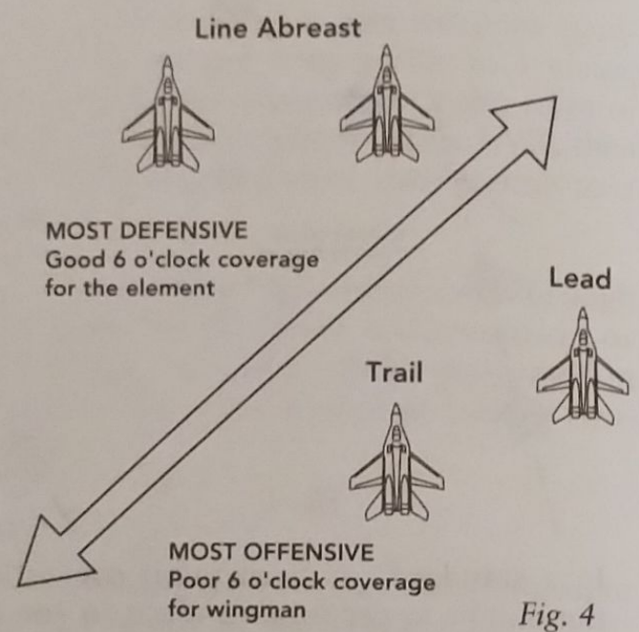


Fig. 4

The offensive formations shown are only used when the element's situational awareness (SA) is high. Should fighter SA deteriorate at any point in the intercept, the fighters should move into a defensive line abreast formation as soon as possible.

As a general rule, when the number of players (good guys and bad) increase, the complexity of any resulting engagement increases exponentially. As the complexity of the engagement increases, the formation and tactics flown become simpler and more defensive. Also the speed of the element should also increase in a complex, lethal environment. Hence the fighter pilot term, "speed is life." Here are a few Fulcrum intercept options.

Single side offset

Fig. 5 shows two Fulcrums making an initial move to get horizontal turning room. In this intercept, both fighters turn away from the target and stay on the same side. This is a direct confrontation, conversion intercept. This type of intercept is the most widely used two-ship conversion intercept. Note that both MiG-29s started in line abreast formation but ended up in wedge. The biggest advantage to this intercept is simplicity. The biggest drawback is that if the bandits detect you and turn into you, neither fighter has turning room to convert to six o'clock.

Bracket

The Bracket intercept (also called the Pincer) is the same as the single side offset except that the fighters split up and run their intercepts on opposite sides. The biggest advantage of this intercept is that if the bandits stay together as a flight and turn into one arm of the bracket, the other Fulcrum will have an entry to six o'clock. The disadvantage of a bracket intercept is that the element loses sight of each other until they get to the merge. Mutual support is lost momentarily and that can lead to trouble. Fig. 6 shows the bandits staying together and turning into one arm of the bracket. This is the most desirable outcome of a bracket intercept.

You and a computer wingman can run a

bracket intercept on your locked target. Do not initiate the maneuver if the bandit is within 15 miles or if the aspect angle is less than 160°. Press Shift Z to tell your wingman to fly the left side of the bracket, or press Shift C to order a bracket right.

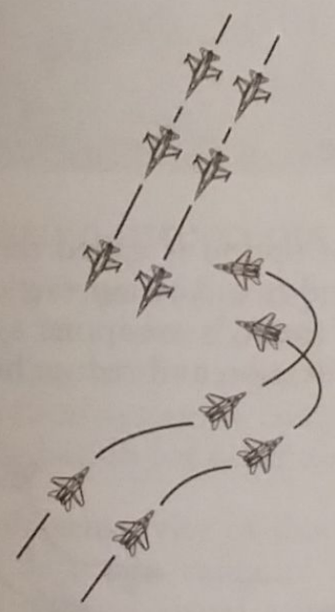


Fig. 5

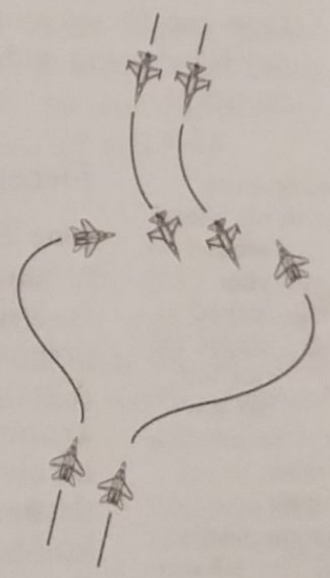


Fig. 6

The other possible result of a bracket is shown in Fig. 7. In this situation the enemy fighters react "inside out" causing two separate merges. This is also a good outcome if you have a maneuverability advantage. In a Fulcrum, you can outmaneuver most other jets so you now end up with two 1v1 engagements. The exception is the Falcon. The F-16 and MiG-29 are roughly equal in maneuverability, so the outcome will depend solely on the skill and cunning of the stick and throttle actuator—you.

The single side offset and bracket intercept are both examples of direct confrontation intercepts. These intercepts are used when you do not fear getting shot in the lips from long range or you have the enemy outgunned (actually, outmissiled). In these intercepts, you just duke it out with the enemy using your superior firepower.

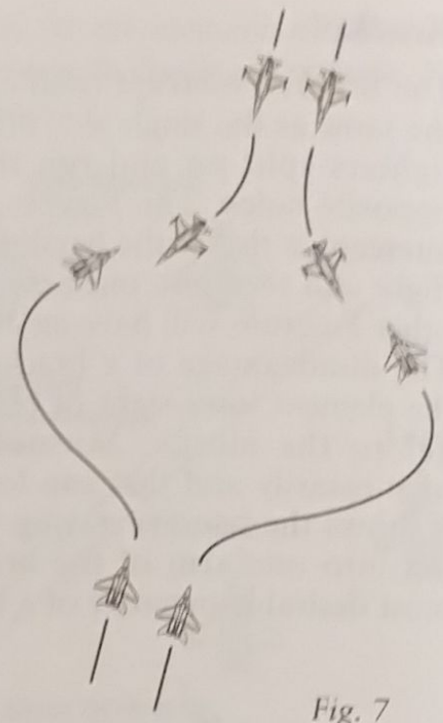


Fig. 7

What happens when you do not have technical advantage? A technical disadvantage can be overcome tactically by using a deception intercept.

Deception intercepts

One deception intercept option is called the lead around. The lead around is a Russian tactic that is designed to get the enemy's weapons system pointed in a harmless direction and reduce his SA. A standard lead around would involve starting the intercept in a lead-trail formation and then turning the leader 90° away. If the bandits are slow to realize that this fighter is no longer a threat and follow him, the trailing fighter in the element will gain an unobserved entry to six o'clock. Fig. 8 shows a lead around.

You and a computer wingman can run a lead around on your locked target. Do not initiate the maneuver if the bandit is within 10 miles, or you risk getting hit by a medium-range missile. Press Shift 3 to tell your wingman to lead the enemy to the right, or press Shift 2 to order a lead around to the left.

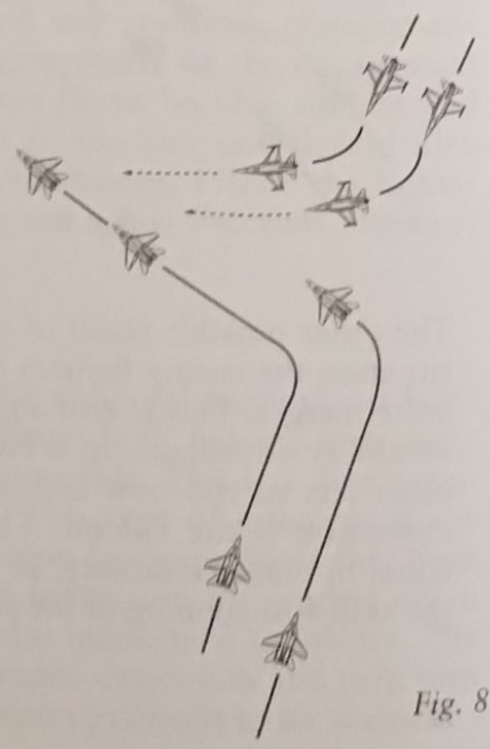
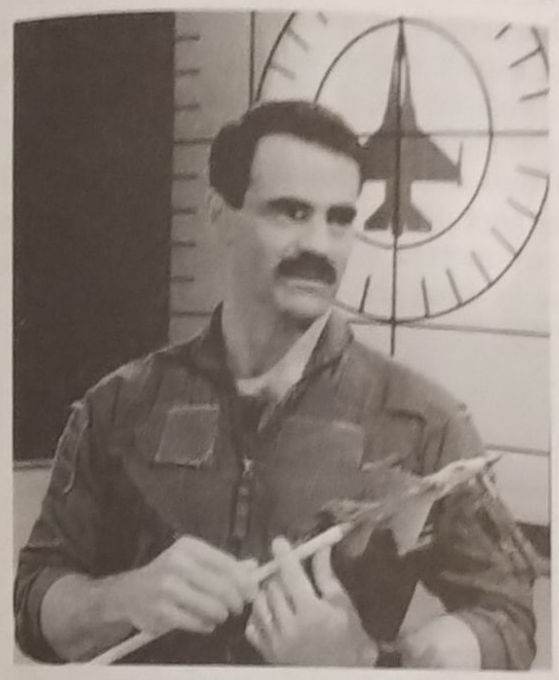


Fig. 8

Remember that tactical intercepts are flown to get you into a position to fire your weapons BVR. In addition, tactical intercepts can also get you into a position of advantage for a visual fight. The visual fight in a Fulcrum (or Falcon) is the subject of the *Fighter Air Combat Trainer* (*FACT*).



In *FACT*, I provide a detailed training package for within visual range maneuvering. *FACT* uses a comprehensive workbook augmented by a videotape and a disk that can be used with *Falcon 3.0* or *MiG-29*. The maneuvers and tactics described in *FACT* apply to all fighters but especially aircraft with the energy maneuverability of the MiG-29 and F-16.

Passive intercepts

In any type of intercept, the key is surprise. The IRST on the MiG-29 allows a firing solution to be achieved with no prior warning to the target. This is due to the fact that the infrared detection system is completely passive. In other words, it emits no radiation for other aircraft to detect.

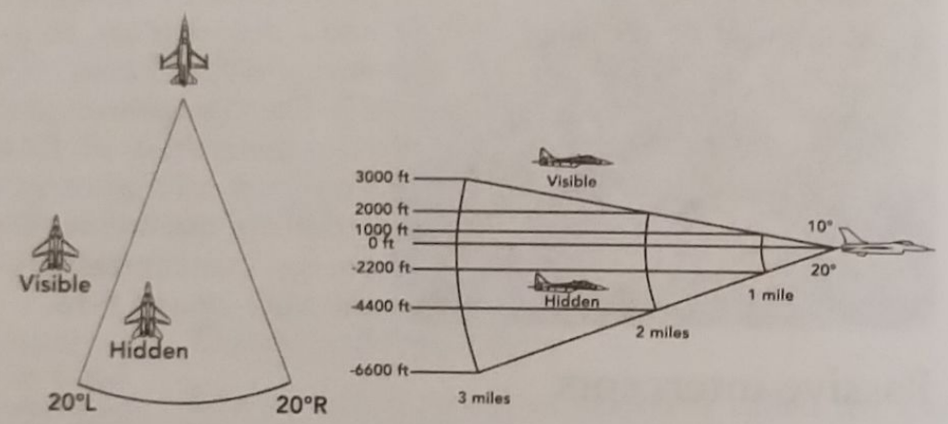
While sensitivity of this system limits its effective detection range to about 10 miles, this is still beyond visual range. With limited use of your radar and proper positioning, you can keep your presence unknown until weapons launch.

While the IRST is limited to 10 miles, a target which is locked up in IRST mode can still be tracked at longer distances because the radar will automatically turn on to maintain the lock. If you turn off the radar (by pressing R), the lock would be lost as soon as the target got more than 10 miles distant.

At first glance, it would seem that if you do not use radar, your "vision" is limited to just 10 miles—quite a handicap against an 80-mile-range APG-66 radar. In actuality, this is not so. Your radar warning receiver will let you know if you have been detected by radar, in which case you simply activate your radar and engage using one of the methods Pete Bonanni described above or whatever other tactic you prefer.

If you see the enemy on IRST before they see you on radar, the advantage is yours. The most important thing is to maintain situational awareness (SA). This means a comprehensive knowledge of the situation. You can achieve SA with the IRST by cycling through all available targets (press [X]). As you lock on to each target, examine its aspect angle, altitude, distance and closure rate.

The aspect angle and altitude tell you if you are where the enemy can see you. The distance and closure rate tell you when you are within weapons parameters. All four figures together give you an accurate picture of what the enemy is doing.



As a rule of thumb, you will probably remain hidden from your opponent visually and on radar if you keep your aspect angle between 20°R and 20°L and keep your relative altitude between 1,000 ft and -2,000 ft for every mile of separation.

Be ready to engage if you see the target maneuvering suddenly—it probably means you've been spotted, especially if the aspect angle increases rapidly (which means he's turning into you or you've overshot). If you can stay undetected long enough to get a firing solution, you've won the encounter.

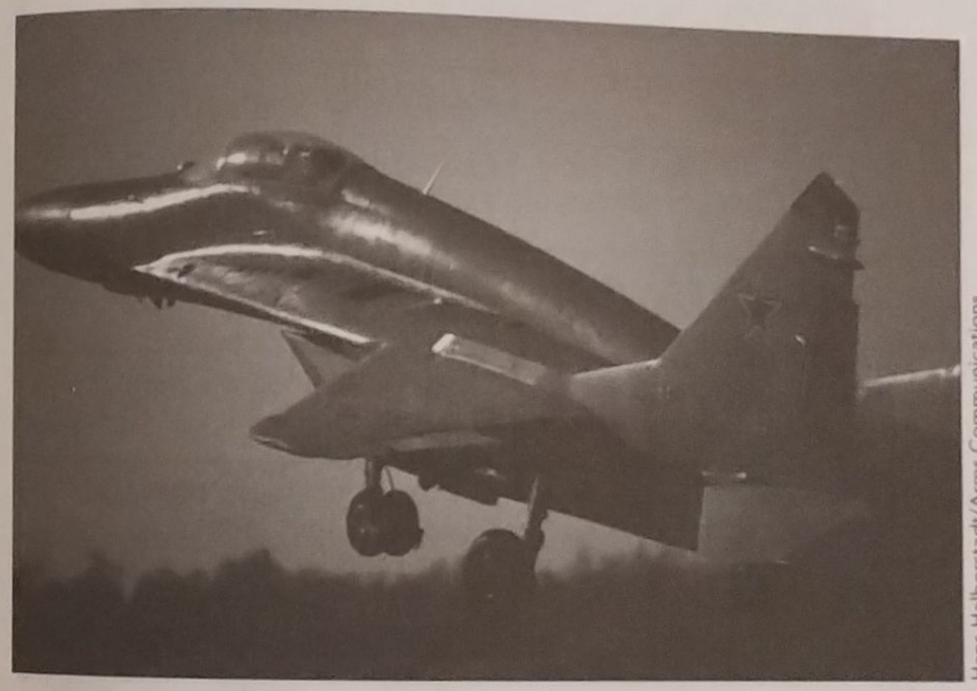
Off-boresight intercepts

The helmet mounted system (HMS) on the MiG-29, together with the design of the R-60, R-73 and R-23R missiles, allows you to engage and destroy a target "off boresight." This means you don't have to have the plane pointed directly at the target to get a weapons lock.

In a 1v1 dogfight, the benefits of this off-boresight capability are obvious. In a turning fight, you will be able to shoot first against an aircraft with a narrower weapons envelope.

In a 1 versus many fight, your job is to get out of there once you've been detected. The sooner you can get your shot off and disengage, the better. Using the HMS to lock up your target allows you to get a firing solution much sooner than with the HUD alone.

In a 2 versus many fight, be sure to take full advantage of the HMS. In a multiship combat, the guy who can take care of his target first can engage the enemy who threatens his wingman. Don't wait to get your target in the HUD; lock it up with the helmet mounted system.



Hans Halberstadt/Arms Communications

AIR-TO-GROUND

The air-to-ground capability of the MiG-29, while limited, is an important part of its mission. The most important elements of air-to-ground tactics, beyond the skillful employment of weapons, are weapons selection, evading air defenses and dodging SAMs.

Weapons selection

Choosing the right weapon for the target is critical to success at air-to-ground missions. With only four pylons available for air-to-ground weapons (only two of which can carry the Kh-23), weapons selection becomes even more important for the MiG-29 than it is for other aircraft.

Consult the chart below to find out which weapon is optimal for each type of target. The weapons are listed in preferential order from best choice to worst; the number in parentheses shows how many hits from that weapon (on average) are needed to destroy the target in question.

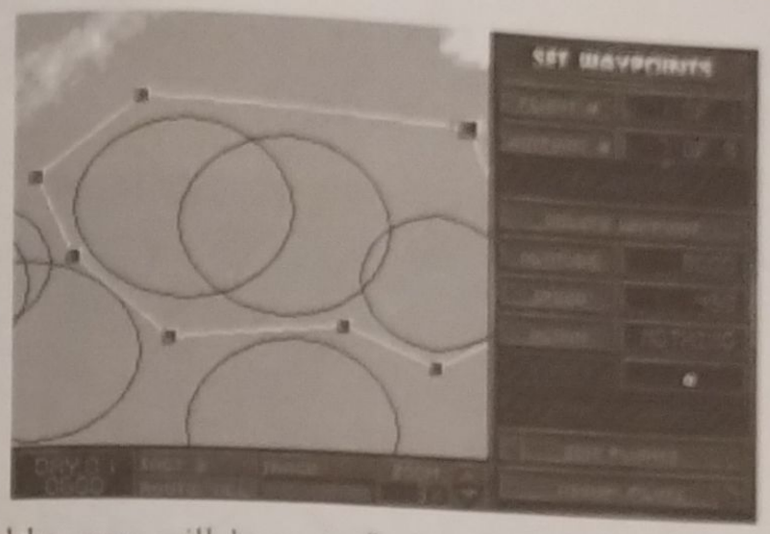
Target	Weapon
Destroyer	FAB-500 (2), FAB-250 (4)
Large building, factory	FAB-500 (4)
Small building	FAB-500 (1), FAB-250 (2)
Hangar	FAB-500 (2)
POL tank	UB-57 (2), RBK-50 (1), Kh-23 (1), FAB-250 (1), FAB-500 (1)
SAM launcher, radar vehicle	Kh-23 (1), UB-57 (4), FAB-500 (1), FAB-250 (2)
Tank	Kh-23 (1), UB-57 (6), FAB-500 (1), FAB-250 (1)
APC	RBK-50 (1), Kh-23 (1), UB-57 (2), FAB-500 (1), FAB-250 (1)
Truck, personnel	RBK-50 (1), UB-57 (2), Strafe (10), Kh-23 (1), FAB-500 (1), FAB-250 (1)

Evading air defenses

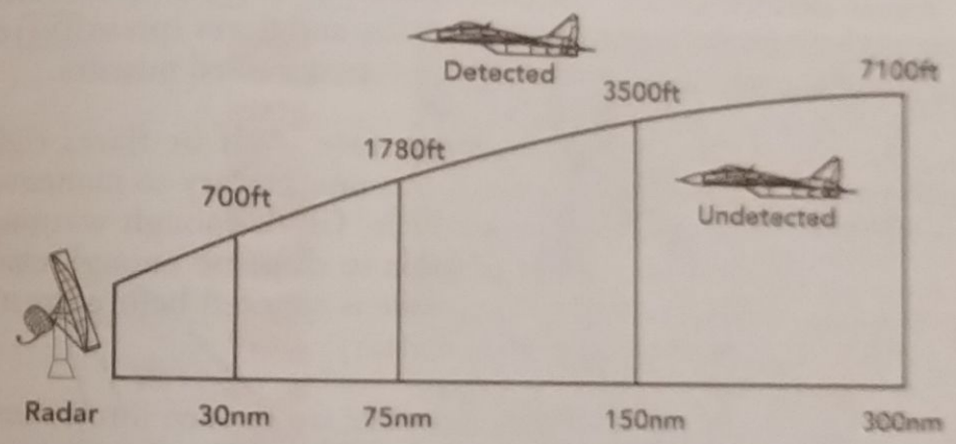
The best way to reduce your vulnerability to enemy air defenses is to avoid them entirely. To do this requires proper mission planning and good radar evasion techniques.

When planning your mission, pay close attention to the air defense threat circles on the campaign map. By adding and moving waypoints, try to create a flight path which goes between the threat circles rather than through them.

Then, when you are in the cockpit, use the navigation display to stick to your flight plan. Used properly, the HSI will guide you back onto the selected approach radial to your waypoint no matter how far astray you may have flown.



Inevitably, you will have to fly into areas defended by enemy anti-aircraft units. When this happens, you can often minimize your exposure to ground-based radar by flying low. You can often pass as close as 20 miles to enemy radar sites if you are below 500 ft AGL.

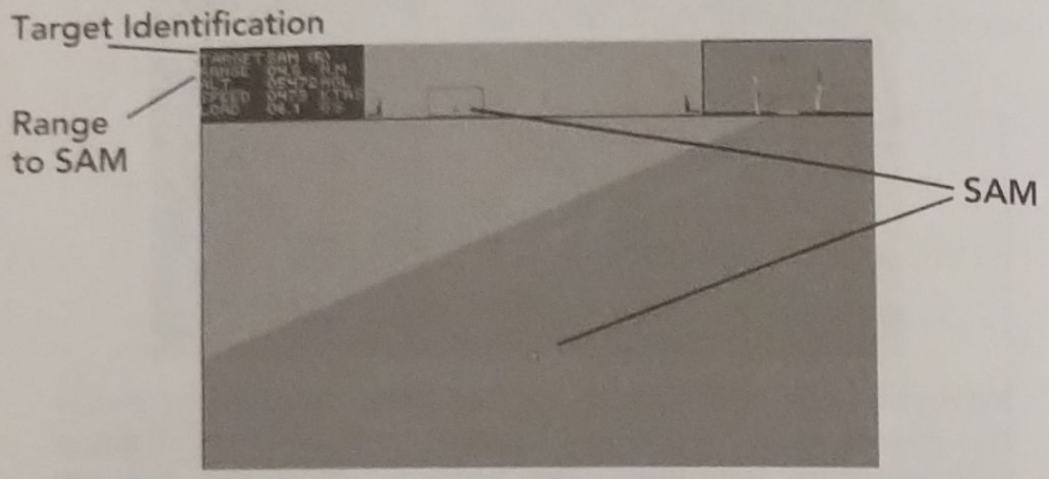


If you are hunting enemy SAM sites with Kh-23s, you will have to approach at an extremely low altitude in order to get within missile range (10 miles) before the SAM site detects you with its radar. Once you are within range, pop up to about 3,000 feet, lock on and launch. Watch your radar warning receiver, however, because a good SAM crew will respond with a missile before yours arrives.

Missile avoidance

The techniques for escaping SAMs are similar to that for dodging air-to-air missiles. Using the right countermeasures and maneuvers, you stand an excellent chance of evading an incoming missile if given sufficient warning.

As soon as you detect a launch on your radar warning receiver (a flashing white light), switch to Padlock View (press [8]). You should automatically be tracking the missile.



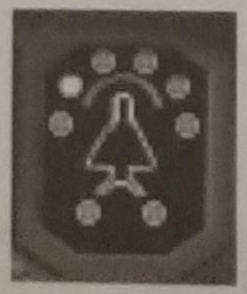
The target identification in the upper left tells you what kind of guidance the missile is using; (R) means radar-guided and (I) means infrared-guided. Use chaff (press [Ins] or [0] on the numeric keypad) against radar-guided missiles and flares (press [Del] or [.] on the numeric keypad) against infrared-guided missiles.

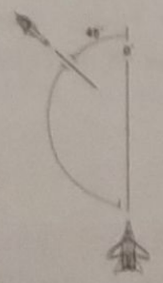
Radar-guided missiles can also be jammed with electronic countermeasures (ECM). Press [E] to activate your onboard ECM gear.

Start off by dropping some chaff or flares right away. Occasionally, it is unnecessary to maneuver wildly to evade a missile. Given enough warning, it is sometimes possible to dispense enough chaff or flares that the missile is spoofed before drastic measures become necessary.

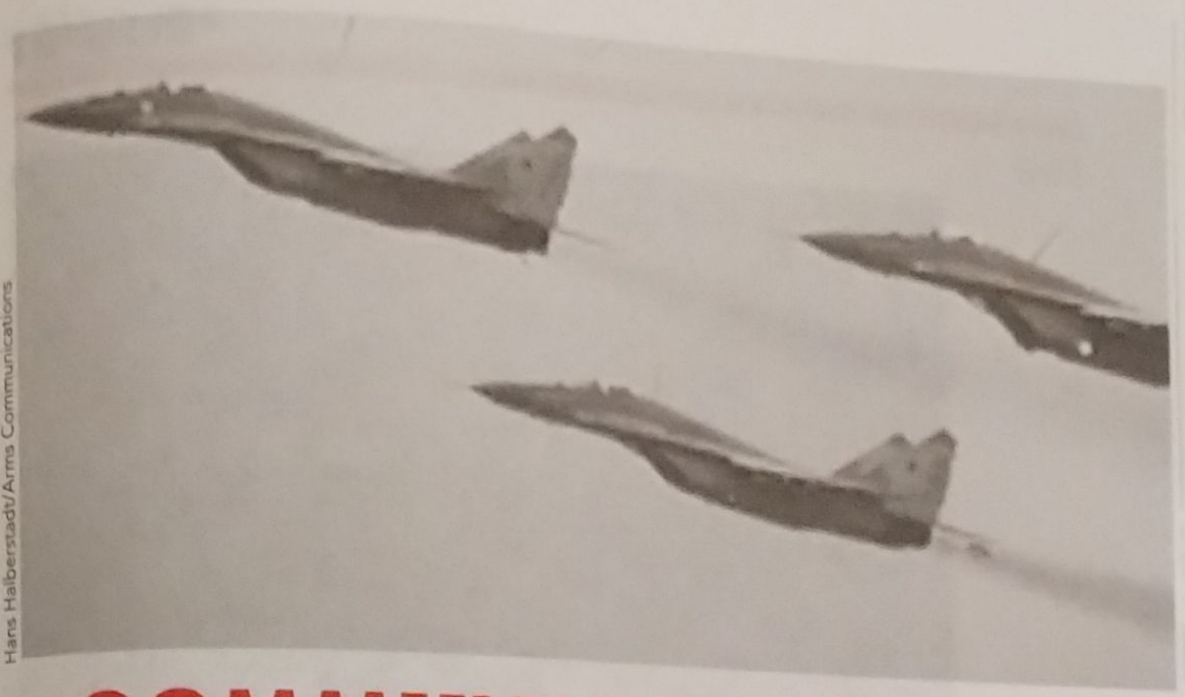
As the missile approaches, try to turn into it until it is approaching from about 45° off your nose (check your radar warning receiver).

Watch how fast it closes by keeping an eye on the distance in the upper left corner in Padlock View. At the last possible moment, break hard across the path of the missile and down while dumping chaff or flares (as appropriate).





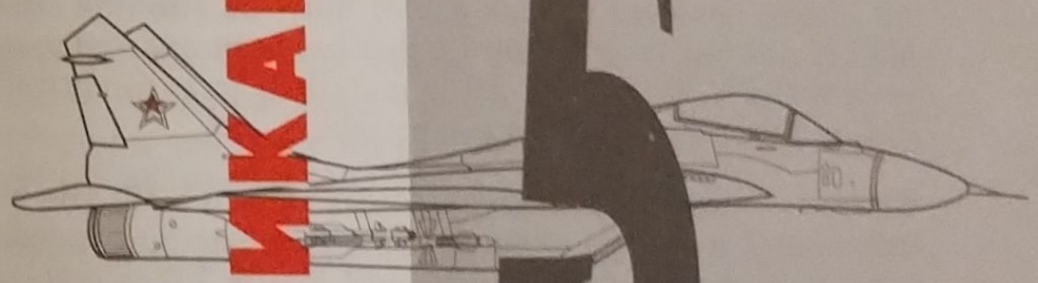
A similar but riskier method is to dive steeply, pulling up at the last second (dropping chaff or flares the whole way).



Hans Halberstadt/Arms Communications

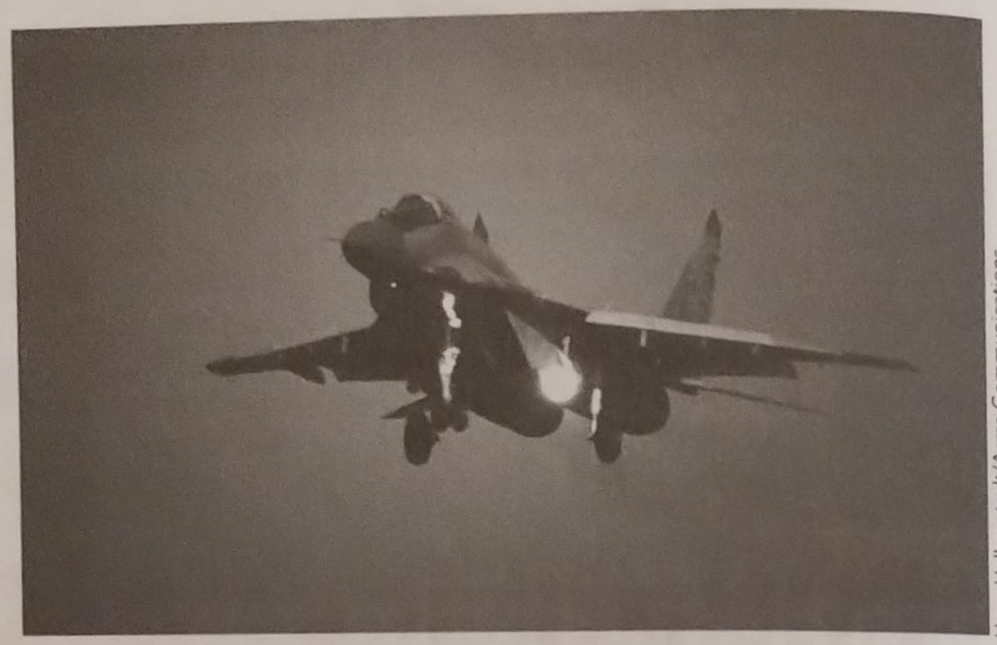
COMMUNICATIONS

КОММУНИКАЦИЯ



5

ПЯТЬ



Hans Halberstadt/Arms Communications

MiG-29: Deadly Adversary of Falcon 3.0 expands the communications options available to *Falcon 3.0* players. Once installed, it adds head-to-head duels between two MiG-29s, between an F-16 and a MiG-29 or between multiple MiG-29s and F-16s. It also allows allied missions involving multiple MiG-29s, as well as multigame campaigns in which one side flies MiG-29s and the other flies F-16s.

All systems involved in such games must be running either *MiG-29* or a version of *Falcon 3.0* that has been updated to run with *MiG-29*. When *MiG-29* is installed, it automatically updates *Falcon 3.0* to make it compatible.

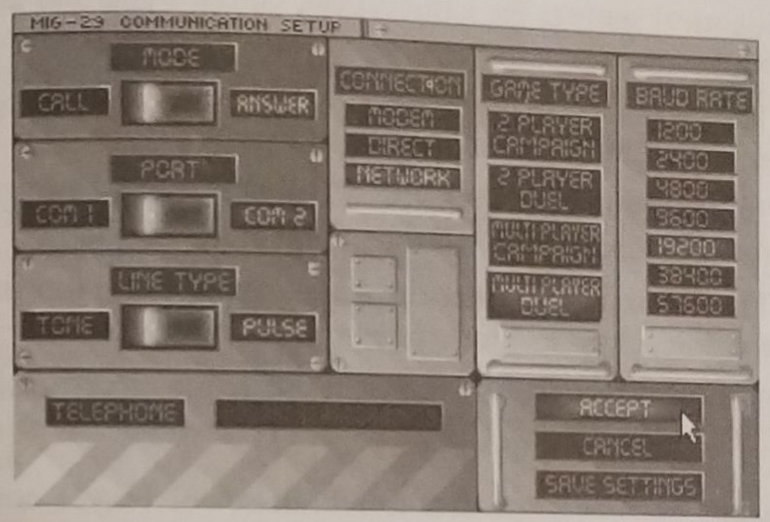
While it is illegal to give away copies of *MiG-29*, you are authorized to use copies of the *Falcon 3.0* update disk included in this package to upgrade your friends' *Falcon 3.0* so that it is compatible with *MiG-29*.

Once all systems are upgraded, communications can be attempted. If all players are using the same game (all are running *Falcon 3.0* or all are running *MiG-29*), follow the instructions for single-game play.

If players are using a combination of *MiG-29* and *Falcon 3.0*, follow the instructions for multigame play. Because of the complexities involved in multigame (*Falcon 3.0* vs. *MiG-29*) communications, it is important to set up your play sessions in exact accordance with the instructions that follow.

MAKING CONTACT

There are some instructions which hold true for all communications. As you read these directions, refer to the Communication Setup screen (click on the Comms monitor in the War Room) and to the *Falcon 3.0* Release Notes and Communications Handbook.



The computer designated as the Caller controls the game. There can be only one Caller; all others must be set to Answer. Since the Caller handles all communications traffic, it should be the fastest computer available.

In multigame sessions (*Falcon 3.0* vs. *MiG-29*), there is a secondary master computer which handles all *MiG-29* traffic. This computer, called the Master MiG, should be the fastest of the computers running *MiG-29*.

Before setting up communications between computers, communicate with the other people. Talk to all other players before setting up the game to coordinate the game session. Decide who is running which software, who is going to be the Caller, who will be the Master MiG, what baud rate everyone will be set to, when and how the connection will be established and so forth.

Once the connection commences, there are two visual cues to let you know what is going on. The first is a small spinning wheel, which is an indication that the program is working and has not locked up or crashed. The second is a series of white and red dots which appear on the screen. A white dot is an expected information packet; a red dot represents a packet which has been sent and received.

Modem play

Two-player duel and two-player allied games can be played via modem. When setting up a modem connection, be aware that the connection process begins as soon as the ACCEPT or SAVE SETTINGS buttons are selected.

- Before connecting, be sure that both players have selected identical settings for Connection, Game Type and Baud Rate.
- The Caller should have Mode set to CALL, and the answerer should be set to ANSWER.
- The Port and Line Type should be adjusted as appropriate for your situation.
- The Caller should enter the Answerer's phone number in the Telephone window.

When all settings are satisfactory, both players should select ACCEPT or SAVE SETTINGS.

Falcon 3.0 and MiG-29 only support the COM 1 and COM 2 serial ports for communications and do not support data compression (v32, v42, v42bis MNP).

The game uses a default set of Hayes commands to establish modem communications. When the Caller hits the ACCEPT or SAVE SETTINGS button, a dialog box pops up asking if you wish to use these modem defaults. If you wish to view or change the default modem commands, press [N]. If the default settings are acceptable, press [Y] and the modem will begin dialing.

If you press [N], you will see the default commands one at a time. To send a command line to the modem, press [Enter]. To change a command, erase it by pressing [Backspace], then type in the replacement and press [Enter] to send it to the modem. When you are done entering modem commands, press [Enter] on a blank line.

Modem problems

If you have trouble connecting, you may need to change the modem initialization commands to suit your modem. When it asks if you want to use the default settings, press [N]. The initialization command lines are the ones which read "ATZ" and "ATE0V0Q0." Replace one or both of these lines as appropriate for your modem:

Modem	Replace	With
Zoom Fax	ATZ	AT&F0
Gateway Telepath at 2400 baud (if you have version 1.0 or 1.3, call Gateway for an upgrade)	ATZ	AT&FX4S37=6N0
Gateway Telepath at 9600 baud (see above)	ATZ	AT&FX4S37=9N0
Practical Peripherals PM 144000 FXS	ATZ	AT&F0, then AT&Q5S37=11N1, then AT&W0
US Robotics Courier HST (most reliable at 2400 baud)	ATZ	AT&K0
	ATE0V0Q0	ATE1V1Q0

If your modem is not listed or the above changes do not help, try the following:

- Use a communications program (such as ProComm or QModem) to dial and establish a communications link between the two computers.
- Quit the communications program. Do not hang up the phone line or otherwise break the modem connection.
- Run *Falcon 3.0* or *MiG-29* as before, choose a squadron and select the Comms monitor to bring up the Communication Setup screen.
- Select DIRECT under the Connection menu and set the Baud Rate and Port to the proper settings for the modem.
- Choose ACCEPT or SAVE SETTINGS to start play.

If you follow this procedure, the game will treat the connection as a direct serial hookup.

Direct connections

To play a two-player game by direct connection, you must connect the two computers through either the COM 1 or COM 2 serial ports using a null-modem serial cable. Be aware that the

connection process begins as soon as the ACCEPT or SAVE SETTINGS buttons are selected.

- Before connecting, be sure that both players have selected identical settings for Connection, Game Type and Baud Rate.
- The Caller should have Mode set to CALL, and the Answerer should be set to ANSWER.
- The Port should be set to COM 1 or COM 2 as appropriate for each computer.

When all settings are satisfactory, both players should select ACCEPT or SAVE SETTINGS.

Network play

When playing over a network, up to six players can fly at once. Since *Falcon 3.0* and *MiG-29* only support Novell's IPX standard (version 3.10 or later), all players must be connected to a Novell NetWare or Novell NetWare Lite network. NetWare users should be sure to use version 3.22 or later of the NETX or EMSNETX programs.

All players should talk to each other beforehand and decide what all the settings will be, when to join, who is flying which game, and so forth. Be aware that the connection process begins as soon as ACCEPT or SAVE SETTINGS is selected.

In a network game, there is only one Caller. All the rest of the players are Answerers. The basic procedure is as follows:

- Before connecting, be sure that all players have selected identical settings for Connection, Game Type and Baud Rate.
- The Caller should have Mode set to CALL, and the Answerers should be set to ANSWER.
- When all settings are satisfactory, the Caller should select ACCEPT or SAVE SETTINGS, followed by the Answerers.
- The Caller will see a box with all the Answerers listed by network name (with NetWare Lite, the other players will be listed as question marks). By choosing a name and pressing OK, the Caller invites other players to join.

Once all Answerers accept the Caller's invitations, the game can begin.

DUELS

A duel is a free-for-all where every pilot is out for himself. Players can opt to play in teams, but the game will last until only one aircraft remains unless the players quit sooner.

The IFF indicator (see chapter 2) will see all similar aircraft as friendly in a duel. MiG-29s will detect other MiG-29s as allies, and F-16s will see other F-16s as allies.

In a duel, all *Falcon 3.0* players will automatically be armed with four AMRAAMs, two AIM-9Ms, two AIM-9Ps and an ECM pod. All MiG-29 players will be armed with two R-27Rs, two R-73s and two R-60s. If the Caller has Limited Arms turned off on the Difficulty Levels screen, all players will have every available weapon in unlimited quantities.

There is no difference between the setup for a two-player duel and the setup for a multiplayer duel, but there are some important differences between the single-game setup (*Falcon 3.0* only or *MiG-29* only) and the multigame setup (*Falcon 3.0* and *MiG-29* simultaneously).

Single-game duels

In single-game duels, follow the directions in "Making Contact," above, for setting up communications.

In a two-player duel, once both players have selected ACCEPT or SAVE SETTINGS, the duel will begin. In a multiplayer duel, the game begins once the Caller has invited all the other players and they have accepted. A duel ends when there is only one survivor or when the Caller ends the duel by pressing Alt Q (to start a new duel) or Alt X (to exit to DOS).

Multigame (*Falcon 3.0* vs. *MiG-29*) duels

When setting up a duel where some players are using *MiG-29* and the others are running *Falcon 3.0*, there are specific steps to be followed when setting up communications.

In a two-player *MiG-29* vs. *Falcon 3.0* head-to-head duel, the *Falcon 3.0* player must be the Caller and the *MiG-29* player is the Answerer. Otherwise, set up communications normally as appropriate for your communication type (see "Making Contact" above).

In a multiplayer, multigame duel, things are a bit more complex. One *Falcon 3.0* player is the Caller. The Caller's computer will control the game and handle all communications traffic for all players. The Caller's computer should, therefore, be the fastest available.

One of the *MiG-29* players will be designated the "Master MiG." This player's computer will handle all the *MiG-29* communications traffic and, therefore, should be the fastest of the machines running *MiG-29*.

Otherwise, multigame duels are just like single-game duels. The game will end when there is only one survivor unless the Caller ends the duel sooner by pressing Alt Q (to start a new duel) or Alt X (to exit to DOS).

Remember that games involving more than two players can only be played on a network.

Setting up a multiplayer, multigame duel

Continuous communication between the players is essential to setting up a multiplayer, multigame duel. Everyone should either be within earshot or reachable by phone during the setup. This way, you can coordinate when each person joins the game.

In multiplayer, multigame duels, the most important thing to remember is the order in which players are added to the game. The order must be as follows:

1) Caller
2) All other *Falcon 3.0* players
3) Master MiG
4) All other *MiG-29* players

First, be sure that everyone agrees beforehand who is flying what game. Then select a Caller and a Master MiG. Once that's settled, follow these steps:

- Before connecting, be sure that all players have selected identical settings for Connection, Game Type and Baud Rate.
- The Caller (who must be running *Falcon 3.0* in a multigame session) should have Mode set to CALL, and all other players should be set to ANSWER.
- The Caller must click ACCEPT on the Communication Setup screen first.

- All *Falcon 3.0* players then choose ACCEPT or SAVE SETTINGS.
- The Caller will see a box with all the Answerers listed by network name. By choosing a name and pressing OK, the Caller invites another player to join. The Caller must invite only *Falcon 3.0* players to join at this point.
- After all the *Falcon 3.0* players have accepted, the Master MiG should then choose ACCEPT or SAVE SETTINGS.
- The Caller invites the Master MiG.
- Once the Master MiG has accepted, the other *MiG-29* players select ACCEPT or SAVE SETTINGS.
- The Caller then invites any remaining *MiG-29* players.
- When all players have accepted, the Caller selects EXIT to begin play. In a six-player game, play will commence immediately after the sixth player accepts.

As usual, all *Falcon 3.0* players will automatically be armed with four AMRAAMs, two AIM-9Ms, two AIM-9Ps and an ECM pod. All *MiG-29* players will be armed with two R-27Rs, two R-73s and two R-60s. If the Caller has Limited Arms turned off on the Difficulty Levels screen, all players will have every available weapon in unlimited quantities.

CAMPAIGNS

Multiplayer campaigns allow up to six players to play one of the campaign scenarios provided with *Falcon 3.0* and *Operation: Fighting Tiger*. In a single-game campaign, the players can be all MiG-29 pilots or all F-16 pilots. In a multigame campaign, some players can fly MiG-29s while their opponents fly F-16s.

In a campaign, each side has a flight leader. This flight leader does all the mission planning for the other pilots on his or her side. The other pilots can do nothing until the mission begins except watch as the flight leader makes all the arrangements.

In single-game campaigns, the Caller is the flight leader. In multigame campaigns, the Caller is the F-16 flight leader, and the Master MiG is the MiG-29 flight leader.

Single-game campaigns

When all players in a campaign session are running the same game, the main thing to remember is that the Caller is the flight leader and, therefore, the Caller's machine will be running the session. For this reason, it is highly recommended that the Caller's computer be the fastest available.

Setting up a single-game multiplayer campaign

Set up communications as appropriate for the method you intend to use (see "Making Contact" above). When all players have joined, a series of campaign news screens scrolls by, and then the flight is given its mission.

The other members of the flight now sit back and watch as the flight leader adjusts the waypoints, then arranges the flights and wingmen and finally assigns weapons. When the flight leader clicks TAKEOFF, the action begins.

Multigame (*Falcon 3.0* vs. *MiG-29*) campaigns

In multigame campaigns, the most important thing to remember is the order in which players are added to the game. The order must be as follows:

1) Caller
2) All other *Falcon 3.0* players
3) Master MiG
4) All other *MiG-29* players

Otherwise, multigame campaigns operate very much like single-game campaigns. Flight leaders control all the waypoints, wingmen and weapons, and each team works together to complete its mission (which usually involves thwarting the opposite team).

The Caller is the flight leader for the F-16s, while the Master MiG is the flight leader for the MiG-29s. Missions in multigame campaigns are specifically designed to bring both groups of players in contact with each other. For example, if the F-16 pilots were assigned to bomb a bridge, the MiG-29 pilots might be assigned to fly CAP (combat air patrol) over that same bridge.

Setting up a multigame campaign

First, be sure that everyone agrees beforehand who is flying what game. Then select a Caller and a Master MiG. Once that's settled, follow these steps:

- Before connecting, be sure that all players have selected identical settings for Connection, Game Type and Baud Rate.
- The Caller (who must be running *Falcon 3.0* in a multigame session) should have Mode set to CALL, and everyone else should be set to ANSWER.
- The Caller must click ACCEPT on the Communication Setup screen first.
- All other players then choose ACCEPT or SAVE SETTINGS.
- The Caller will see a box with the Answerers listed by network name. By choosing a name and pressing OK, the Caller invites another player to join. The Caller must invite only *Falcon 3.0* players to join at this point.
- After all the *Falcon 3.0* players have accepted, the Caller invites the Master MiG.
- Once the Master MiG has accepted, the Caller then invites any remaining *MiG-29* players.
- When all players have accepted, the Caller selects EXIT to begin play. In a six-player game, play will commence immediately after the sixth player accepts.

Each of the two teams (MiG-29 pilots and F-16 pilots) will receive a different news briefing and setup for the campaign and mission. Each flight leader will then set up the waypoints, wingmen and weapons for the mission. When both flight leaders have clicked TAKEOFF, the mission begins.

Saving campaigns

There is no explicit method in the game for saving campaigns, but a way to save one campaign at a time has been provided. After the session has terminated, exit to DOS. Type **SAVECAMP** to run the SAVECAMP.BAT batch file provided with *MiG-29*.

This batch file will copy all the campaign information on top of your first squadron. Note that this will wipe out that squadron,

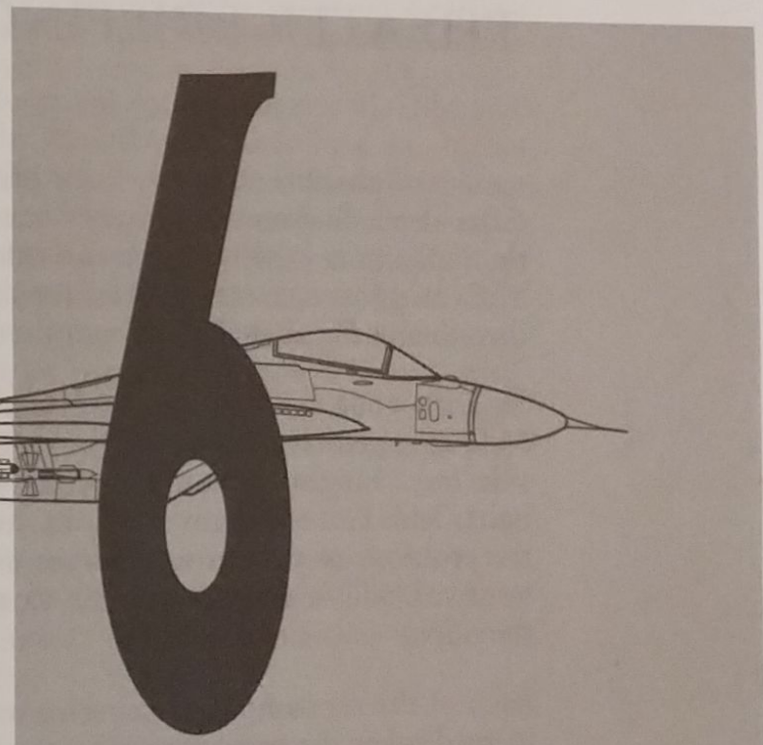
replacing it with the squadron from the campaign. To avoid confusion, always use the first squadron when you start a campaign you intend to save.

To save and resume the campaign:

- The Caller must run the SAVECAMP.BAT batch file after the play session ends.
- When setting up the next session to resume play, the same computer must be the Caller and must use the first squadron which appears when the Squad monitor is selected in the War Room.
- Set up communications as before, and play will pick up where it left off.

Hans Halberstadt/Arms Communications

THEATER BRIEFINGS



Hans Halberstadt/Arms Communications

THEATER BRIEFINGS

The MiG-29 serves in the air forces of many nations all over the world. As a pilot of the MiG-29, therefore, you face a wide variety of possible missions. Your primary goals will, of course, differ depending on what type of campaign is being waged, but the Fulcrum is usually seen in an offensive role. It is the job of MiG-29 pilots to clear the skies for the strike aircraft to follow, then join in the airstrikes and put the enemy to rout.

As with your F-16 counterpart in the Coalition forces, your MiG-29 squadron is the focal point in each conflict. How your side fares hinges on you and your squadron's success on the battlefield. Fail too many missions, and the enemy will overrun the positions of your ground forces. A sharp MiG-29 squadron, however, will be able to stop the Coalition's advances and turn the tide.

Each of the six campaign scenarios and briefings you are about to read takes the opposite position of its counterpart in *Falcon 3.0* and *Operation: Fighting Tiger*. The first three are from *Falcon 3.0*, while the latter three are from *Operation: Fighting Tiger*. Please consult these respective manuals if you are interested in reading the other viewpoints of these conflicts.

The following campaign briefings correspond to each of the six nations you will fly for: Iraq (Kuwait theater), Syria (Israel theater), Cuba (Panama theater), Russia (Kurile Islands theater),

North Korea (Korea theater) and India (Kashmir/Pakistan theater). *Remember that these accounts are fictionalized for purposes of this simulation and are not meant to be taken as the actual doctrines or positions of the respective countries.*

IRAQ

The Kuwait situation is a classic example of the problems caused by the intervention of European powers in areas where they were neither needed nor welcome.

During World War I, European nations included the Arab world in their fighting. Germany, Turkey, France and Great Britain waged war across northern Africa and the Middle East. When they were done, they partitioned the region into states as they saw fit. The separation of Kuwait from Iraq was one such division.

After World War I, Great Britain remained in control of Iraq and many other Arab regions with the approval of the League of Nations. In 1932, when the British troops finally left Iraq, they retained control of one of the provinces. This 19th province eventually became known as Kuwait. At the time, there was little that could be done to dispute the continued British occupation of Kuwait. Great Britain and its army supported the local ruling family and ran Kuwait as a British protectorate.

In the 1940s, oil was discovered in Kuwait. British petrochemical firms moved in quickly to exploit the rich oil fields. As a result of this economic interest, the British delayed their exit from Kuwait until 1961.

Through the British-held petrochemical companies, Kuwait became a major oil producer and a key player in OPEC (the Organization of Petroleum Exporting Countries).

Kuwait consistently ignored OPEC production limits, however. The flooding of the market with Kuwaiti oil lowered the world market price, which has cost Iraq an estimated $14 billion in lost revenue.

The huge petroleum production of Kuwait has made it a very wealthy nation, especially for its size. In the war against Iran, Kuwait provided Iraq with a substantial amount of monetary aid ($12 billion). Considering that a successful Iranian invasion of Iraq would certainly have included snapping up tiny Kuwait, this was money well spent. But when the war was over, Kuwait demanded that Iraq pay back the full amount.

Further, Kuwait has used a drilling technique in the Rumaila oil field (which lies on the border of Iraq and Kuwait) that allows Kuwait to pump oil from the Iraqi side of the field. Despite many protests from Iraq, this exploitation continues. The estimated value of the oil Kuwait has removed from the Iraqi portion of the Rumaila field is over $2.5 billion.

Historically, the entire Rumaila oil field belongs in Iraqi territory. While Iraq has been tolerant about this in the past, Kuwait's flagrant exploitation of the Rumaila oil reserves owned by Iraq left President Saddam Hussein with no choice. He demanded that Kuwait vacate the area and return the rest of the Rumaila oil field to Iraq.

The situation had developed to a crisis point. The rulers of Kuwait refused to pay back the oil money owed to Iraq from Kuwaiti drilling practices in the Rumaila fields and overproduction of oil. In addition, they were unwilling to forgive the $12 billion in wartime aid given to Iraq for Kuwait's protection. Finally, Kuwait's leaders refused to return the Rumaila fields to Iraq.

Iraqi diplomats were sent to Kuwait to discuss these matters, but with admittedly little hope of success. As a show of resolve, several divisions of Iraqi troops were deployed near Kuwait to conduct military exercises.

As feared, the Kuwaiti government flatly refused all attempts to resolve its differences with Iraq. It became clear that the government in Kuwait was corrupt, interested only in gathering wealth for its members, and that negotiations would be fruitless.

Enough was enough. President Saddam Hussein ordered Iraqi troops into Kuwait August 2, 1990 to remove the corrupt government and restore Kuwait, historically and culturally a part of Iraq, to its rightful position as the 19th province. Fighting was minimal, and in less than a day Kuwait had joined the rest of Iraq.

Saudi Arabia, claiming to need protection from Iraqi troops still in Kuwait after the annexation, became a staging area for the United States military. Using the defense of Saudi Arabia as a pretext, America has built an offensive army to invade Iraq.

Other powers external to this strictly Arab matter have gathered alongside the United States in Saudi Arabia with the rubber-stamp approval of the United Nations. U.S. influence in the U.N. brought about an embargo against Iraq, which is really more symbolic than effective.

It is clear that the U.S. intent here is to invade the 19th province and restore the old, corrupt Kuwaiti regime, which was one of the United States' chief suppliers of oil. Sovereignty is not a real issue; Kuwait has historically and culturally always been a part of Iraq. The U.S. motivation is oil.

On January 16, 1991, U.S.-led air forces began airstrikes throughout Kuwait and other neighboring provinces of Iraq. This relentless bombing of both military and civilian targets has continued unabated for a month now, but the Iraqi front-line forces have held. It is believed that the Americans will begin a ground assault soon.

Strategy

The key to stopping the foreign invaders is to maintain control of the skies. If the United States can be denied the ability to bomb with impunity, the defensive fortifications in southern Kuwait will be sufficient to stop the American advance. Once the attack has bogged down, elite Republican Guard units can strike the enemy flanks. When the casualties have mounted high enough, the U.S. will withdraw as it did in Vietnam.

There are several important factors to consider before engaging the American coalition:

- The U.S. Air Force has a significant edge in equipment and numbers. U.S. combat aircraft in the region number in the thousands, including hundreds of the best fighter aircraft available, whereas Iraqi combat aircraft total just over 800 (many of which are seriously outdated). The battle for the skies will be a difficult one.
- U.S. ground forces, while outnumbered, possess the latest equipment and are highly trained. Air defenses and mobility are excellent.

- Much of Iraq's battlefield advantage comes from the prepared positions in the south and from other defensive arrangements. Thus, Iraqi forces will be most effective when on the defensive.

Campaign objectives

The primary goal of the Iraqi forces in Kuwait is to defend Kuwait. To that end, there are three basic objectives that must be accomplished:

- Contain the invading air force. If Iraq can maintain a credible air threat, the ground forces can at least fight on an even footing.
- Slow the advancing U.S. ground forces as much as possible. Airstrikes on command centers and advancing troop formations will be used to hinder the invaders and break morale.
- Inflict maximum casualties. If the losses become high enough, public pressure in the United States will force a withdrawal.

SYRIA

Syria's autonomic origins stem from World War I. Formerly part of Turkey, the state of Syria was created by the Treaty of Sevres in 1920. It was managed by a French League of Nations mandate until 1941 when it was proclaimed a republic by France. It achieved complete independence on April 17, 1946.

Before and during World War II, Jewish refugees and immigrants poured relentlessly into the area known as Palestine, which was controlled by Great Britain. The division of the state of Palestine into Arab and Jewish territories in 1948 and the formation of Israel as an independent nation angered Arab purists.

Almost immediately, Syria joined Egypt, Jordan, Lebanon, Saudi Arabia and Jordan to forcibly remove the Jewish presence from the region. The short-lived war of 1948–1949 saw Israel emerge as the winner and actually consume more Arab territory than

before. Although on the losing side, Jordan received the West Bank and the old city of Jerusalem while Egypt occupied the Gaza Strip area.

Throughout the next 20 or so years, hostilities between the Arab states and Israel increased. Israel's insistence on retaining its newly absorbed territory and refusal to grant independence (despite the U.N. mandate) to a separate Palestinian nation enraged Syria and its neighbors.

In 1956, responding to presumed "terrorist activities," Israel invaded Egypt's Sinai Peninsula aided by the Christian forces of the French and British. A week later the U.N. stopped the hostilities, but Israel was allowed to retain its latest conquest.

The United Nations withdrew their peacekeeping troops from the Palestine disputed regions in May 1967. Egypt moved back into its rightful territory in the Sinai. Almost immediately, Israel again continued its warlike ways by sending troops in a brutal six-day invasion. During this time Israel annexed the Gaza Strip, Sinai Peninsula, Syria's Golan Heights and Jordan's West Bank. Again, the U.N. entered to stop the fighting, but didn't return the territories to their rightful owners.

The 1973 Yom Kippur War saw Syria and Egypt attempting to regain their lost territories once more. Israel counterattacked and again garnered new territories, this time across the Suez Canal. While other Arab nations did not contribute troops to the effort, the oil-producing Arab states agreed in 1974 to give $1 billion a year to Syria to help them achieve their territorial rights and to counteract Israeli aggression.

During the next eight years, the political might of the Arab nations pressured the U.N. to remove Israel from their conquered lands. Israel retreated from the Sinai and the west bank of the Suez, but this was merely a ruse for their ultimate goal: the invasion of Lebanon.

In 1978, Israel marched into southern Lebanon, under the all-too-familiar guise of "terrorist attacks" from that region. Israel eventually withdrew from the territory in favor of yet another U.N. force in the region, but continued to illegally bomb Palestine Liberation Organization (PLO) strongholds deep within Lebanon.

Finally, in 1982, Israel used a combination of land, sea and air attacks to muscle its way into Lebanon. Syrian troops skir-

mished with Israelis in the Bekaa Valley three days later, but Israel, fearing Syrian retaliation, quickly agreed to a truce.

Since that time, Syria sided with Kuwait in the Persian Gulf War and sent troops to help in that conflict.

After the end of the Persian Gulf War, Israel felt it could flaunt its military might and flatly refused to come to the negotiating table to resolve the territorial differences. In the meantime, Israel continued to flagrantly settle the Syrian Golan Heights.

Fearing further territorial acquisitions by the increasingly aggressive Israel, Syria purchased defensive fighter aircraft and deterrent surface-to-surface missiles from the Soviet Union in early 1991.

Concerned by troop movements northward in Israel, Syria arms 200,000 of its elite reserves and further arms deals bolster their anti-aircraft batteries and fortify their tank reserves.

Syrian President Assad, sure that the Israelis will not risk any more political fallout, calls on Israel to peaceably remove themselves from the Golan Heights and to honor the U.N. mandate by granting Palestinian independence.

Much to the dismay of Syria, Israel hotly refuses and furthermore threatens Syria in a private communiqué to disband its troops and dismantle its new equipment or be invaded.

Syria, seeing the Israeli troop positions in the Golan Heights strengthen, calls on the Moslem groups and PLO forces in Lebanon for aid. The groups readily agree, seeing their chance to be rid of Israeli rule. Syria then moves its 300,000 troops in through Lebanon and south towards Israel.

Meanwhile, Assad sees his chance to execute a pincer maneuver on the Israeli army. While Syria's southern neighbor Jordan wishes to be uninvolved in the conflict, Jordan is more than willing to allow troop movements through their country to the occupied West Bank.

Assad realizes that this is the best opportunity to be rid of the Israeli thorn that has been embedded in the Arab skin for so long. With a quick strike, he could bring peace to the region and establish a Palestinian homeland all in one fell swoop. Opposing him, however, is the threat of U.S. troops and weapons being rushed to the area.

Strategy

Syria must strike quickly and precisely. A slow advance into Israel is prone to certain disaster as additional troops from the United States and her allies could swing the battle. The Israeli and U.S. air forces must be suppressed and denied air superiority.

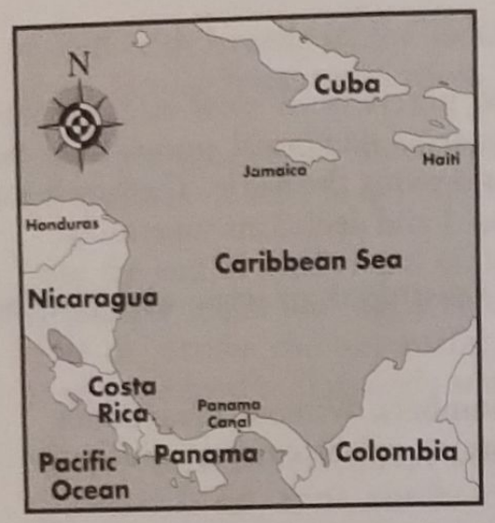
There are certain factors one must think of when encountering Israel and her allies:

- Syria has the strength in numbers. Without squandering troops, certain Israeli positions can be overrun with sufficient air support.
- Speed is of the essence. U.S. troops and equipment are sparse from the recent encounter with Iraq. Moving troops quickly and easily through enemy positions is vital.
- Jerusalem must be taken. If the city is brought back into Arab hands and fortified, Israel will be unlikely to attack the region for fear of heavy civilian casualties.

Campaign objectives

To achieve Assad's strategies and emerge victorious from this conflict, Syria must accomplish three vital objectives:

- Keep the skies clear of Israeli aircraft, especially around important troop convoy routes. Israel will assuredly try to deny ground access to the Golan Heights by destroying transportation rails and bridges. In addition, Israeli command centers in northern Israel must be eliminated.
- Inflict heavy casualties on major Israeli cities. The destruction and loss of life will be physically and mentally draining on the Israeli people and will divert Israeli fighters from the battlefront to help defend the cities.
- Repeatedly pound the U.S. and Israeli ground forces attempting to withstand the Syrian army. The success of these close air support missions is vital to a Syrian victory.



CUBA

The history of U.S. colonialism in Central and South America is rife with profiteering and subordination of local interests to the bottom line.

Since the Spanish-American War in 1898, the United States and its allies have had their hand in the Central American profit pie. Throughout the years, the U.S. has used such tactics as espionage, assassination, bribery and threats of nuclear annihilation to achieve its economic ends in the region.

In 1952, the democratic regime of former president Fulgencio Batista turned autocratic, as he seized control of Cuba through a ruthless and bloody coup. His newly-formed dictatorship was extremely brutal, violent and corrupt. For four years, the once-strong citizens of Cuba suffered and labored under the harsh confines of Batista's iron will.

There was a shining star to these four years of night. In 1956, a freedom fighter emerged from the gloom that settled over Cuba. His name was Fidel Castro. With his gritty and determined band of liberators, Castro began a guerrilla war against the Batista government that shook the nation.

Two years later, Castro and his ever-increasing popularity had the Batista regime on the run, and on January 1, 1959, Batista left the country. A little more than one month after that, Castro took the reins of power that he has retained ever since.

A number of attempts to overthrow Castro by the U.S. failed in the 1960s, including the mishandled Bay of Pigs invasion of 1961 and President Kennedy's international incident with the Cuban Missile Crisis of 1962. Cuba maintained cordial relations with the Soviet Union throughout that time.

In the early 20th century, the U.S. wanted to build a canal from the Atlantic to the Pacific. Colombia was approached with the idea, but their leadership rejected the plan. The United States wasn't going to take this sitting down, however, and found a group of Panamanian rebels willing to declare independence from Colombia and form their own nation. In 1903, they did

just that, and Colombia was prevented from retaking their lawful territory by a U.S. naval blockade.

As a final stab in the back of the Colombians, in 1978 the U.S. gave Panama all rights to the canal starting in 1999. It also declared the Canal Zone completely neutral.

Panama has been in bed with U.S. interests since the turn of the century, and nothing demonstrates this fact more than the illegal and immoral removal of General Manuel Noriega from Panamanian power. The U.S. invasion of the country, and once again interference with Central American politics, was a slap in the face to self-willed governments everywhere.

Common scuttlebutt is that the CIA and U.S. government officials felt that Noriega had outlived his usefulness as a puppet and needed to eliminate him before he could point the finger at other illegal U.S. activities. Since Noriega is quietly locked up in a United States prison, the verdict on these rumors is still out.

In 1994, a hastily-called Central American Summit is called to quench the infighting and re-establish the peace agreement called for in a 1987 accord. The Central American states, represented by a majority of the primary countries in the region, pass a military alliance treaty which calls for removal of all foreign military forces and to deny external interference in what are rightly Central American affairs.

Cuba, seeing the need for peace and stability in the region, joins the alliance, although it is blocked (primarily by Panama) from attending the summit. After consulting with Cuba on the terms of the treaty, Colombia also decides to participate in the alliance. Both are more than willing to provide troops to keep the peace.

Of the countries participating, only Honduras, El Salvador and Panama balk on signing the treaty. Once Cuba and Colombia join, Honduras and El Salvador see the light and move to ratify. This leaves Panama as the only dissenting nation.

President Endara of Panama refuses to join the alliance, as he fears U.S. repercussions. Panamanians break out in riots and looting, demanding Endara join the alliance. With the Orwellian head of the United States watching, President Endara imposes martial law, blaming the unrest on Colombian and Cuban espionage. Panama and the U.S. sign a "defense treaty" and America sends in advisors and elite shock troops to quell the riots.

Alliance troops move quickly to the Panamanian borders and call for the U.S. presence to cease and for the people of Panama to decide their own fate. The U.S. counters that the Colombian drug lords have duped the people into abandoning their long-standing democracy and refuses to leave, claiming Panama "needs U.S. help."

Cuba, at this time, has purchased a number of elite MiG-29 fighters from the former Soviet Union and is moving them into strike position if this conflict turns to war. Meanwhile, the Alliance has amassed nearly 100,000 troops in Costa Rica and Colombia and stands ready to liberate Panama.

The United States, on the other hand, is having its own problems throughout the world. Its own government sees the immorality of its actions and is reluctant to have its sons and daughters embroiled in a war it believes is unjust. The CIA's quasi-illegal activities are brought up as a reason for retreat, but the President is adamant about "defending Panama from invasion."

Almost 5,000 U.S. Marines are sent to the region along with the 8,000-man (U.S. trained) Panamanian Defense Force. In addition, there are a number of F-16s and other U.S. Air Force units in the region. B-52s from Texas can also be launched if need be.

The conflict is turning into a war of liberation; if the Alliance is successful and Panama is liberated, Central America will be finally free of foreign interference. If the Alliance fails, the dark shadow of Big Brother America will permeate the Central American skies for years to come.

Strategy

The Central American Alliance is in a strong military position, but its political position is somewhat shaky. World opinion has not swung one way or the other, but a long, protracted war will look bad for the Alliance. In addition, the people of Central America want an end to all bloodshed and may decide to withdraw their support of the Alliance if the conflict doesn't end soon. One final note: the United States is sending in reinforcements and, if the U.S. can hold on until then, the Alliance is doomed.

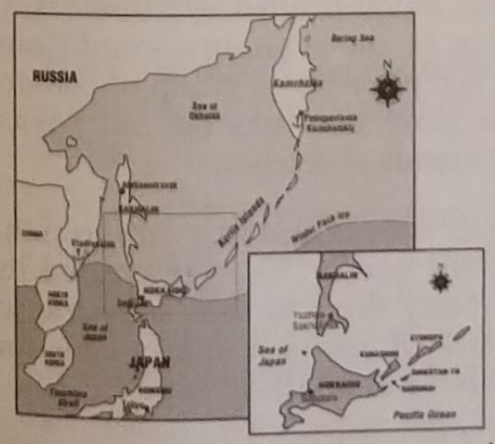
These three items must be considered when pursuing victory against the U.S.-Panamanian forces:

- Strike quickly and effectively. U.S. and Panamanian troops are outnumbered and, if they can be overrun swiftly, the Alliance has a greater chance of ultimate victory.
- Keep the skies clear of enemy aircraft. Cuban MiG-29s will be essential in pursuing this role.
- Limit civilian casualties on both sides. World opinion can be easily swayed and with the news media acting as a "global village," a bloody war will look bad for the Alliance.

Campaign objectives

The Alliance must overrun Panama and seize control of important command centers, while sustaining minimal casualties. They can accomplish this by achieving the following three objectives:

- Protect vital supply and reinforcement lines. The United States will most likely attempt to cut off supply routes by bombing roads, bridges, railyards, etc. Keeping these routes open will keep the ground pressure heavy on the Panamanians and their allies.
- Deny Panamanian control of the skies. Although the Alliance has the advantage of sheer aircraft numbers, the U.S. has sent in its ace pilots to Panama. Alliance MiG-29s and other aircraft are too valuable to be squandered carelessly and are nearly irreplaceable.
- Defend Alliance command posts and control centers. The elite military minds of Central America are in the field coordinating Alliance troop movements and air strikes. They must be protected at all costs.



RUSSIA

The defeat of Japan in World War II was a tremendous victory for Russia and her allies. The hostilities that Japan initiated on Mother Russia were not soon to be forgotten. Therefore, the homeland must be protected from further aggression at all costs. A major factor of this defense is the Kurile Island group.

Japan's imperialist practices are notorious through the past two decades. Not content to remain on its islands, Japan pushed forward, invading vast expanses of both China and Russia. Japan was only stopped by the powerful might of the Russian military during World War II.

After World War II, Russia was finally recognized as the legitimate owner of the Kurile Islands. At last, Russia could be safe from the imperialistic aspirations of the Japanese Empire.

However, Japan was not about to take this lying down. They realized the importance of the Kurile Islands to Russia and were constantly scheming to win them back somehow. The years after World War II were difficult for Japan. Demilitarization and rebuilding of their economic infrastructure left no room for their innate imperialism. They had to bide their time.

As the Soviet Union gained in military strength throughout the 1960s and 1970s, Japan feebly relied on the protection of others for its defense. Japan's capitalistic ways flourished in the 1980s and bolstered their economy to the point where they could afford to revamp their military. Finally, in the early 1990s, Japan saw their chance to retake the islands.

The collapse of the Soviet Union saw the Soviet military fragment into numerous parts. Without the Soviet military threat staring Japan in its face, Japan used its economic leverage in an attempt to ram through economic aid to the new Commonwealth with regaining the Kurile Islands as part of the deal.

The year is 2002. Russia, literally starving for economic relief, relinquished the four pivotal islands in the Kurile group to Japan in 1995 in exchange for much-needed economic aid. However, even with this influx in the economy, conditions worsen. The ineffective Yeltsin is replaced by a group of conservatives intent on restoring the glory that was the old Soviet Union. Their first item on their agenda is defense.

Conceding the four Kurile Islands is perceived as a serious military weakness by the new Russian leaders. The gap created by the loss of the islands allows enemy fleets to slide unrestricted into the Sea of Okhotsk. This lack of defense is inexcusable for the security of Russia and must be re-established.

Russia's leaders now have halted the withdrawal of their troops from the island group and have dug in for the inevitable battle.

Meanwhile, Japan has also established troops on the main island of Kunashiri and is beginning to bolster its presence on the islands.

The unforeseen factor is the United States. Russian spies in Washington have uncovered the fact the decision makers wish this conflict to remain a purely Japanese/Russian affair, but that the Pacific Commander has other ideas. He realizes the strategic importance of these four Kurile Islands and has been authorized to send U.S. squadrons to the area.

To counteract this, Russia has developed the Unmanned Fighter (UMF). The UMF is a highly maneuverable aircraft capable of incredible turn rate and high-G maneuvers. Russian leaders expect this aircraft to be the deciding factor in this conflict and lead their way to victory.

At the beginning of the campaign, Russia and Japan have had some minor skirmishes and air battles, with neither side gaining the upper hand as yet. The U.S. Air Force is stationed in the area and is primarily flying support for the Japanese. It is extremely important not to get the U.S. embroiled heavily in this conflict, so rules of engagement (see *Operation: Fighting Tiger,* page 74) apply when engaging U.S. aircraft. This includes all aircraft except the Japanese FSX and F-1. Of course, if they fire first....

Strategy

It is crucial to regain the control of these four disputed Kurile Islands. This must be accomplished by a combined land, air and sea attack on the Kuriles and the Japanese island of Hokkaido. Once one island is secure, the rest of the islands can be taken from there.

There are three key factors the Russian military must remember when facing the combined forces of Japan and the United States:

- Avoid U.S. intervention. The United States must be prevented at all costs from entering this conflict in a major way. This means strictly following the rules of engagement when encountering U.S. aircraft. All Japanese aircraft and ground troops, however, are fair game.
- Protect the Russian fleet in the Sea of Okhotsk. Because the Japanese realize an amphibious assault on the islands can be launched from the Russian fleet, they will be eager to

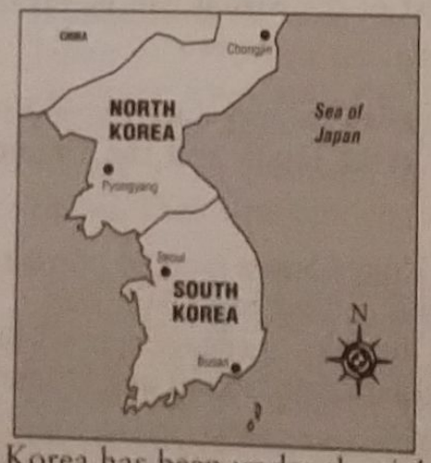
stop this threat. Your MiG-29 squadron may be called on to provide CAP in the skies over the fleet or for the landing craft on the islands.

- Use the highly advanced UMF to control the skies. The UMF is a key to air superiority in this conflict. Once the skies are clear of enemy aircraft, the Japanese ground forces can be pounded with impunity.

Campaign objectives

While there are many key things the Russians must do to defeat the Japanese, these three are the most important:

- Pound the Japanese ground forces on the Kurile Islands. Using heavy bombers and pinpoint surgical strikes, the Russians must neutralize the effectiveness of the Japanese army. To do so, the Russians must first achieve control of the skies.
- Support the Russian fleet while they launch amphibious attacks. The aircraft carrier and cruiser group will be a prime target for the Japanese air force and must be protected at all costs. Support for the Lebed hovercraft as they reach the shores is also essential.
- Inflict heavy damage on the Japanese island of Hokkaido. Much of the Japanese forces will be launched from this island, possibly leaving an opening for invasion. Regardless of whether an invasion is possible or feasible, severe losses must be delivered on the island to weaken Japan's strategic position.



NORTH KOREA

The people of Korea are tired of being a puppet for foreign governments. The time has come to rise up and assume our independence under one leadership. A unified Korea is within our grasp!

Throughout its modern history, Korea has been under the tight grip of foreign colonial imperialism. In the early 1900s, Japan invaded Korea and forced its

workers to produce food and industrial items for its missions of conquest. This grueling slavery lasted nearly 40 years, until Japan was defeated in World War II.

In the final weeks of World War II, the Russians swept south through Korea, liberating the northern area from Japanese occupation and establishing a border at the 38th parallel. Meanwhile, the United States flooded in from the south, forcing the Japanese to surrender. While this alleviated one form of oppression, it merely transferred control of Korea from one foreigner to another.

The Soviet Union treated the Koreans better than the Japanese did, giving North Korea an economic and military boost with the ultimate goal being an attempt to unify the country. Their chance came in 1949, when the protective blanket of the U.S. Army left South Korea. The time was ripe for attack.

In June 1950, that attack was realized. Armies in North Korea marched through the south, taking the capital of Seoul in less than three days. Little resistance was put up in the next two weeks as the North Korean army gained control of half the south.

Then, once again, outsiders interfered in Korean matters. The United Nations, fearing what a unified Korea would mean to Southeast Asia both militarily and politically, sent in U.S. troops to quell the fighting. Still the tide of unification swept onward, pushing nearly to the southern end of Korea.

A lapse in tactical judgment led to some U.S. troops invading the shores at Inchon, and they moved through the North to cut off crucial supply routes. The stricken North Koreans, deprived of their vital food and ammunition, panicked under poor leadership. The U.S. then initiated a counterattack from the south, and in less than two weeks the North was routed.

Meanwhile, another neighboring country wanted to carve a piece of the Korean pie. China, seizing the opportunity left by the void created by U.S. troops, used its massive numbers to march through North Korea and take Seoul. By this time, the strength and heart of the stalwart Koreans were shattered by the constant ravaging of their native lands.

The United States again moved in through the south to thwart the progress of the Chinese and hold them at the 38th parallel. Finally in 1953, after two years of sporadic fighting, the battles

ceased and the Demilitarized Zone (DMZ) was established along the 38th parallel.

Between the 1950s and early 1980s, military actions between the two Koreas remained at a virtual standstill. North Korea was ruled by an autocratic leader, while South Korea has had a number of different governments from democracy to military rule.

The 1980s saw an economic revival in South Korea, while North Korea began to receive less and less help from their biggest ally, the Soviet Union. With the ever-downward spiral of the Soviet Union's economy, North Korea was beginning to lose the economic and military edge it had in recent years. In addition, South Korea was beginning to become indoctrinated in Western ways and political leaders were succumbing to the capitalist machine.

The collapse of the Soviet Union and the loss of North Korea's biggest importer of high-tech weaponry and machinery forced North Korea's hand. Realizing that the brainwashed political leaders of the South will never agree to a rightful unification, the North unleashes a masterful plan in late 1994.

Under the guise of nationwide unification elections, North Korea offers to disband their forces along the DMZ. Meanwhile, the North trains insurgent commandos for raids at key military facilities and political headquarters. On election day, riots break out in the North as word spreads that South Korea is trying to fix the election. Within hours, an attack is launched by the commandos to remove the South Korean policy makers and military leaders duped by the lure of capitalism.

The attack is successful—many key figures are eliminated while strategic military facilities are destroyed. The "disbanded" forces at the DMZ rush over the border and in the confusion caused by the loss of communications centers, the North launches an overwhelming three-prong attack on the South.

North Korean armies sweep to the south, facing little resistance from overmatched Southern armies. Seoul is the next stop for the North. The only thing that threatens their defeat is the intervention of the United States and its substantial air power. If the skies can be controlled by the North, the flags of a unified Korea will be flying as far south as Busan.

Strategy

The forces of North Korea are strong and numerous but must be used judiciously to defeat the combined forces of South Korea and the United States. With very few advanced fighter aircraft, the North must be careful not to squander them needlessly.

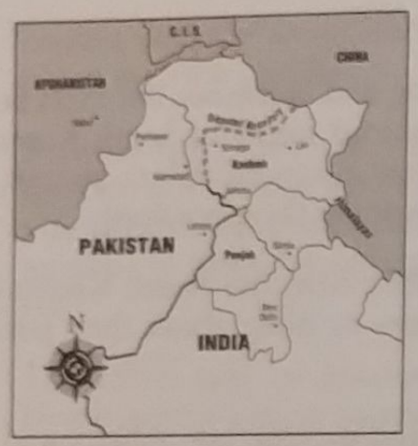
With the element of surprise on their side, the North Koreans must employ the following strategies to bring about their goal of unification:

- Accelerate the ground assault and overrun key military and civilian sites. The longer the combat troops are in the field, the more exposed they are to enemy attack.
- Bomb crucial factories and the battle area deep within South Korea. Destroying factories essential to the economy of the South will reduce their ability to wage war, while constant pounding of ground positions will dishearten their troops and soften them for the Northern charge.
- Protect the North's valuable strike aircraft. The combined air power of the United States and South Korea is more than a match for the North. MiG-29s must be used sparingly and only on crucial missions to avoid losing the skies.

Campaign objectives

In order to unify Korea under one leader and one government, the following objectives must be fulfilled:

- Defend North Korean command centers from air strikes by Southern aircraft. The North's superior radar defenses are essential to the defense of the area. Without them, the North is extremely vulnerable to surprise Southern air attacks.
- Support front-line troops with combat air patrol missions. The South must not be allowed to thwart the progress of the Northern army. Combat aircraft must clear the skies of Southern fighters and provide uninterrupted movement for the North Korean army.
- Protect valuable supply lines running to the front. Key to the success of the North Korean assault is the continuous operation of the supply lines from the north. If these lines are severed, Northern armies will be left out to dry and will surely be defeated.



INDIA

Since the formation of India as an independent nation in 1947, the area known as Kashmir has been hotly disputed between the nations of Pakistan and India.

When the British relinquished control of the Indian region, each of the provinces under their control was allowed to join the nation of India. All but one chose to join with a united India. The renegade maharaja of Kashmir, acting in his own self-interest without authority of his people, chose to create Kashmir as a separate nation.

This decision angered Kashmir's neighbor, Pakistan, which immediately rushed troops into the area to seize valuable land. India, responding to the needs of the Hindu population in the Jammu region, sent a number of defending forces in to avert the Pakistani onslaught.

The buildup of enemy forces led to a two-year war which was finally halted by the interference of the United Nations. The end result of this conflict was a shaky cease-fire and an unstable division of Kashmir into two sections: Pak-Occupied Kashmir in the northeast and the State of Jammu and Kashmir in the southwest. This unsteady partition held until the mid-1960s.

Pakistan invaded Kashmir again in 1965. Their successes were short-lived, as Indian forces halted the Pakistani troops and forced a stalemate after only 17 days. Popular opinion was obviously against Pakistan, since the United States—once Pakistan's closest ally—refused to help in this ill-conceived attack.

1971 saw East Pakistan, geographically separated from the rest of Pakistan by India, rise up in an attempt to assume independence. Pakistan then mobilized their army to stop the rebellion and regain control. The civil war that ensued caused the death of over 300,000 East Pakistanis.

Realizing East Pakistan's right to self-government, India committed heavily to the struggle. Fierce battles commenced, and while war casualties were low on both sides, the India/East Pakistan side was victorious. East Pakistan then became the independent nation of Bangladesh.

Since that time, there have been no large-scale hostilities between Pakistan and India. (Small skirmishes in the Himalayas between tiny branches of the two armed forces result in a constant stalemate.) However, Pakistan has been continually training and aiding a minor faction of Kashmiri guerrillas waging a brutal terrorist campaign against the government of India. The Jammu and Kashmir Liberation Front (JKLF) stages acts of murder and kidnapping to pressure India to relinquish Kashmir.

The 1980s saw Pakistan renew ties with the United States and devote nearly 50% of the Pakistan budget to building up the military. Concerned about another Pakistani invasion, India purchases sophisticated manufacturing facilities and weaponry from Europe and the Soviet Union, among which is the advanced MiG-29 fighter aircraft.

The unrest aided by Pakistan bears fruit in mid-1995 when JKLF rebels ambush and slaughter members of an Indian security force just north of Srinagar. The terrorists flee back into Pak-Occupied Kashmir, hotly pursued by Indian soldiers. Shortly after entering northeast Kashmir, the Indians encounter Pakistani regulars and open fighting breaks out.

Seeing the chance to crush the JKLF terrorists, India mobilizes its northern armies and assaults rebel positions beyond the cease-fire line with armored columns and air support. Pakistan quickly sends its regular army to engage Indian forces and the conflict intensifies, soon breaking out into full-scale war.

Pakistan has the early upper hand, being able to mobilize the majority of their troops in a short time. They are also receiving help from the United States military, who have sent aircraft, weapons and a carrier group to the region.

Since India can feel the hot breath of Burma and China on its back, it can ill afford to send its entire army to the Pakistani front. While India has the strength in numbers, it must slow the Pakistani assault and defend its lengthy borders until such time as it can muster additional troops from the south. If Pakistan can be delayed in its invasion, India can take the region by force, finally bringing Kashmir back into the fold of India's rightful rule.

Strategy

Because Pakistan's ground forces are already in the battle area and marching south, India's primary strategy should be to cor-

ral their progress and prevent them from acquiring as much territory as possible. The war must be drawn out enough so that Indian reinforcements can be brought in from the south.

In order to stop Pakistani aggression once and for all and assimilate Kashmir, India must utilize the following strategies:

- Dig in on the front line to prevent Pakistani advances. Entrenched positions and cover from artillery and air support should be enough to withstand a charge from the Pakistani army.
- Neutralize the effectiveness of Pakistan's armored divisions. Bombers and surgical strikes must be employed to slow the movement of Pakistan's tank columns. If the full force of the tanks reach the front lines before Indian reserves can be brought up, the battle could be over before it's begun.
- Protect advancing Indian troops and support teams. Reinforcements are being mustered from the south and sent out as quickly as possible, but Pakistan is on the watch for them. Use India's huge superiority of numbers in air defense weapons to deter and destroy any Pakistani threats.

Campaign objectives

To successfully complete the campaign and regain control of Kashmir, India must follow and attain three specific objectives:

- Eliminate Pakistan's key military production facilities and supply centers. Using India's strike aircraft, penetrate Pakistan's air defenses and destroy these essential centers. Without them, Pakistan's ability to make war will slow to a crawl.
- Deny Pakistani air superiority. Although United States F-16s are engaged in the conflict, Indian MiG-29s and Mirage 2000s must be used effectively to control the skies.
- Sever supply lines leading to the front. If their supply centers are destroyed and the supply routes cut, the Pakistani army will be locked down and unable to move. Then, when the Indian reinforcements arrive, the Pakistanis will be sitting ducks.

Hans Halberstadt/Arms Communications

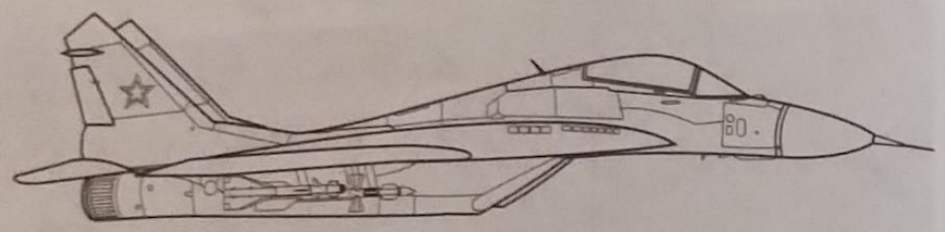
REFERENCE

СПЕЦИФИКАЦИЯ

7

СЕМЬ

TECHNICAL SPECIFICATIONS

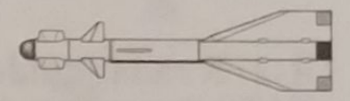


MiG-29 Fulcrum-A Specifications

Type:	All-weather counter-air fighter
Length:	56 ft 10 in
Wingspan:	37 ft 3 in
Wing area:	415 ft^2
Aspect ratio:	3.43
Combat weight:	28,000 lb
Maximum takeoff weight:	38,000 lb
Maximum payload:	8,500 lb
Engines:	Two Isotov RD-33 turbofans each rated at 18,300 lb static thrust using afterburner
Top speed:	Mach 1.06 (700 kts) at sea level, Mach 2.3 (1,320 kts) at height
Maximum climb:	65,000 ft/min
Ceiling:	56,000 ft
Instantaneous corner velocity (at 15,000 ft):	350 kts
Turn radius:	2,100 ft
Sustained corner velocity (at 15,000 ft):	435 kts
Turn radius:	3,800 ft
Maximum range:	1,130 nm
Combat radius:	400 nm
Fuel capacity:	1,153 U.S. gallons
Flight controls:	Hydraulic actuated system
Equipment:	Pulse doppler radar, IRST, laser range finder, SRZO-2 IFF, Sirena-3 360° radar warning system, HUD, cruciform brake chute and helmet-mounted sight.

AIR-TO-AIR WEAPONS

R-60 IR missile

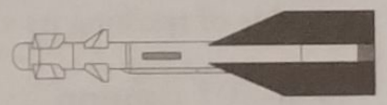


NATO designation: AA-8 Aphid

Range:	3.5 nm
Guidance:	All-aspect infrared
Speed:	Mach 2
Weight:	120 lb
Warhead:	18 lb blast fragmentation
Length:	6 ft 10 in

Until the development of the R-73, the Aphid was the top-of-the-line Russian heat-seeker. Although rated as an all-aspect weapon, combat experience has shown that it strikes most reliably from the rear quarter.

R-73 IR missile

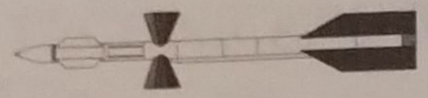


NATO designation: AA-11 Archer

Range:	5 nm
Guidance:	All-aspect infrared
Speed:	Mach 2.5
Weight:	275 lb
Warhead:	45 lb blast fragmentation
Length:	7 ft 4 in

Although its seeker technology is probably a little behind the latest American designs, the R-73 incorporates some features which surpass any other short-range missile in service. Foremost of these features is the close integration of the missile tracking system with the MiG-29's own IRST and radar. While Western fighters commonly slave the seeker head of an IR missile to the aircraft radar, the MiG-29's combination of radar, IRST and helmet-mounted sight allows the MiG pilot to fire at targets much farther off his nose than can his Western counterpart. In a close-in knife fight, this advantage can be decisive.

The R-73 design also employs a few other features to maximize the missile's capabilities. The seeker head has an exceptionally wide gimbal limit to allow it to follow the helmet-mounted sight beyond the limits of the HUD. Two sets of control fins, front and rear, combined with a vectored thrust exhaust nozzle make the R-73 highly maneuverable. When launched in a turn or at a high angle off the nose, the R-73 flight control circuits are fed an "english" bias to optimize the missile's pursuit course during the early, critical portions of flight.



R-27R radar-guided missile
NATO designation: AA-10 Alamo

Range:	55 nm
Guidance:	Active radar
Speed:	Mach 2
Weight:	550 lb
Warhead:	45 lb blast fragmentation
Length:	13 ft 2 in

The R-27 is the MiG-29's big stick. It is a large missile with long range, packing a powerful warhead. The missile has both proximity and impact fuzes. The available version of the R-27 has an active radar guidance package resistant to ECM and capable of tracking its victims against ground clutter.

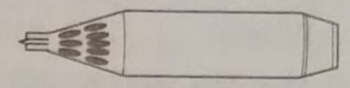
GSh-301 cannon

Gun weight:	110 lb
Magazine capacity:	150 rnds
Projectile weight:	.9 lb
Fire rate:	1,800 rnds/min
Muzzle velocity:	2,000 ft/sec

The MiG-29's GSh-301 cannon is the lightest gun of any fighter in service. The trade-off for this weight savings is a short barrel life (2,000 rounds). But then, barrel life should not be much of an issue. Because of the information provided by the laser range finder incorporated into the IRST, the gun sighting mechanism is extremely accurate. The gun system is so accurate that the aircraft's designers claim five rounds should be sufficient to destroy an aerial target (each round weighs almost a pound). Such accuracy also means that the 150 round magazine is not as small as it first appears.

AIR-TO-GROUND WEAPONS

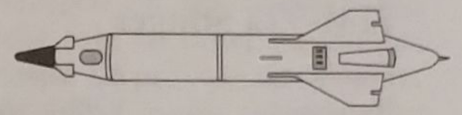
UB-32-57 unguided rocket pods with S-5 rocket



Rocket diameter:	57mm
Rocket weight:	8.8 lb
Rocket warhead:	1.1 lb high explosive
Rocket speed:	311 kts

Unguided rockets provide a method for putting controlled amounts of ordnance on the target. They are only effective against soft targets, but are useful for close air support missions against infantry units.

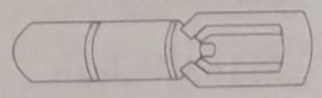
Kh-23 A-G missile
NATO designation: AS-7 Kerry



Range:	9 nm
Guidance:	Radio beam rider
Speed:	Mach 1
Weight:	770 lb
Warhead:	220 lb high explosive
Length:	11 ft 4 in

The Kh-23 provides the MiG-29 with precision ground attack capabilities for taking out enemy armor and hardened facilities. Because the missile follows a radio command beam to the point of impact, the target must be kept locked until the missile hits.

FAB-250 and -500 general purpose bombs



Weights: 550 and 1,100 lb

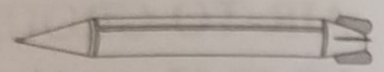
Fugasnaya awiatsyonna bomba, or general purpose bombs, can be used for virtually any air-to-ground mission. Pilot skill is extremely important in putting this ordnance on target.

FAB-250, high drag

Weight: 550 lb

A high-drag model of the FAB-250 is available for missions involving low-altitude bombing passes. After release, high drag fins pop open decelerating the bomb so that it falls significantly behind the aircraft. Without drag fins, a bomb dropped at low altitude and low dive angle could explode too close to the attacking aircraft or even ricochet off the ground.

RBK-500
cluster bomb dispenser

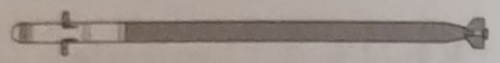


Weight: 950 lb

The RBK-500 consists of a casing containing about 200 2–3 lb bomblets. After release, the casing breaks apart. The bomblets spread as they fall to pepper a wide area with explosives. The RBK-500 is effective against personnel, lightly armored vehicles and other soft targets.

ENEMY ANTI-AIRCRAFT WEAPONS

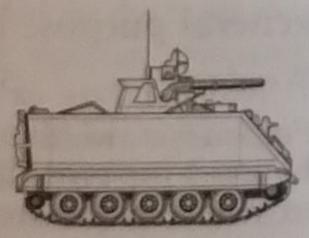
FIM-92a Stinger



Guidance: All-aspect infrared
Range: 1,980–16,500 ft
Altitude limit: 11–15,750 ft
Length: 5 ft
Weight: 35 lb
Warhead: 6.6 lb

The Stinger is a man-portable system commonly used by U.S.-supplied forces to provide air defense against low flying aircraft and helicopters. It is a very effective system capable of locking onto a target from any aspect angle. The Stinger is also rugged and easy to use. During the war in Afghanistan, U.S.-backed rebels used the Stinger to shoot down more than 100 aircraft and helicopters.

M163 Vulcan



Crew: 4
Armament: 6 barrel 20mm cannon in power turret
Gun range: 8,700 ft
Fire rate: 3,000 rnds/min in 10–100 round bursts
Traverse: 360°
Elevation: -5°–80°
Armor: 12–38mm
Length: 15 ft 11 in
Width: 9 ft 4 in
Height: 9 ft
Road speed: 42 mph

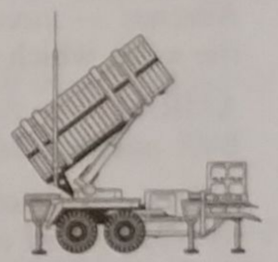
The M163 Vulcan must be considered mediocre compared to the ZSU-23-4, but anything that can put a concentrated quantity of lead in the air must be regarded as a threat. Still, it isn't much of a threat except at extremely close range. The U.S. intended to phase out the Vulcan, but the weapon remains in service because its replacement, the Sgt. York, was canceled.

Roland

Guidance: Radar or optical
Missile range: 5 nm
Warhead: 15 lb

The Roland SAM system was developed for the German and French armies, but has been widely sold to other nations as well. It is a highly mobile system used to provide air defenses for front-line troops and for high value, point targets. The Roland is an effective design, but suffers from a short range and a relatively small warhead.

Patriot



Guidance: Track via missile
Missile range: 30–44 nm
Warhead: 198 lb fragmentation

The Patriot missile must be considered the enemy's primary anti-aircraft threat. Designed with dual tracking and guidance channels, the system is very resistant to ECM and extremely accurate. The best defense against the Patriot is to fly low in order to minimize radar exposure.

GLOSSARY

AA-8 Aphid — NATO designation for the R-60 IR missile.

AA-10 Alamo — NATO designation for the R-27 radar missile.

AA-11 Archer — NATO designation for the R-73 IR missile.

ACM — Air Combat Maneuvers — dogfighting tactics involving more than two aircraft.

ACMI — Air Combat Maneuvering Instrumentation — a facility developed in *Falcon 3.0* for recording missions. Named for the U.S. Air Force's equipment for accomplishing the same feat in real aircraft.

ADI — Attitude Director Indicator — a cockpit instrument which displays an aircraft's pitch and bank angles relative to an artificial horizon.

AGL — Altitude above ground level.

Ailerons — aircraft control surfaces, located along the back of the wing, which produce rolling moment.

AMRAAM — Advanced Medium Range Air-to-Air Missile — a BVR missile carried by the F-16 Fighting Falcon.

AOA — Angle of Attack — the angle between an aircraft's velocity vector and a reference line pointing straight ahead along the fuselage.

AS-7 Kerry — NATO designation for the Kh-23 A-G missile.

Aspect Angle — the angle formed by a line running directly to another aircraft and a line parallel to the other aircraft's longitudinal axis.

Attitude — the state of an aircraft in terms of pitch, bank and yaw.

Bank — the angle of the wings about the longitudinal axis referenced to horizontal.

Beaming — flying perpendicular to the emissions from a threat radar.

Bracket — an ACM tactic in which two (or more) aircraft fly to opposite sides of a threat formation.

BVR — Beyond Visual Range.

Chaff — packets of foil used to decoy or obscure radar systems.

Concussive — with violent, if not explosive, force.

Corner velocity — the velocity at which an aircraft achieves its best turn performance.

Drogue chute — a parachute released to slow the MiG-29, usually when landing.

Element — two aircraft working together as a team, possibly as an element of a larger flight.

Elevators — aircraft control surfaces, located at the back of the horizontal stabilizers, which provide pitching moment.

FAB — Fugasnaya Awiatsyonna Bomba — Russian designation for "general purpose bomb."

FBW — Fly-by-wire — a flight control system which transmits flight commands via wires to servo actuators which drive the ailerons, rudders or other control surfaces. In most FBW systems, pilot input is processed and possibly negated by a flight computer before being sent to the control actuators.

FCS — Flight Control System. See also *HOTAS*.

Flaps — aircraft control surfaces, located at the rear of the wing, which alter the wing profile to provide greater lift at lower speeds.

Fuze — a device implanted in a missile, bomb or other piece of ordnance which triggers the actual explosion.

G — the force of gravity.

HOTAS — Hands On Throttle And Stick — a control setup which allows a pilot to perform all actions necessary to dogfighting without removing his hands from the stick or throttle. To achieve this with a computer simulation requires a more sophisticated input setup than a two-button joystick.

HSI — Horizontal Situation Indicator — a cockpit indicator which combines a compass with information from an inertial reference system or navigation beacons to indicate relationship of the aircraft with the planned course.

HUD — Head Up Display.

IFF — Identify Friend or Foe.

Indicated Airspeed — the airspeed shown by an airspeed indicator and not corrected for error due to air density variations caused by altitude and temperature.

IRST — Infrared Search and Track — a MiG-29 sensor system capable of tracking other aircraft using the heat generated by engines and surface friction.

Kh-23 — a radio-command A-G missile.

Knots — nautical miles per hour.

Load factor — the force acting on an aircraft as a multiple of the force of gravity.

Mach — a velocity given as a multiple of the velocity of sound under the same atmospheric conditions.

MiG — Mikoyan Gurevich — the founders of one of four major Russian aircraft design bureaus.

NATO — North Atlantic Treaty Organization.

Nautical mile — 6,076 ft.

NOE — Nap Of the Earth — very low altitude.

OKB — Russian abbreviation for "Aircraft Design Bureau."

Ordnance — bombs, missiles, bullets and other offensive hardware.

Pincer — see *Bracket*.

PIO — Pilot Induced Oscillation — an oscillating perturbation in an aircraft's flight path or attitude caused by a pilot failing to compensate for the lag time between pilot input and aircraft reaction.

Pipper — a small dot at the center of the aiming reticle.

Pitch — the angle of an aircraft's nose above or below the horizon.

R-27 — an advanced Russian A-A radar-guided missile.

R-60 — a second-generation Russian heat-seeking missile.

R-73 — a third-generation Russian heat-seeking missile.

Redout — the experience of pulling too many negative Gs and having the blood pressure in your head increase tremendously, coloring all objects red.

ROE — Rules Of Engagement — rules governing the conditions under which a fighter can engage or fire upon a bogey.

Roll — rotation around an aircraft's longitudinal axis.

RSBN — a radio navigation beacon.

SA — Situational Awareness — the amount of awareness a pilot has about the tactical environment around him.

Sideslip — motion of an aircraft to the right or left perpendicular to its longitudinal axis.

Six o'clock — directly behind, a fighter's most vulnerable area.

Stall — an abrupt loss of lift occurring when the AOA increases to the point that the airstream separates from the wing surface.

Statute mile — 5,280 ft.

Transponder — a combined radio transmitter and receiver, which transmits a signal when it receives one. Commonly used for IFF systems.

Trim — setting aircraft controls or trim devices so that the aircraft maintains a desired attitude.

Trim tab — a small control surface attached to an aileron or other larger control surface for the purpose of making small trim corrections to that surface's position.

True airspeed — indicated airspeed corrected for installation, compressibility and air density errors.

Vertical velocity — the sink or climb rate of an aircraft.

WCS — Weapon Control System. See also *HOTAS*.

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"Mikoyan MiG-29 'Fulcrum.'" Jon Lake. *World Air Power Journal*, 4 (Winter 1990/91), pp. 44–91.

In-depth article with many nice color photos.

OKB MiG: A History of the Design Bureau and its Aircraft. Piotr Butowski and Jay Miller. Stillwater, MN: Specialty Press, 1991.

A survey of Mikoyan-Gurevich's products up to the MiG-31. Many interesting, albeit black-and-white, photos.

A Quick & Dirty Guide to War. James Dunnigan and Austin Bay. New York, NY: William Morrow and Company, Inc., 1991.

A fairly accurate and thorough guide to current and likely conflicts around the world.

The World's Great Interceptor Aircraft. ed. Chris Marshall. New York, NY: Gallery Books, 1989.

Published before much definitive information was available about the MiG-29. Still provides a lot of detail, however, and covers many aircraft besides the MiG-29.

When creating this product, there were a number of decisions to be made. How best to model the Fulcrum? Metric or English measurements? Russian or American gauges? What keyboard commands to add, and which to remove? And so on. In the end, our desire for realism yielded only to ease of play and verifiable research.

We have tried to make our simulators as complex and flexible as the real aircraft. Ideally, a desktop pilot will find our simulators as challenging as a real jet jockey finds his aircraft. But if the simulator is harder to fly than the real plane, something's wrong. In such cases, adjustments were made to compensate for the differences between our customers and real pilots.

For example, Russian aircraft instruments use the metric system. Then again, so do Russian pilots. *MiG-29* uses English measurements because that's what our customers use. The added difficulty of using the metric system is unnecessary. Likewise, adjustments were made to the flight model and blackout model due to the fact that a simulation pilot can't feel the Gs and other sensations which warn a real pilot of dangerous flight conditions.

Another problem with doing a simulation based on a Russian plane is obtaining verifiable research. In some cases, we simply had no information about a given part of the aircraft. A case in point is the American-style horizontal situation indicator (HSI). The navigational display in the MiG-29 has a similar function, but though we had photos of the instrument, we couldn't get enough information on how it worked. Rather than just invent an instrument, we adapted an HSI used in many American aircraft.

All in all, we feel that this simulator is to *Falcon 3.0* what the *MiG-29* is to the F-16: well-armed, slightly more agile, more work to fly, and definitely an adversary to watch out for.

THE MIG-29 TEAM

Front: *Anthony Chiang, Dan Deremiah, Gary Stottlemyer, Michael Mancuso, Larry Webber, Carrie Galbraith*. Back: *Kus Pranawahadi, Kurt Boutin, Lawrence Kevin, Derwin Wyatt, Steve Blankenship, Erol Otus, Robert Cieri, Joe Ward, Kyle Brink, Ray Gresko, Robert Giedt, Russel Reiss, Peter Ward, Evan Birkby.*

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